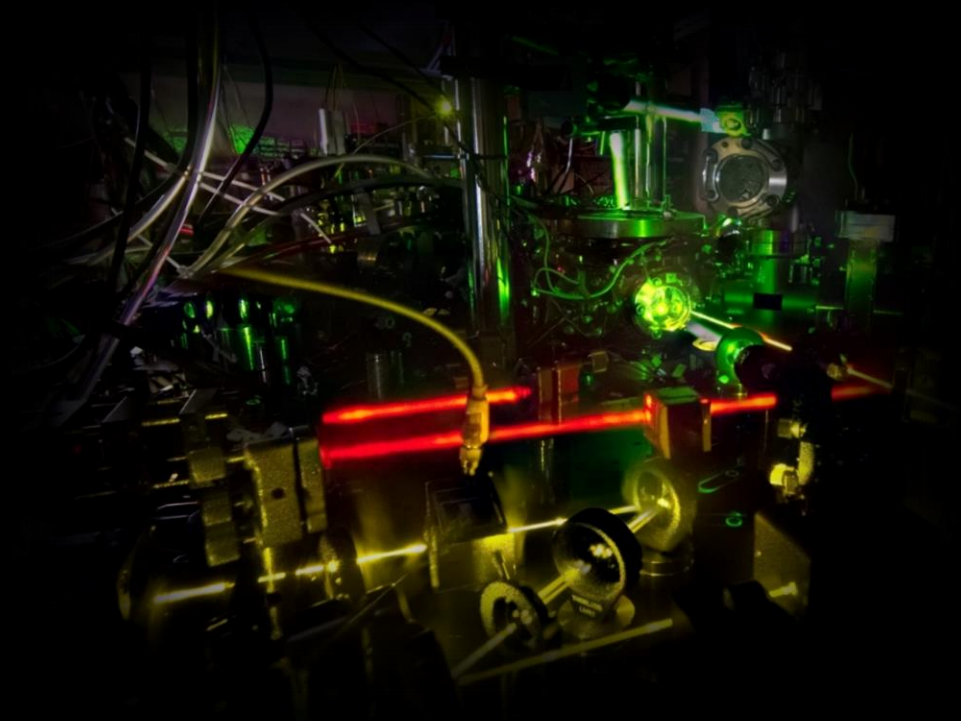


Atomic clocks

Navigating the precision frontier in atomic physics

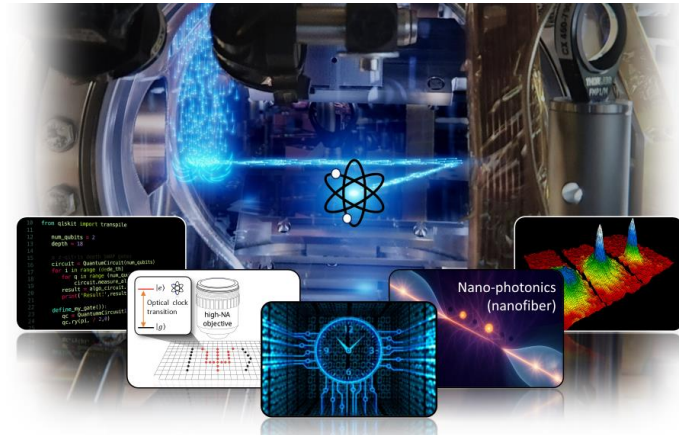


Chun-Chia Chen (陳俊嘉)

Quantum Technology Based on Atomic, Molecular, and Optical Systems

2025/ 0930

徵才



中研院(台大校本部) 原子與分子科學所 原子-奈米光子複合量子技術實驗室 / 量子精密感測實驗室 誠徵博士生，碩士生，大學部專題生，研究助理

我們的團隊專注於為中性原子（鹼土金屬原子）開發創新的原子控制技術，目標是實現下一代代的量子科技應用，例如量子模擬與量子資訊處理的實驗平台與次世代超精密原子鐘

聯絡資訊：

聯絡人：陳俊嘉 博士 / 肖恩 博士

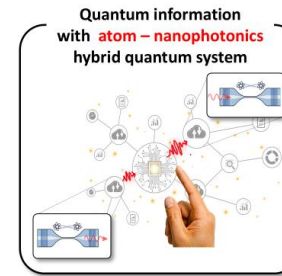
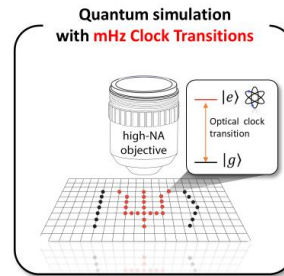
地址：中央研究院 原子與分子科學研究所110實驗室（台大校園內）

Email：chunchia.chen@g.iam.s.sinica.edu.tw / s.p.bennetts@g.iam.s.sinica.edu.tw

Website：https://iamsquantum.github.io/



SCAN ME



實驗工作內容：（許多工作內容具高度研究性，未來可銜接碩士或博士學位進修發展）

- 參與中性原子量子控制實驗與高穩定度雷射技術的開發
- 設計搭建原子光譜光學架設及雷射穩頻系統
- 搭建超高真空（UHV）實驗系統
- 開發控制系統與資料擷取系統（包含電子電路設計，電子儀器製造）
- 奈米光纖製備與測試

應徵資格：

1. 對於實驗物理技術開發與量子科技應用具高度興趣與研究熱忱
2. 對動手做、解決問題有興趣，具團隊合作精神、積極主動之工作態度
3. 歡迎有志留學或攻讀碩士學位者加入，累積實驗經驗與研究實績

工作地點：中央研究院原子與分子科學研究所 Lab 110（位於台大校園內）



Quantum Metrology Lab



Hybrid Atom-Nanophotonics Lab

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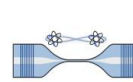
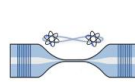
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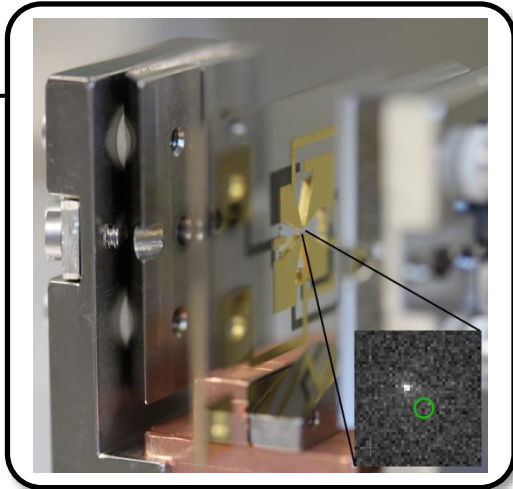
Institute of Atomic and Molecular Sciences, Academia Sinica

Tel: +886-2-2366-8236 | Address: No. 1, Roosevelt Rd. Sec. 4, Taipei 10617, Taiwan

Outline

- Introduction
- Two-level system
 - Atom light interaction
 - Highly accurate two level systems – mHz forbidden optical transitions
- How atomic clock works:
 - Cs Atomic frequency standards (atomic fountain)
 - Physics of optical atomic clocks – (optical lattice clock)
- Current research directions
 - How to improve state of the art atomic clocks (optical lattice clock) ?
 - Next generation optical clocks ?

離子阱原子鐘 vs. 量子電腦



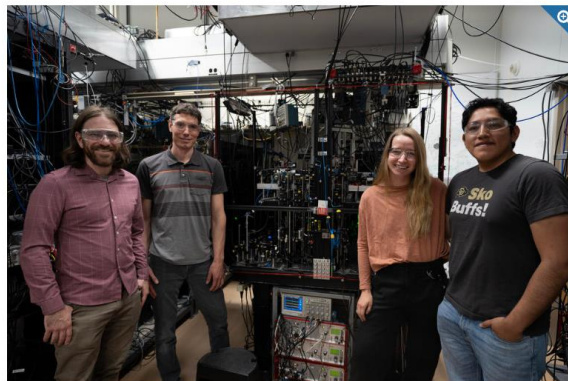
July 14, 2025

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NIST Ion Clock Sets New Record for Most Accurate Clock in the World

July 14, 2025

- NIST researchers have made the most accurate atomic clock to date — one that can measure time down to the 19th decimal place.
- This “quantum logic clock,” under continuous development for 20 years, relies on quantum computing techniques that pair an electrically charged aluminum atom (ion) with a magnesium ion.
- This new result contributes to the international effort to define the second with a much greater level of accuracy than before, enabling new scientific and technological advances.



(From left to right) Mason Marshall, David Hume, Willia Arthur-Dewar and Daniel Rodriguez Castillo stand in front of the aluminum ion clock at NIST. With its recent improvements, the clock can pave the way for the campaign to redefine the second as well as explore new ideas in physics.
Credit: R. Jacobson/NIST

MEDIA CONTACT

Rebecca Jacobson
rebecca.jacobson@nist.gov
(303) 497-4789

ORGANIZATIONS

Physical Measurement Laboratory
Time and Frequency Division
Ion Storage Group

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Quantinuum's private placement, led by [JPMorgan Chase & Co.](#), significantly exceeds the previously reported \$400M. Confirmed investors include [Quanta Computer Inc.](#) 廣達電腦, [Honeywell](#) Cambridge Quantum, [Amgen Mitsui & Co., Ltd.](#) [IBM](#) and [JSR Micro, Inc.](#) with [NVIDIA](#) potentially participating.

中性原子光晶格鐘 vs. 量子電腦

The Asahi Shimbun > Sci & Tech > article

Shimadzu's super accurate clock eyed for quake prediction

By YOKO HIBINO / Staff Writer

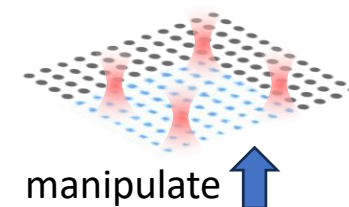
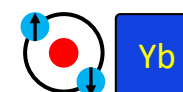
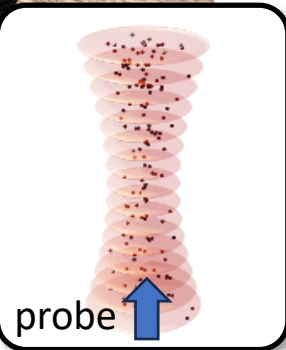
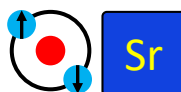
April 6, 2025 at 07:00 JST



日本島津製作所



Aether Clock, a compact optical lattice clock developed by Shimadzu Corp., is shown at the Seika, Kyoto Prefecture. (Yoko Hibino)



Denmark's QuNorth to Acquire 50-Logical-Qubit Magne Quantum Computer from Atom Computing and Microsoft

July 17, 2025

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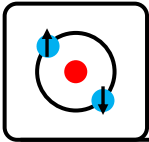
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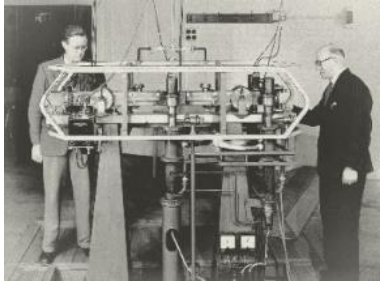
> D-Fine and Planqc Consortium



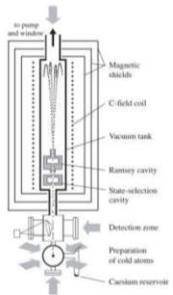
EIFO, the export and investment fund of Denmark, and the Novo Nordisk Foundation, an independent Danish enterprise foundation, have announced €80 million (\$92.7 million USD) investment to establish QuNorth. This new Nordic quantum initiative aims to acquire and operate a quantum computer named Magne, provided by Atom Computing and Microsoft. Magne is projected to be the first commercially available Level 2 quantum computer when it is ready for use by the end of 2026.



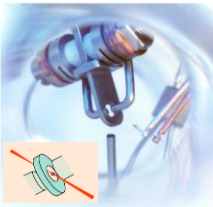
Defining moments in atomic timekeeping



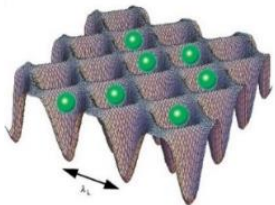
1955 First **caesium atomic clock**



1993 First **caesium-fountain clock**



1999 First optical-frequency measurement with femtosecond combs



2005 First **optical lattice clock**

19XX **Rabi**

1949 **Ramsey's** separated oscillatory field technique

1960 **Hydrogen** maser

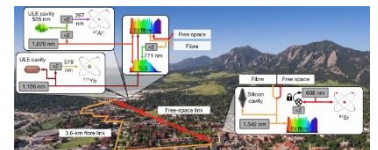
1975 Proposals for laser cooling of atoms and ions

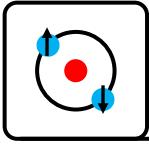
1978 Laser cooling of trapped ions

1985 Laser cooling of atoms

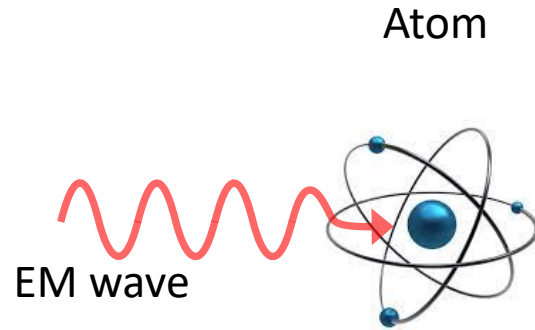
2001 **Concept of an optical clock demonstrated**

2021 Frequency ratio measurements at 18-digit accuracy using an optical clock network

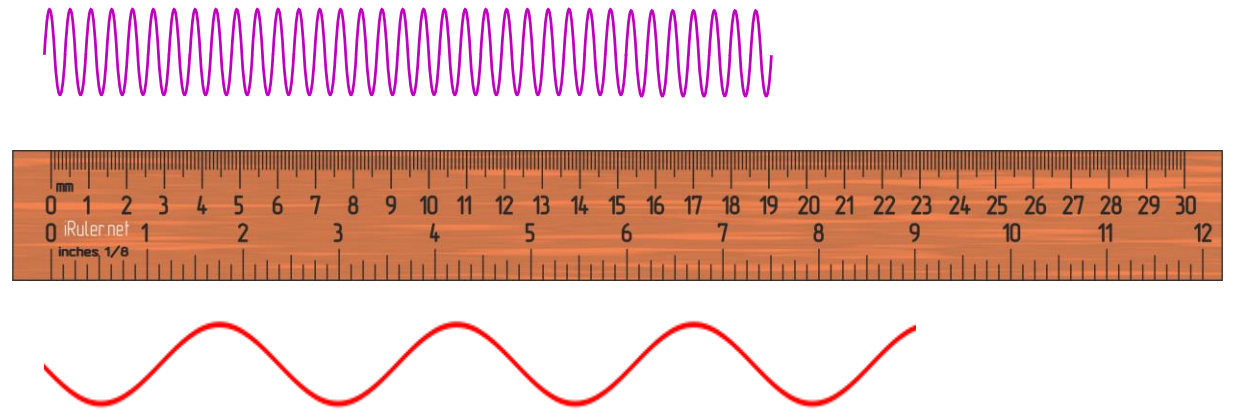
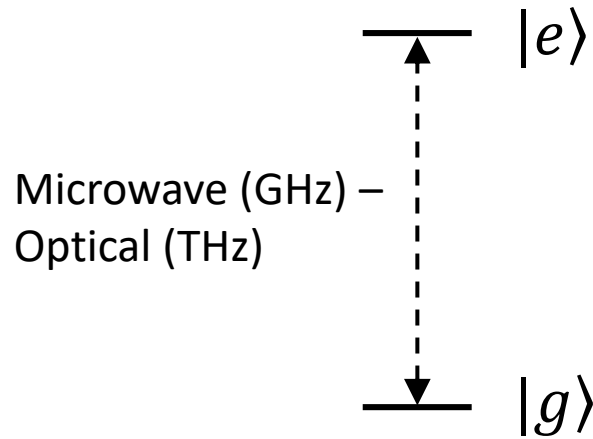


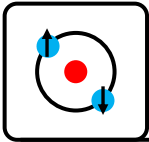


Atomic frequency standards: RF vs. Optical Frequencies

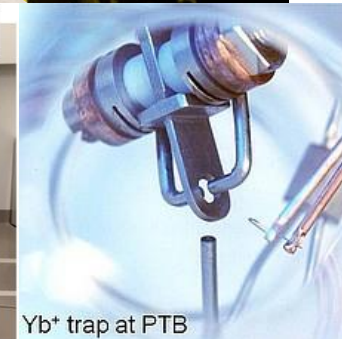
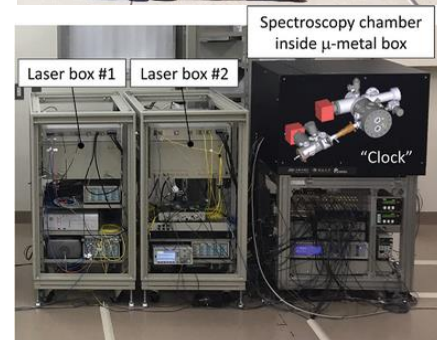
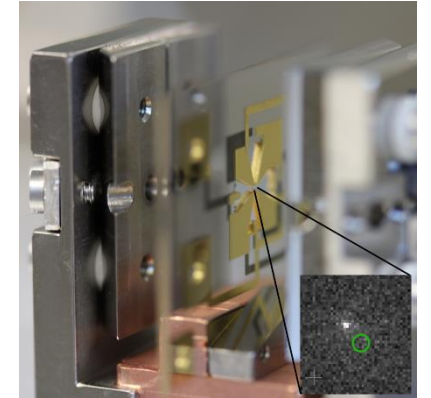
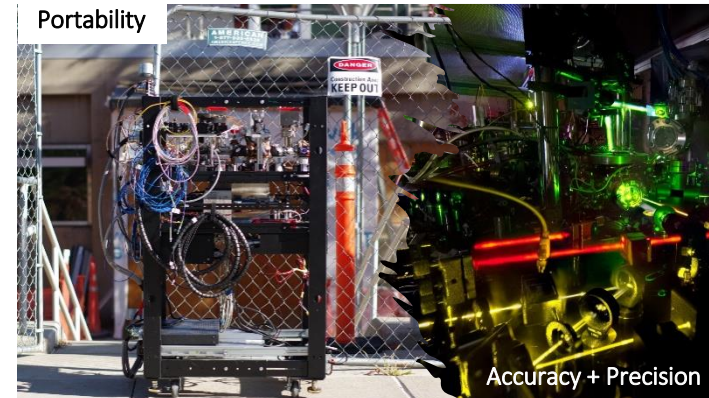


- High operating frequency : **optical** frequency
- Well-defined transition frequency
- States that have **low sensitivity** to and/or can be isolated from the environment



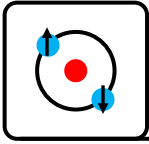


Atomic frequency standards: RF vs. Optical Frequencies



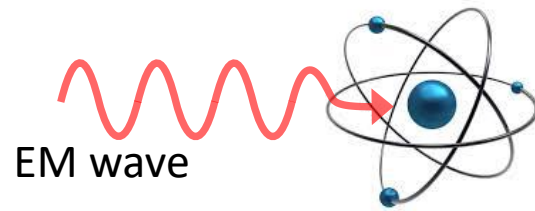
Cs: 9192631770Hz

Yb $^1S_0 - ^3P_0$: 518 295 836 590 863.71(11) Hz
Sr $^1S_0 - ^3P_0$ / NIST Al+ $^1S_0 - ^3P_0$ / PTB YB+ $^1S_0 - ^3P_0$

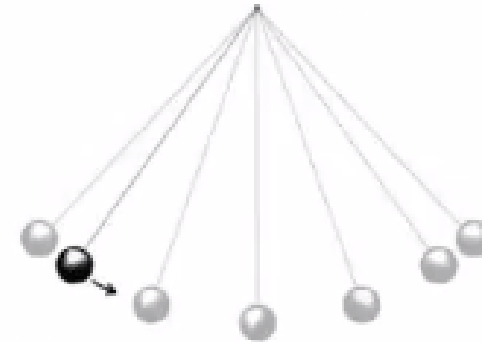
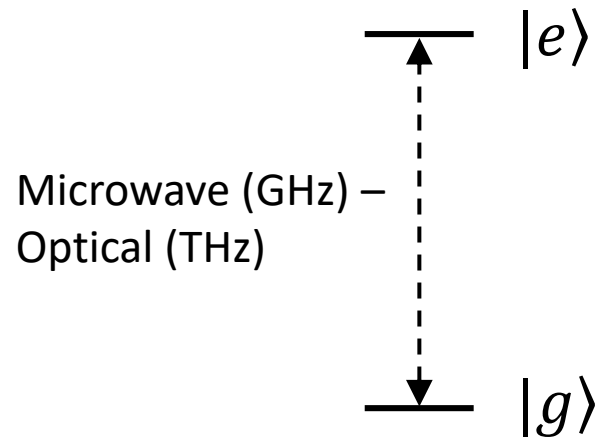


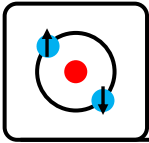
Atomic frequency standards: Atomic clock

Atom



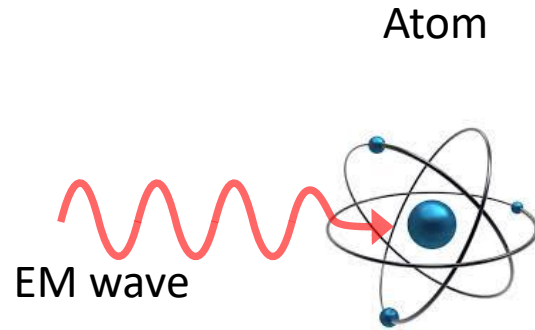
Clock



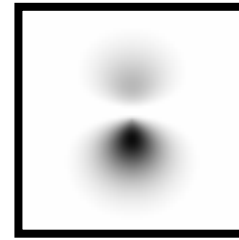


Atomic frequency standards

Atomic energy state **superposition** act like pendulum clock



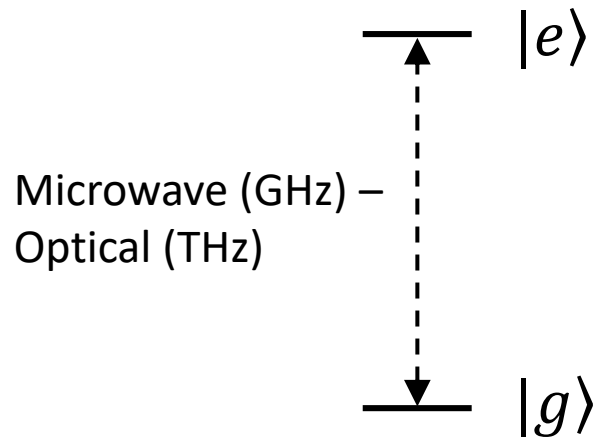
— $|2\rangle$



$$\psi = |1\rangle + e^{-i2\pi f_0 t} |2\rangle$$

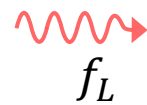
— $|1\rangle$

$$f_0 = \left(\frac{E_2 - E_1}{h} \right)$$



Practical mode of operation

— $|2\rangle$



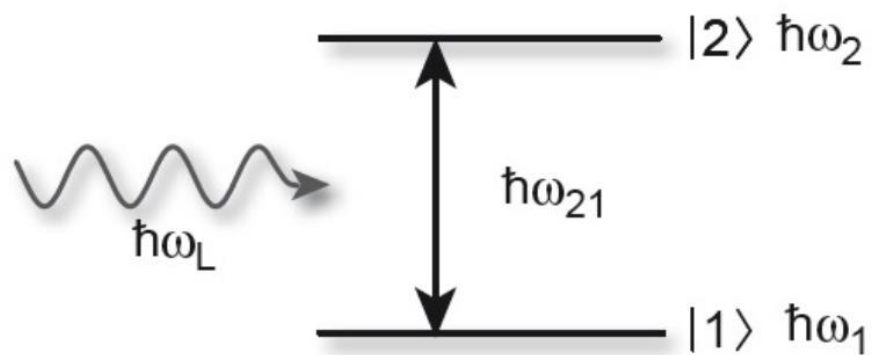
— $|1\rangle$

1. Start in $|1\rangle$

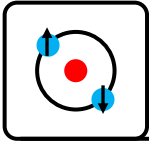
2. Apply radiation at frequency f_L for short time

3. Measure probability of $|2\rangle$

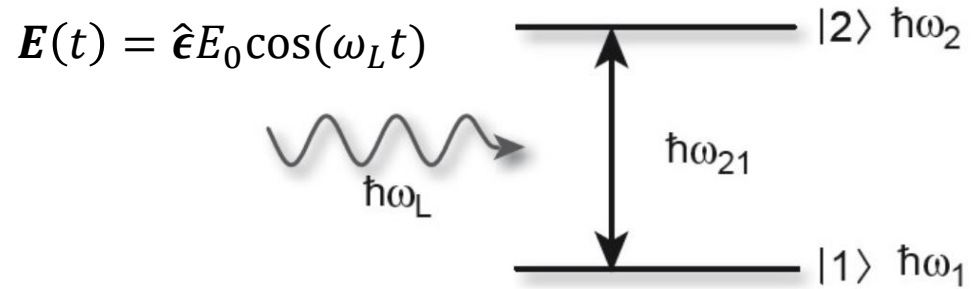
maximum $|1\rangle \rightarrow |2\rangle$ transition probability when $f_L = f_0$



Light atom interaction



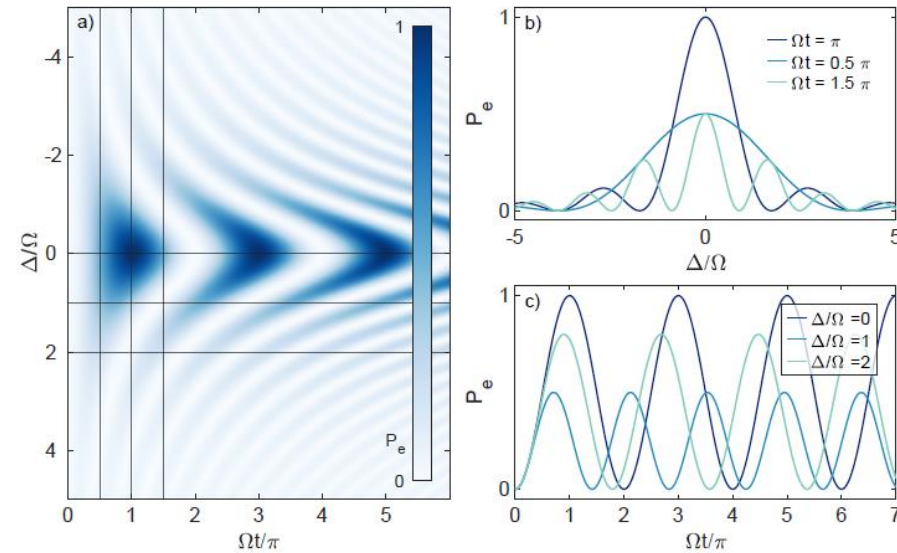
Light atom interaction



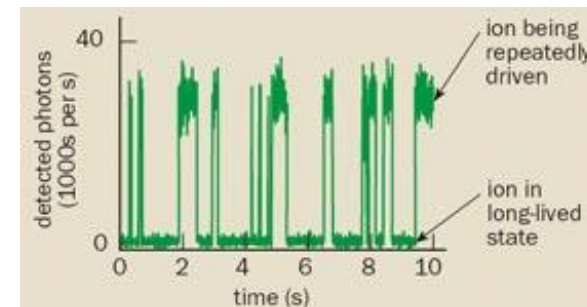
$$H = H_{at} + V(t)$$

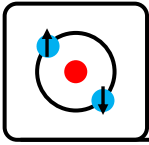
Total Atom Light Hamiltonian

$$V = -e\vec{r} \cdot \mathbf{E}(t) = \epsilon E_0 \cos(\omega_L t)$$

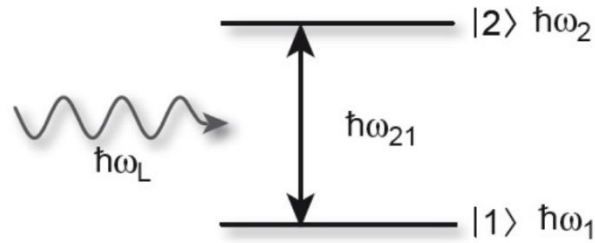


Type equ





Light atom interaction – derivation - 1



$$\Omega \equiv \frac{-eE_0}{\hbar} \langle e|r|g \rangle$$

- Effective Rabi frequency:

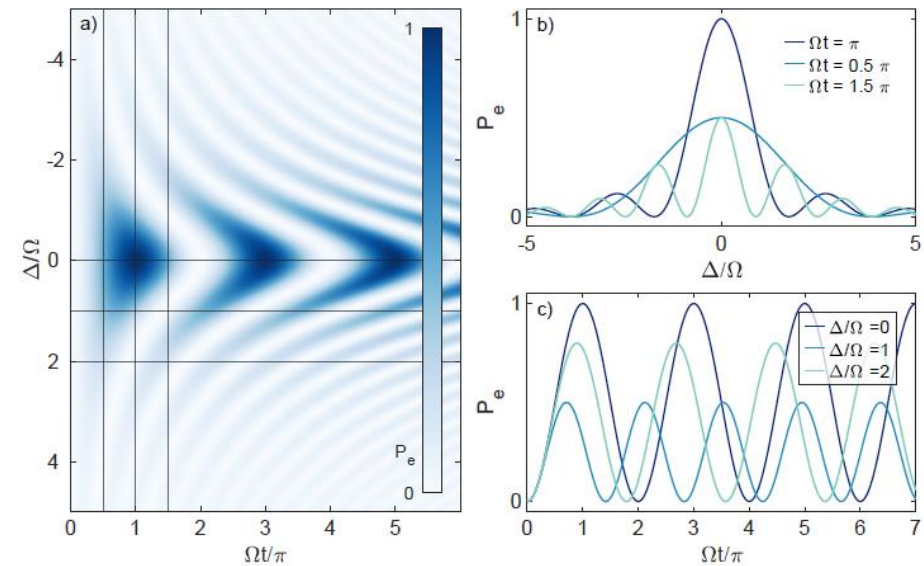
$$\Omega' = \sqrt{\Omega^2 + \Delta^2}$$

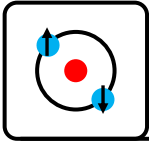
- Excited-state probability:

$$P_e(t) = \frac{\Omega^2}{\Omega'^2} \sin^2\left(\frac{\Omega' t}{2}\right)$$

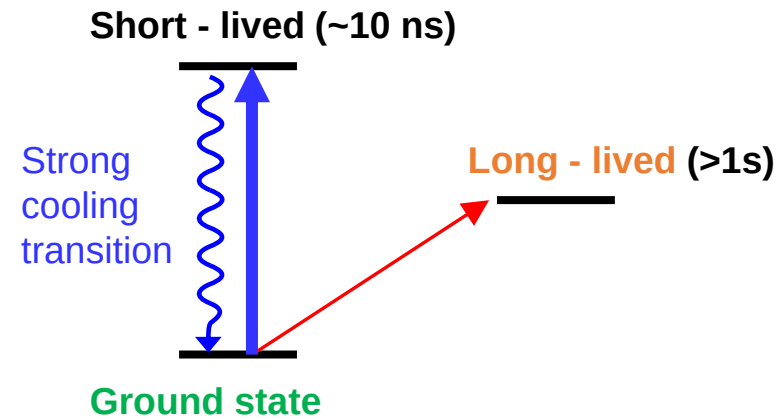
On resonance ($\Delta=0$): full oscillations at frequency Ω

Off resonance ($\Delta \neq 0$): faster oscillations ($\Omega' > \Omega$), reduced amplitude (Ω^2/Ω'^2)

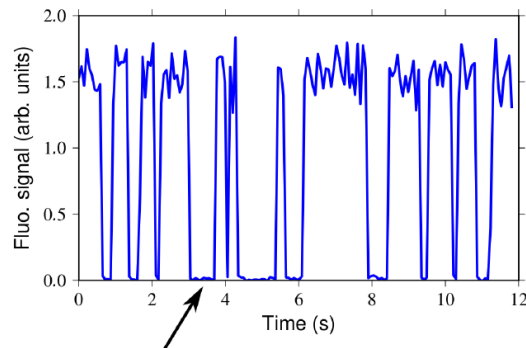




Quantum state of a single particle – quantum jumps



Quantum Jumps



Long-lived state
decay time

Physical Review Letters 1986

Shelved optical electron amplifier: Observation of quantum jumps

Warren Nagourney, Jon Sandberg, and Hans Dehmelt
Phys. Rev. Lett. **56**, 2797 (1986) – Published 30 June 1986
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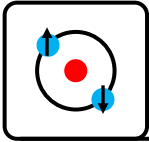
Observation of Quantum Jumps in a Single Atom

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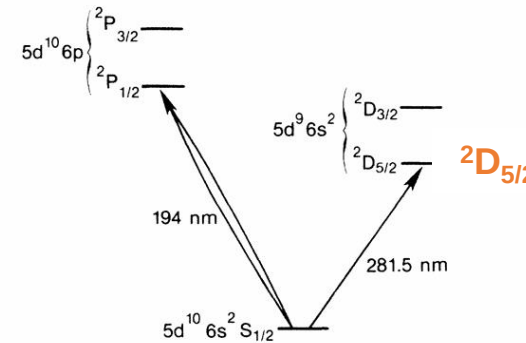
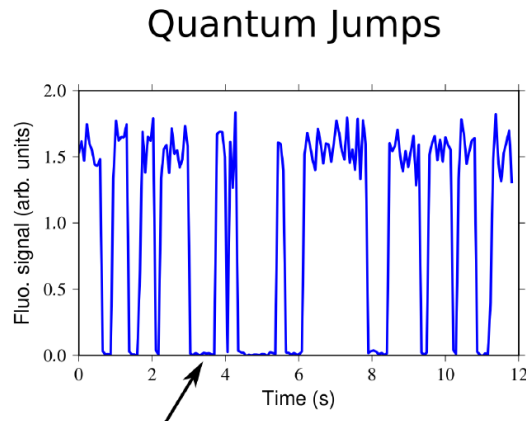
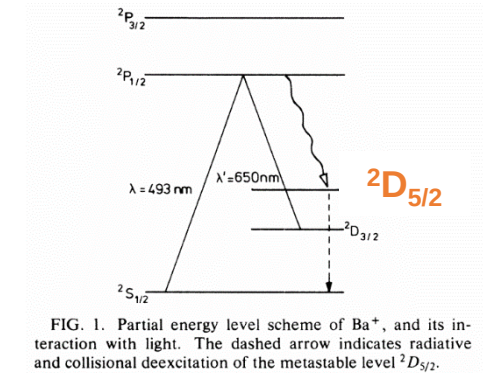
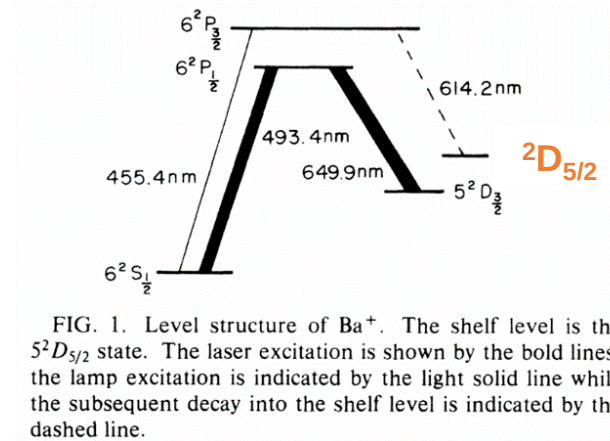
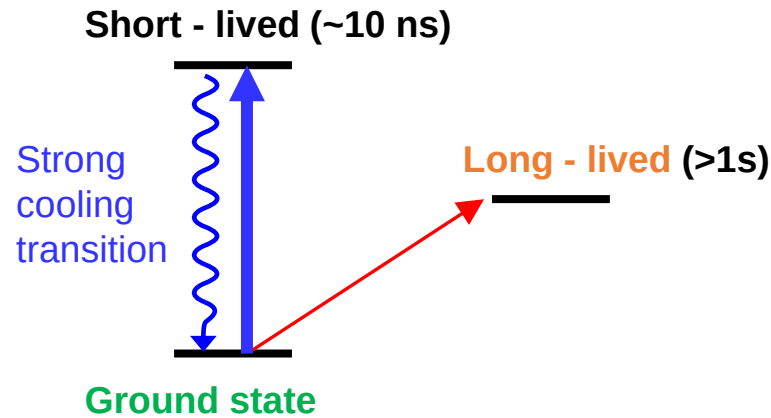


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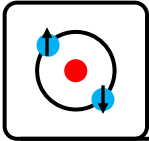
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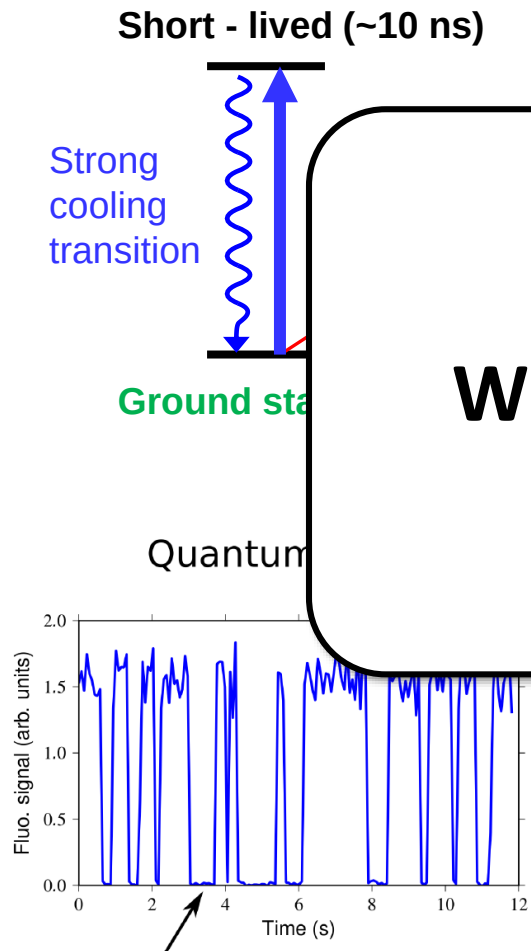
Quantum jumps - Key Experimental Demonstrations



Nagourney, Sandberg & Dehmelt (1986), *Shelved optical electron amplifier: Observation of quantum jumps*. *Phys. Rev. Lett.* 56, 2797–2799 (1986).
 Sauter, Neuhauser, Blatt & Toschek (1986), *Observation of quantum jumps*. *Phys. Rev. Lett.* 57, 1696–1699 (1986).
 Bergquist, Hulet, Itano & Wineland (1986), *Observation of quantum jumps in a single atom*. *Phys. Rev. Lett.* 57, 1699–1702 (1986).



Quantum jumps - Key Experimental Demonstrations



Why these states are long-lived?

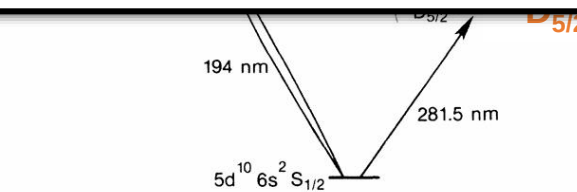
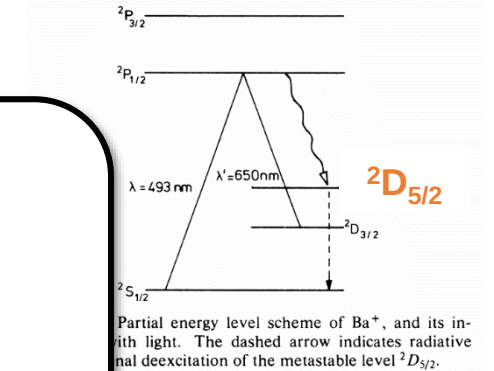
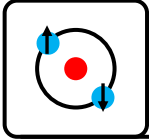


FIG. 1. Simplified optical energy-level diagram for Hg II.

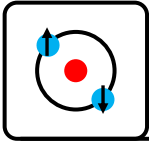
Long-lived state
decay time

Nagourney, Sandberg & Dehmelt (1986), *Shelved optical electron amplifier: Observation of quantum jumps*. *Phys. Rev. Lett.* 56, 2797–2799 (1986).
Sauter, Neuhauser, Blatt & Toschek (1986), *Observation of quantum jumps*. *Phys. Rev. Lett.* 57, 1696–1699 (1986).
Bergquist, Hulet, Itano & Wineland (1986), *Observation of quantum jumps in a single atom*. *Phys. Rev. Lett.* 57, 1699–1702 (1986).

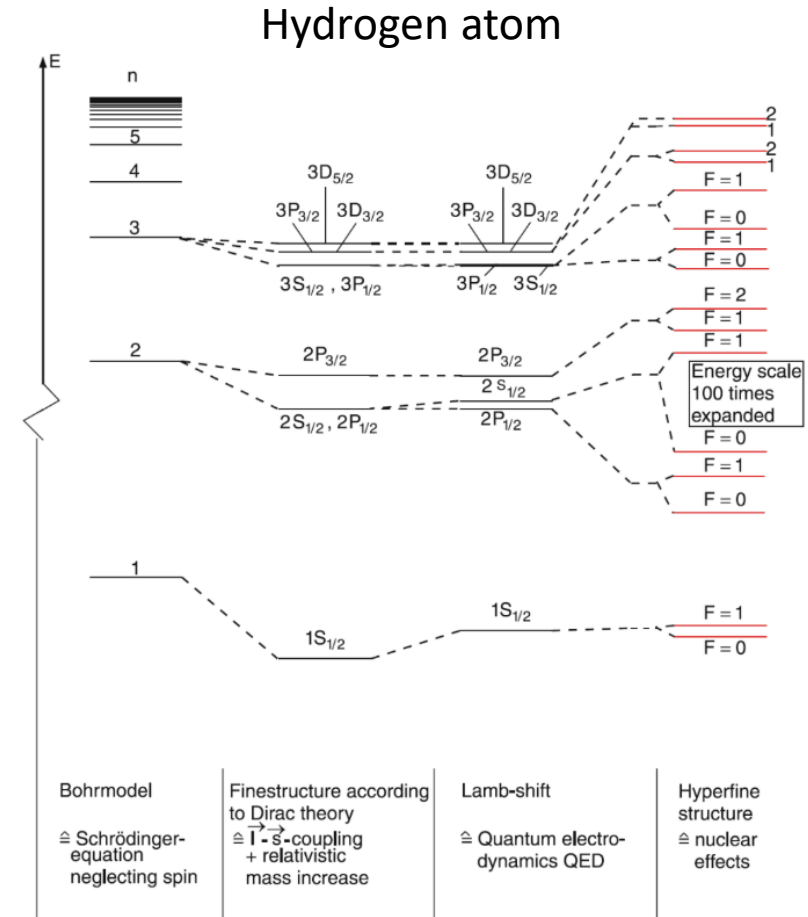
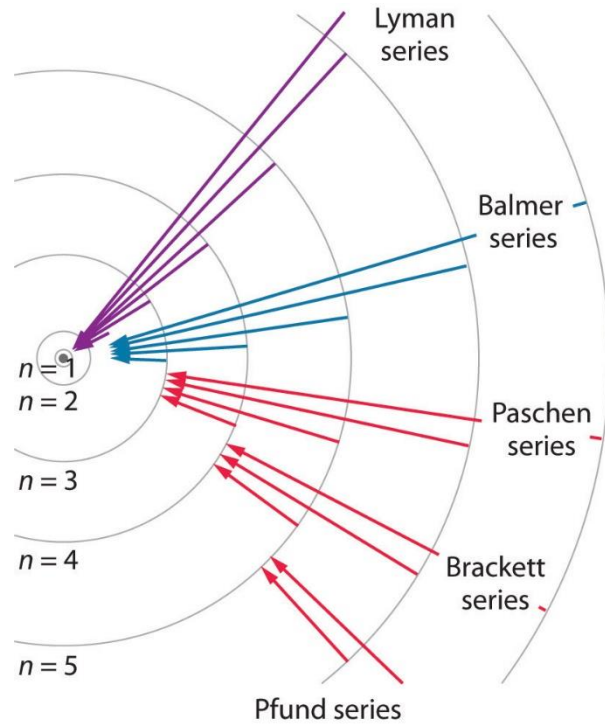


Narrow-line transitions : Long-lived states

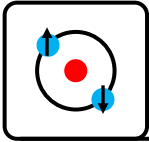
- Excited state lifetime τ has natural linewidth
 - $\Delta\nu = \frac{1}{2\pi\tau}$ (Rb: D2 $2S_{1/2} \rightarrow 2P_{3/2}$ vs. Sr: $1S_0-3P_1$)
- Long-lived state (metastable states, forbidden transitions)
 - Why forbidden (spin, parity, angular momentum)
 - Dipole-allowed transitions (E1) transitions
 - Typical atomic fluorescence transitions
 - Dipole forbidden transition (M1, E2, Higher-order)



Atom's fingerprint (hydrogen)



From Demtroder's book
(see Ming-Shien's lecture notes)



Atom's fingerprint (hydrogen)

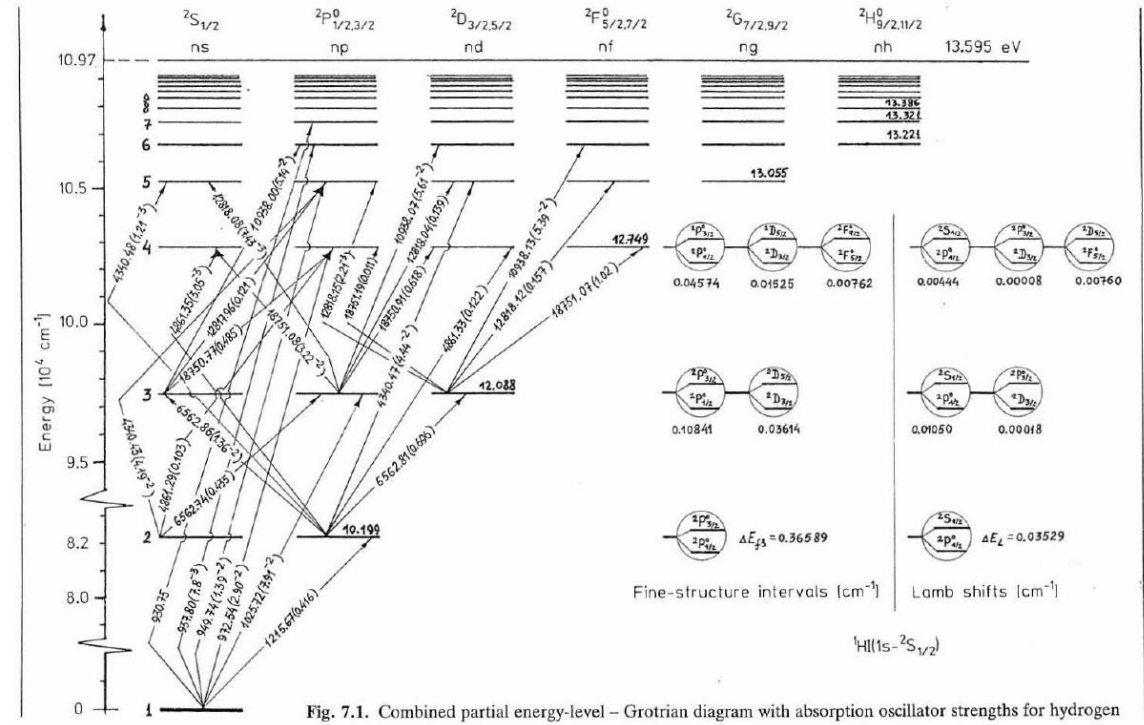
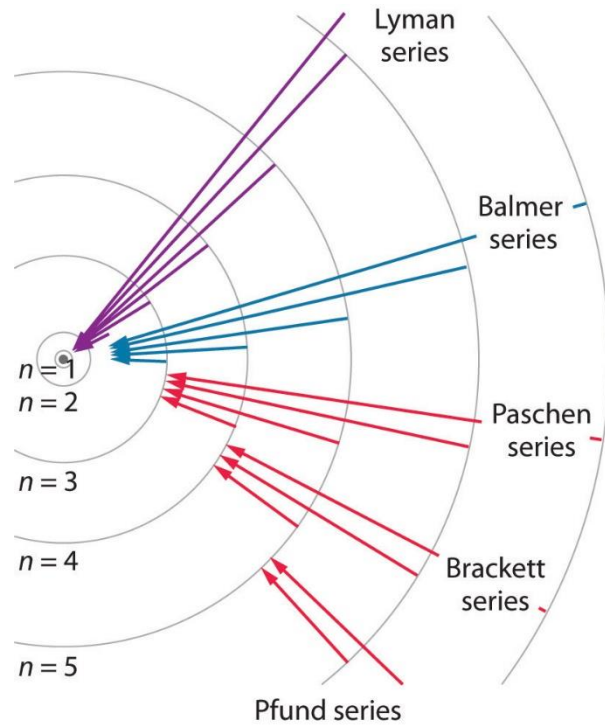
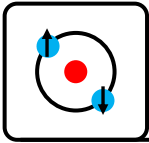


Fig. 7.1. Combined partial energy-level - Grotrian diagram with absorption oscillator strengths for hydrogen

A. A. Radzig B. M. Smirnov
Reference Data on Atoms, Molecules, and Ions
Springer-Verlag, Berlin Heidelberg New York Tokyo





Atom's fingerprint (hydrogen)

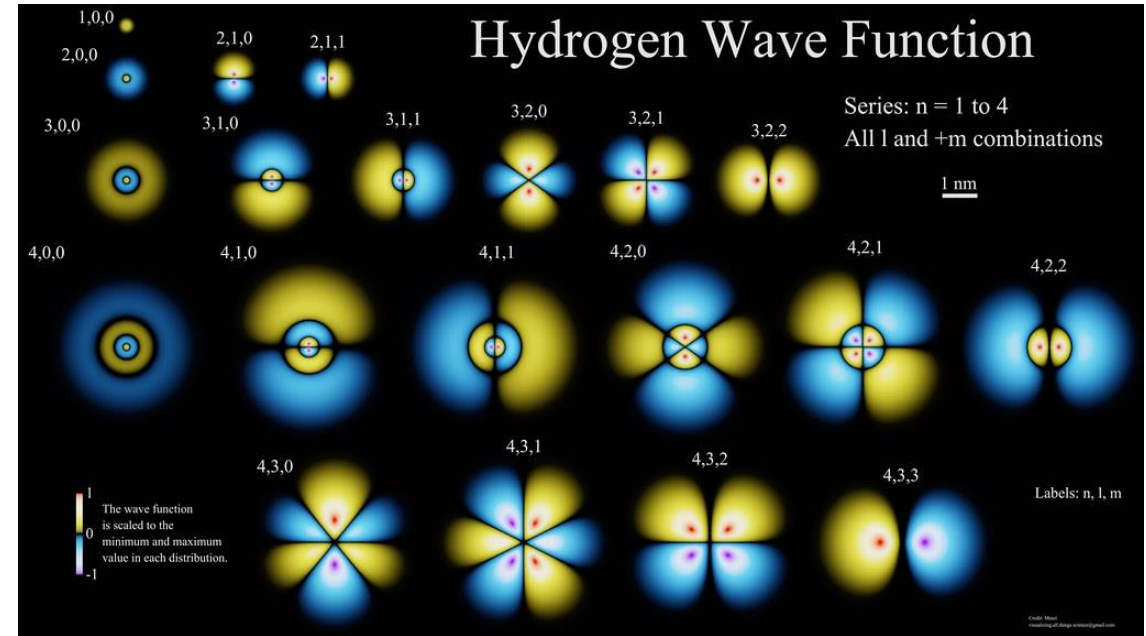
- In hydrogen, Schrodinger equation

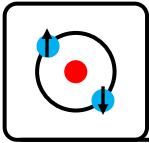
$$\left(-\frac{\hbar^2}{2m_e} \nabla^2 + V(r) \right) \psi = E\psi$$

- $\Psi_{nlm} = R_{nl}(r)Y_{lm}(\theta, \phi)$

- Transition

- Rate $\propto \int \Psi_{n_2 l_2}^* (\hat{r} \cdot \hat{e}) \Psi_{n_1 l_1} d^3 r$
 $\propto \int_0^\infty R_{n_2 l_2}(r) r R_{n_1 l_1}(r) r^2 dr \int_0^{2\pi} \int_0^\pi Y_{l_2 m_2}^*(\theta, \phi) (\hat{r} \cdot \hat{e}) Y_{l_1 m_1}(\theta, \phi) \sin\theta d\theta d\phi$





Selection rules

- Transition

- Rate $\propto \int \Psi_{n_2 l_2 m_2}^* (\mathbf{r} \cdot \hat{\mathbf{e}}) \Psi_{n_1 l_1 m_1} d^3 r$
 $\propto \int_0^\infty R_{n_2, l_2}(r) r R_{n_1, l_1}(r) r^2 dr \int_0^{2\pi} \int_0^\pi Y_{l_2 m_2}^*(\theta, \phi) (\hat{\mathbf{r}} \cdot \hat{\mathbf{e}}) Y_{l_1 m_1}(\theta, \phi) \sin\theta d\theta d\phi$
 $= \text{Radial integral} + \text{Angular integral}$

$$\hat{\mathbf{r}} = \frac{1}{r} (x\hat{\mathbf{e}}_x + y\hat{\mathbf{e}}_y + z\hat{\mathbf{e}}_z)$$

$$= \sin\theta \cos\phi \hat{\mathbf{e}}_x + \sin\theta \sin\phi \hat{\mathbf{e}}_y + \cos\theta \hat{\mathbf{e}}_z$$

$$\propto Y_{1,-1} \frac{\hat{\mathbf{e}}_x + i\hat{\mathbf{e}}_y}{\sqrt{2}} + Y_{1,0} \hat{\mathbf{e}}_z + Y_{1,1} \frac{-\hat{\mathbf{e}}_x + i\hat{\mathbf{e}}_y}{\sqrt{2}}$$

$$\hat{\mathbf{e}}_{\text{rad}} \propto A_{\sigma^-} \frac{\hat{\mathbf{e}}_x - i\hat{\mathbf{e}}_y}{\sqrt{2}} + A_{\pi} \hat{\mathbf{e}}_z + A_{\sigma^+} \left(-\frac{\hat{\mathbf{e}}_x + i\hat{\mathbf{e}}_y}{\sqrt{2}} \right)$$

$$\hat{\mathbf{r}} \cdot \hat{\mathbf{e}}_{\text{rad}} \propto A_{\sigma^-} Y_{1,-1} + A_{\pi} Y_{1,0} + A_{\sigma^+} Y_{1,+1}$$

$$\text{Rate} \propto \int_0^{2\pi} \int_0^\pi Y_{l_2 m_2}^* Y_{1,m} Y_{l_1 m_1} \sin\theta d\theta d\phi$$

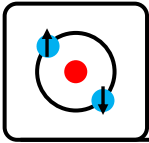
- Cylindrical symmetry

π - transitions

$$\Delta m = 0$$

σ - transitions

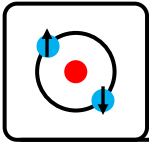
$$\Delta m = \pm 1$$



Selection rule - Parity

- In quantum mechanics, **physical observables must be consistent with spatial symmetries**.
- **Parity inversion** ($\hat{P} : r \rightarrow -r$) is a symmetry operation: the whole physical system (Hamiltonian and light-atom interaction rules) remains unchanged under it.
- Now consider a transition matrix element for an electric dipole transition:
 - Transition
 - Rate $\propto \langle \Psi_f | r | \Psi_i \rangle \propto \langle Y_{l_f, m_f} | r | Y_{l_i, m_i} \rangle$
 - $\hat{P} Y_{l, m} = (-1)^l Y_{l, m}$
 - Rate = $\hat{P}$ Rate = $(-1)^{l_f + l_i + 1} \langle Y_{l_f, m_f} | r | Y_{l_i, m_i} \rangle$

$$\Delta l \neq 0$$



Atom's fingerprint (more than one electron)

- In Helium, Schrodinger equation

$$H = \left[-\frac{\hbar^2}{2m} \nabla_1^2 - \frac{Ze^2}{4\pi\epsilon_0 r_1} \right] + \left[-\frac{\hbar^2}{2m} \nabla_2^2 - \frac{Ze^2}{4\pi\epsilon_0 r_2} \right] + \frac{e^2}{4\pi\epsilon_0 r_{12}}$$

electron-electron repulsion
(manybody-effect)

Ignoring electron–electron repulsion gives –109 eV, but the real helium ground state is –79 eV, showing that electron repulsion is crucial.

- Symmetry and spin
 - Spatial wavefunction**: both electrons occupy the same 1s orbital → symmetric.
 - But electrons are **fermions**, so the total wavefunction must be **antisymmetric**.

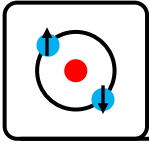
$$\Psi_{total}(r_1, \sigma_1; r_2, \sigma_2) = \Psi_{spatial}(r_1, r_2) \chi_{spin}(\sigma_1, \sigma_2)$$

- Therefore the spin part must be the antisymmetric combination (singlet, S=0)

$$\chi_{S=0} = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

- Hence the ground state of helium has the **term symbol**:

$$1s^2 \ ^1S_0$$



Term symbol – Grotrian diagram for He

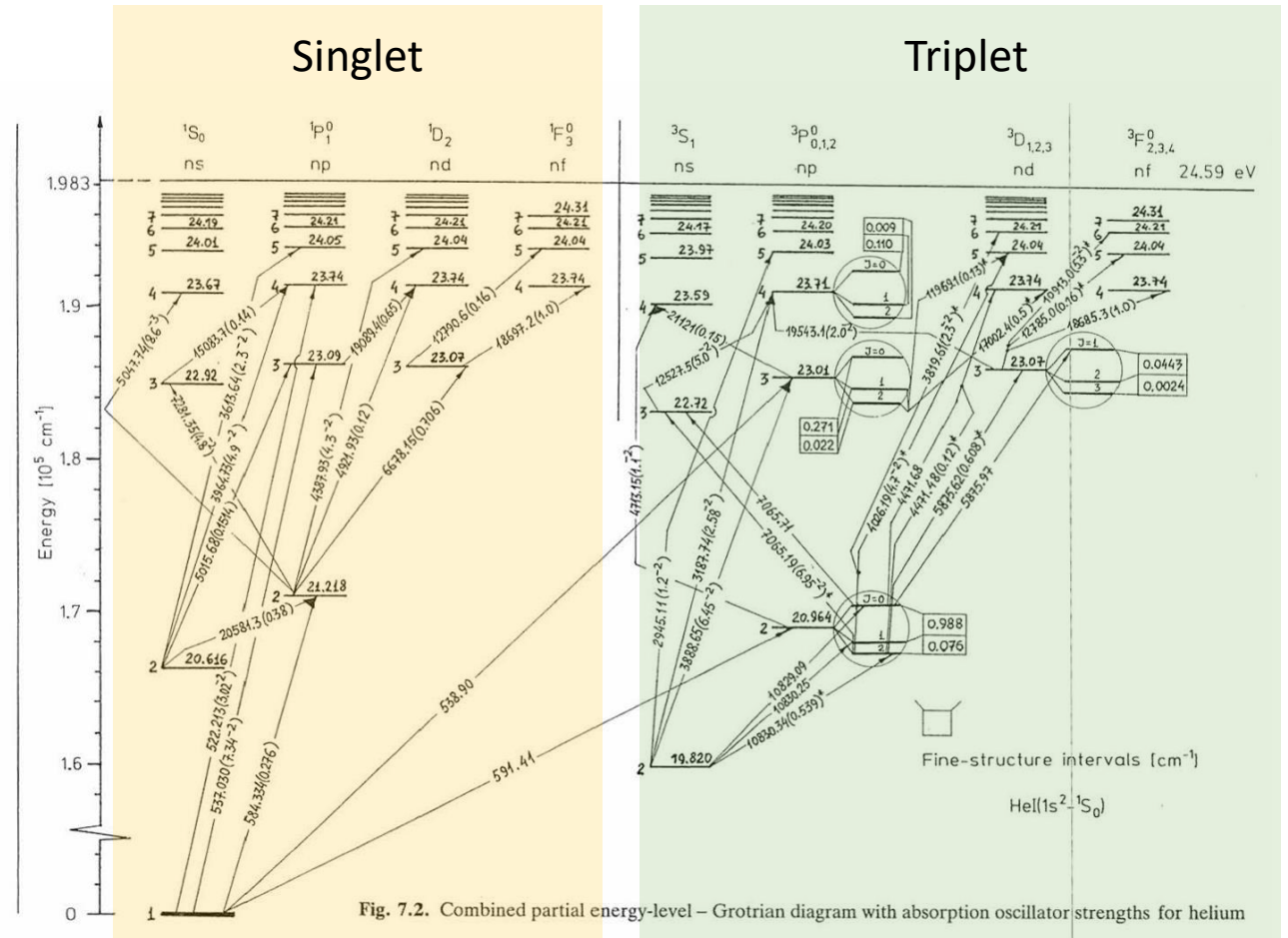
Total orbital angular momentum
 $L = l_1 + l_2 + l_3 + \dots$

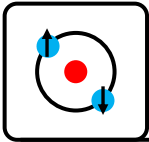
Total spin degeneracy
 $S = s_1 + s_2 + s_3 + \dots$

n $2S+1$ L J
 ↑ ↓ ↓ ↑
 Principal quantum number Total electronic angular momentum

^4He in ground state:

$$\left. \begin{array}{l} s_1 = s_2 = 1/2 \\ l_1 = l_2 = 0 \rightarrow L=0 \\ J=L+S \end{array} \right\} J = S = s_1 + s_2$$





Term symbol – Grotrian diagram for He

Singlet vs Triplet Systems

The total wavefunction must be antisymmetric, so the spatial and spin parts must complement each other.

- **singlet, $S=0$:**

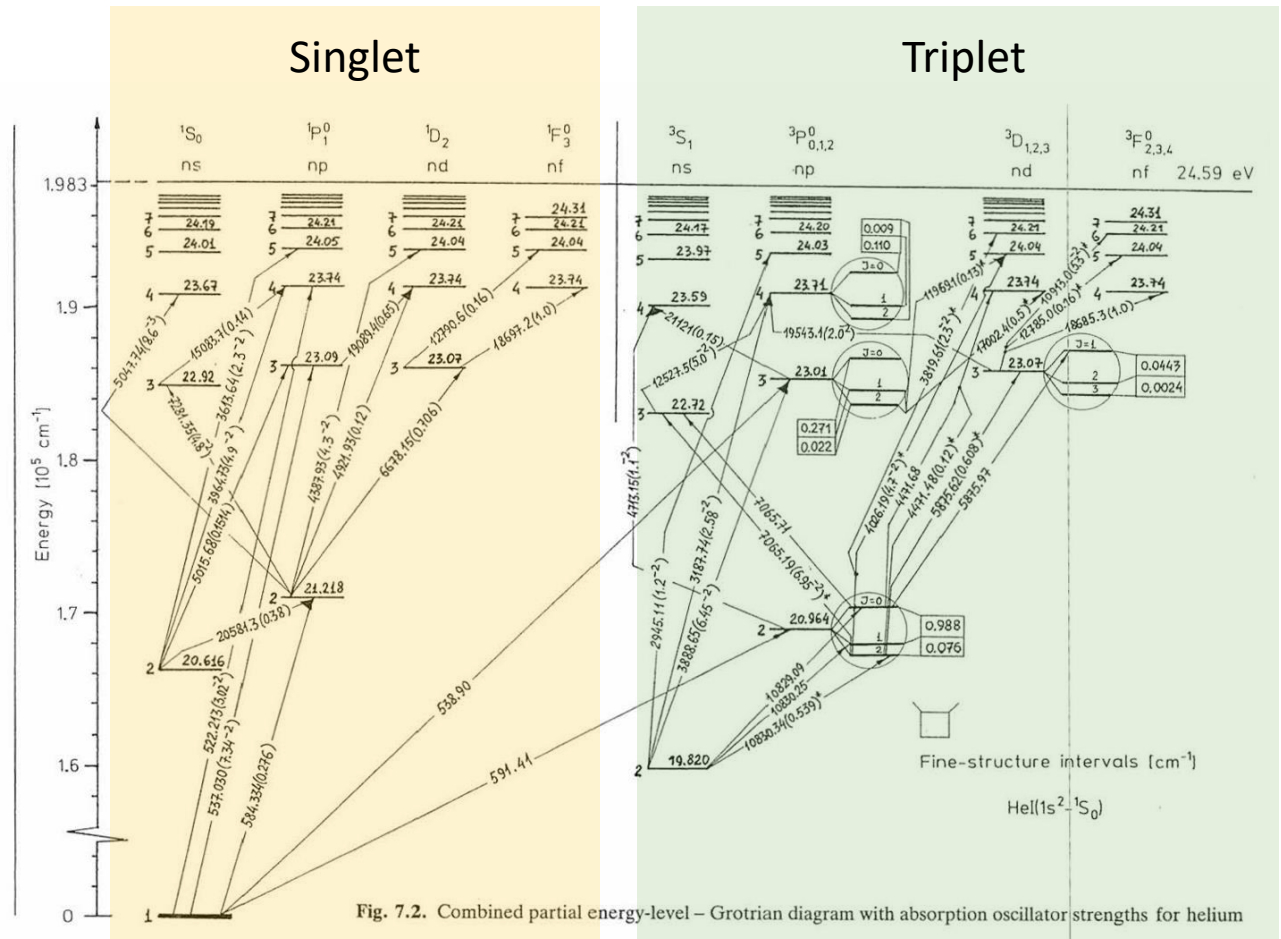
$\Psi_{spatial}$ (symmetric) χ_{spin} (antisymmetric).

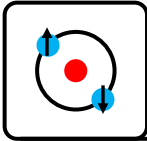
- **triplet, $S=1$:**

$\Psi_{spatial}$ (antisymmetric) χ_{spin} (symmetric).

Spectroscopic Selection Rules

- Electric dipole transitions must satisfy $\Delta S=0$
transitions between singlet and triplet systems are **forbidden** (approximate selection rule).
- Triplet states have long lifetimes (metastable).





Atom's fingerprint

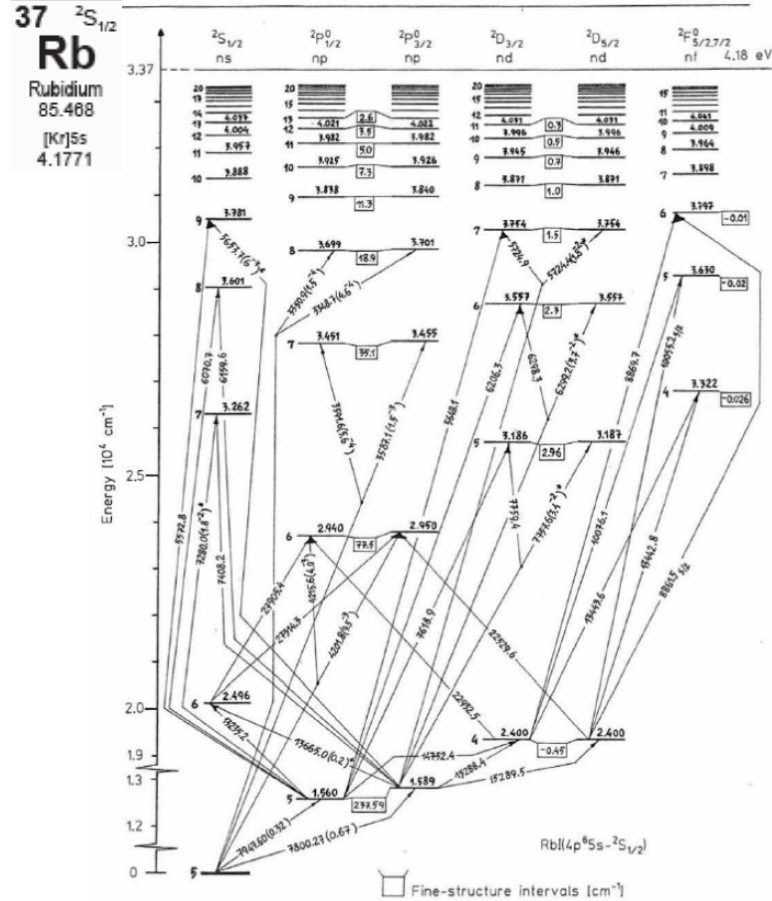


Fig. 7.35. Combined partial energy-level – Grotrian diagram with absorption oscillator strengths for rubidium

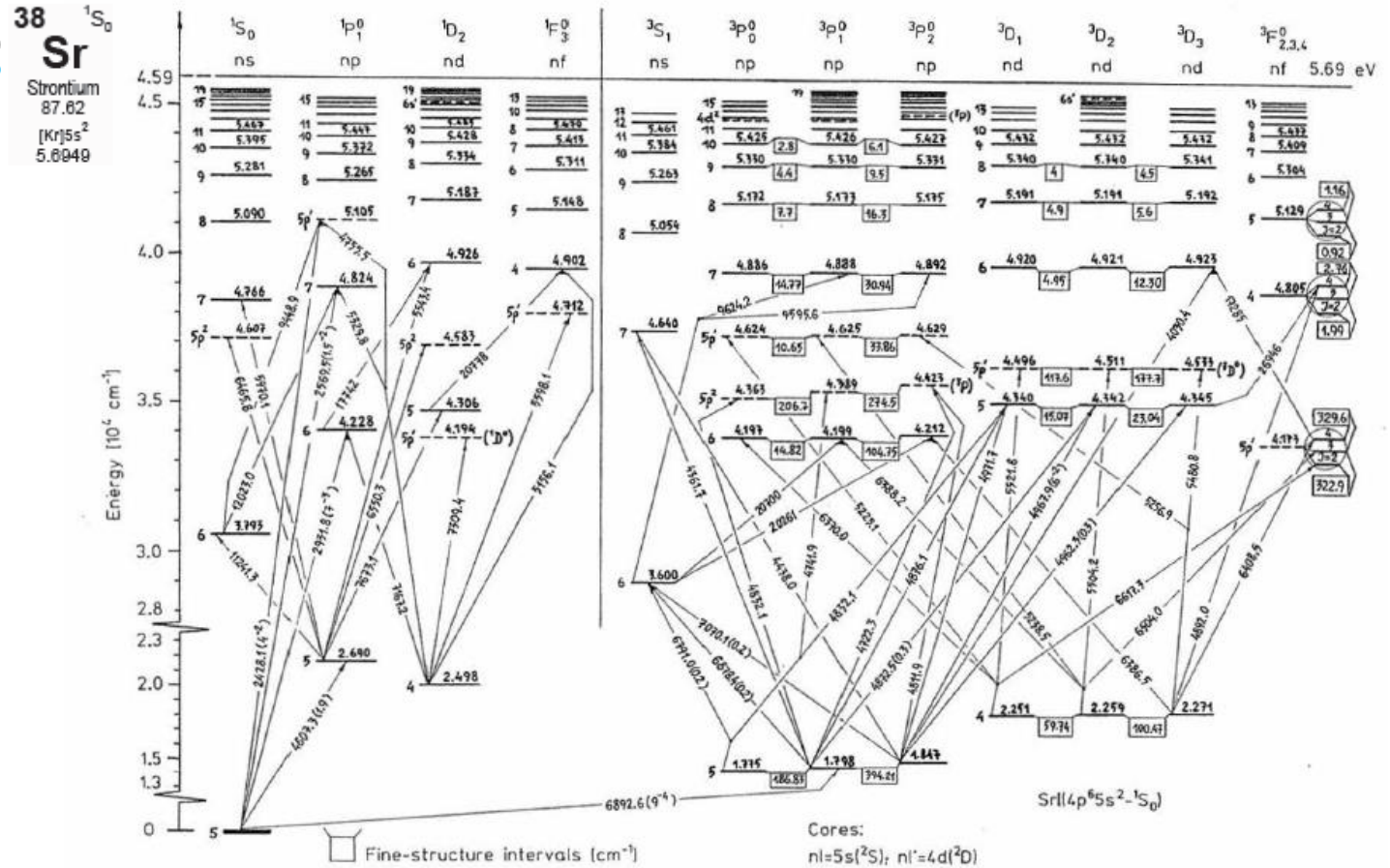
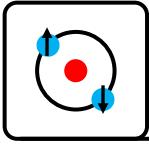


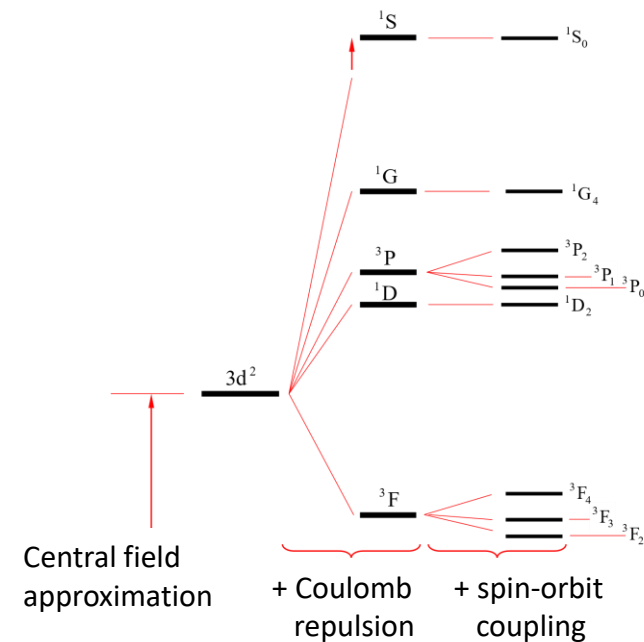
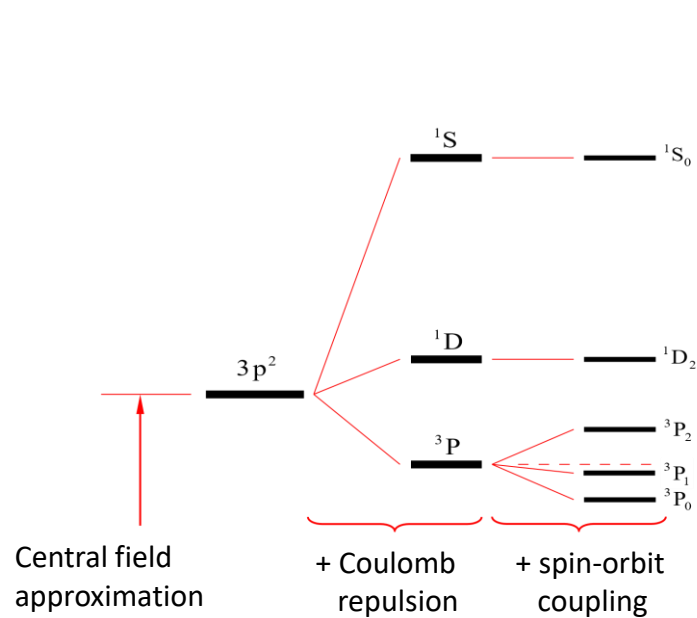
Fig. 7.36. Combined partial energy-level – Grotrian diagram with absorption oscillator strengths for strontium

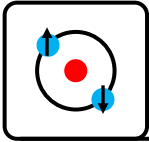


L-S coupling to jj coupling

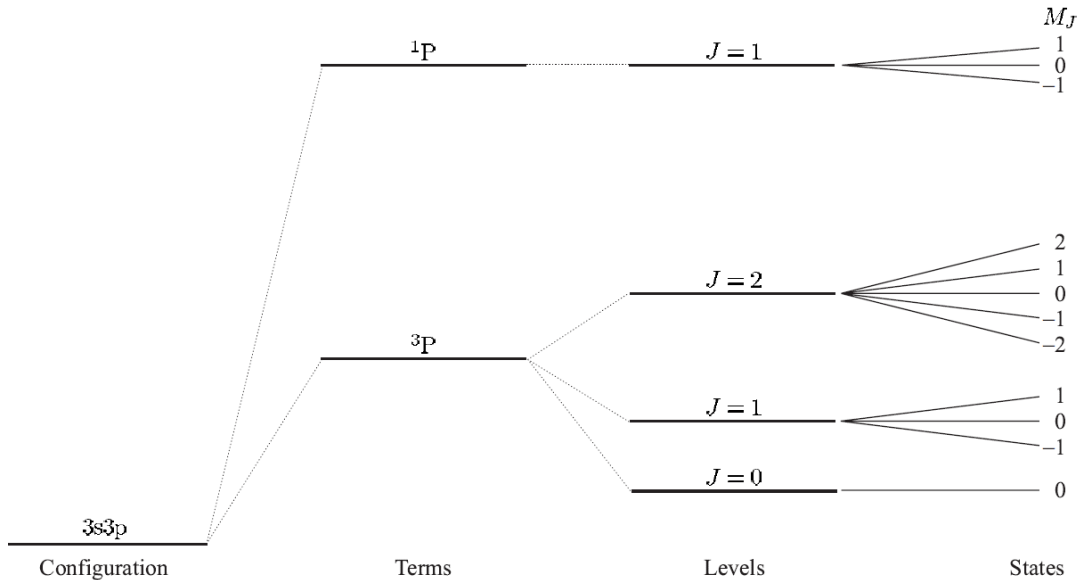
- In many electron system, Schrodinger equation looks like

$$H = \sum_{i=1}^N \left[\underbrace{-\frac{\hbar^2}{2m} \nabla_i^2 + V_{CF}(r_i)}_{\text{Central field}} + \underbrace{\left\{ \sum_{j>i}^N \frac{e^2}{4\pi\epsilon_0 r_{ij}} - S(r_i) \right\}}_{\text{Residual electrostatic interaction (H}_{re})} \right] + \underbrace{\beta_{LS} \langle S \cdot L \rangle}_{\text{Spin-orbit interaction}}$$





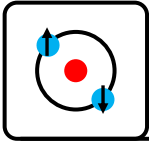
Selection rules



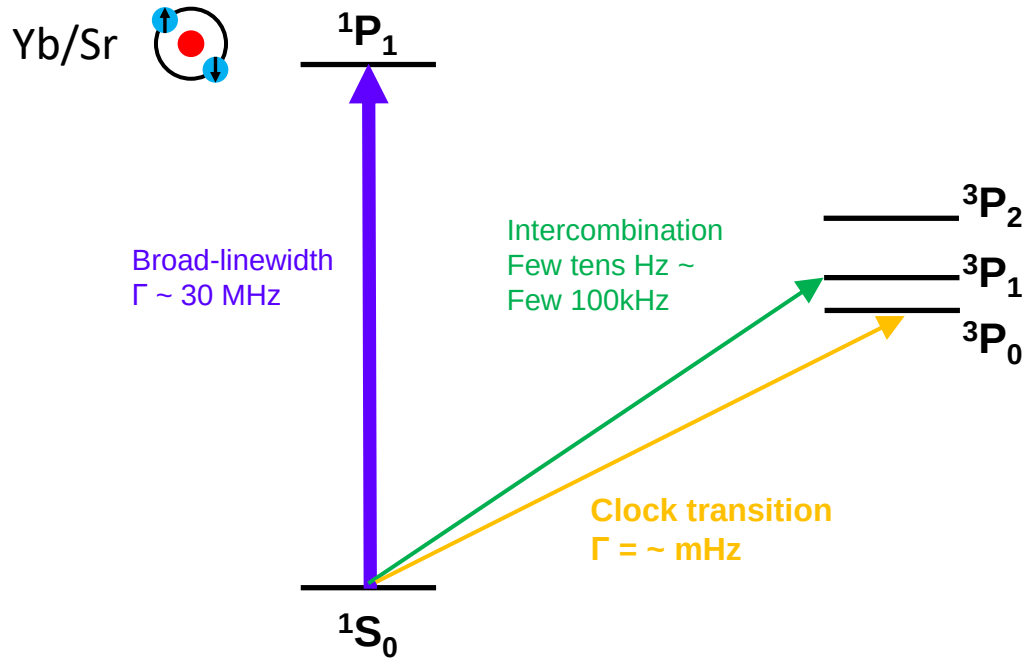
- Dipole allowed transition : **selection rules for electric dipole (E1)** when atoms are described in the **LS-coupling scheme**.
 - Some rules are **absolutely strict** (coming from angular momentum conservation and parity).
 - Others hold only when **L and S are good quantum numbers** (i.e. when LS-coupling is a good approximation).

1	$\Delta J = 0, \pm 1$	$(J = 0 \leftrightarrow J' = 0)$	Level
2	$\Delta M_J = 0, \pm 1$	$(M_J = 0 \leftrightarrow M_{J'} = 0 \text{ if } \Delta J = 0)$	State
3	Parity changes		Configuration
4	$\Delta l = \pm 1$	One electron jump	Configuration
5	$\Delta L = 0, \pm 1$	$(L = 0 \leftrightarrow L' = 0)$	Term
6	$\Delta S = 0$		Term

When does it break the rule?



Selection rules - Alkaline-earth (-like) elements

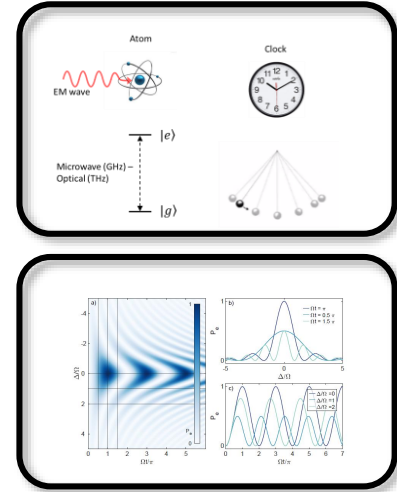


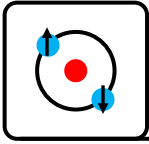
- Dipole allowed transition : **selection rules for electric dipole (E1)** when atoms are described in the **LS-coupling scheme**.
 - Some rules are **absolutely strict** (coming from angular momentum conservation and parity).
 - Others hold only when **L and S are good quantum numbers** (i.e. when LS-coupling is a good approximation).
- In the case of pure LS coupling these intercombination transitions are dipole or even doubly dipole forbidden
 - $\Delta S = 0$ (singlet to triplet)
 - $1S_0 - 3P_0 / 3P_2 \quad \Delta J \neq \pm 1$
- For heavy elements LS coupling is not perfectly realized and a **small admixture of the state $1P_1$ to the triplet states** enables the optical transition, resulting in narrow linewidths

Atomic species	Mg	Ca	Sr	Yb	Hg	Cd
$1S_0 - 3P_1$ cooling λ (nm)	457	657	689	556	253	326
$\Gamma/2\pi$	40 Hz	400Hz	7.4 kHz	182.3 kHz	1.3 MHz	67 kHz

Outline

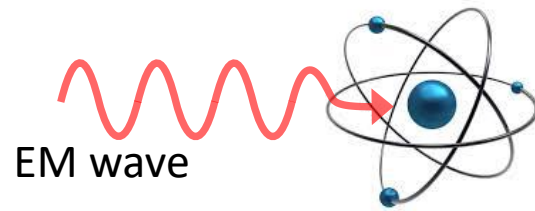
- Introduction
- Two-level system
 - Atom light interaction
 - Highly accurate two level systems – mHz forbidden optical transitions
- How optical atomic clock works:
 - Cs Atomic frequency standards
 - Physics of optical atomic clocks – (optical lattice clock)
- Current research directions
 - How to improve state of the art atomic clocks (optical lattice clock) ?
 - Next generation optical clocks ?



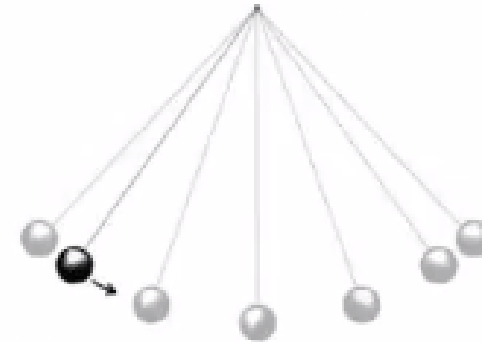
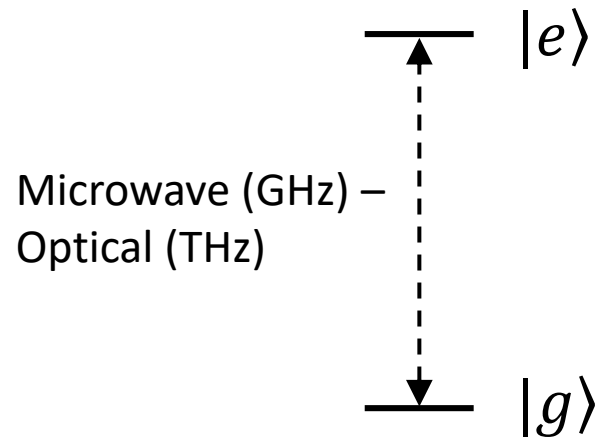


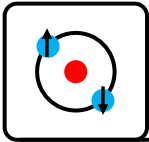
Atomic frequency standards: Atomic clock

Atom

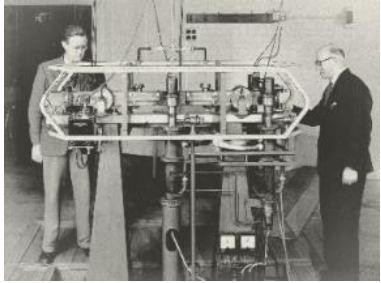


Clock

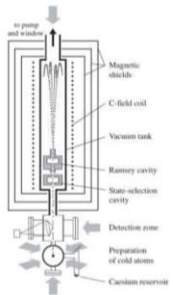




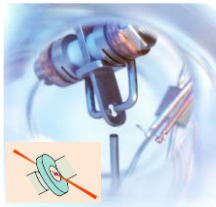
Defining moments in atomic timekeeping



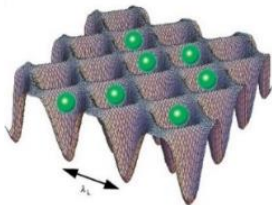
1955 First **caesium atomic clock**



1993 First **caesium-fountain clock**



1999 First optical-frequency measurement with femtosecond combs



2005 First **optical lattice clock**

19XX **Rabi**

1949 **Ramsey's** separated oscillatory field technique

1960 **Hydrogen** maser

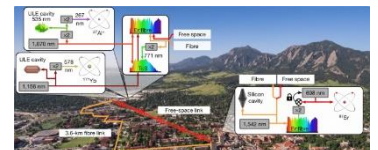
1975 Proposals for laser cooling of atoms and ions

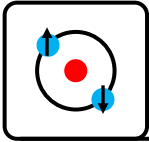
1978 Laser cooling of trapped ions

1985 Laser cooling of atoms

2001 **Concept of an optical clock demonstrated**

2021 Frequency ratio measurements at 18-digit accuracy using an optical clock network

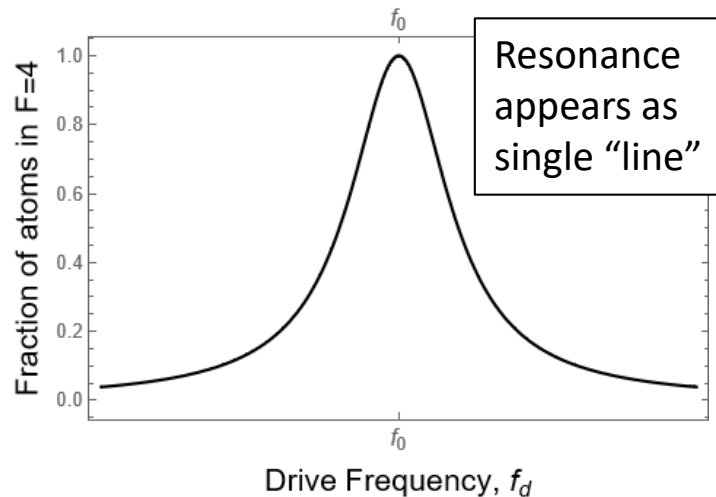
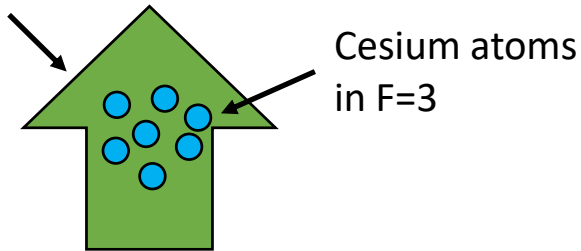




Two ways to observe atomic resonance

Rabi measurement

Oscillating
EM field

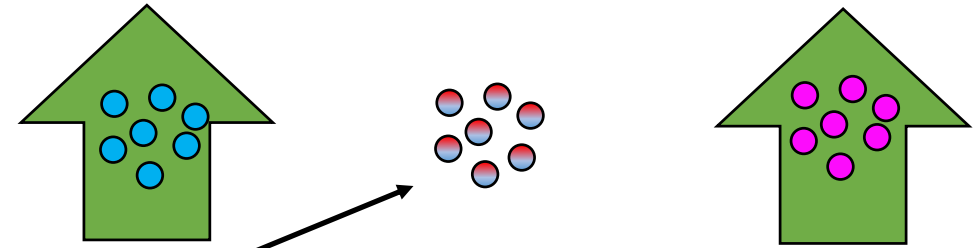


Ramsey measurement

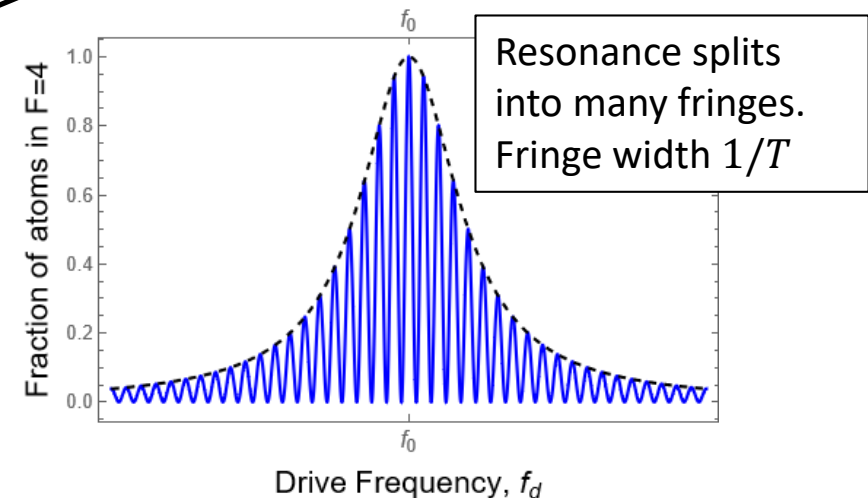
Atoms interact
with field once

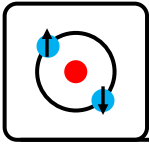
Freely evolve
for a time T

Atoms interact
with field again

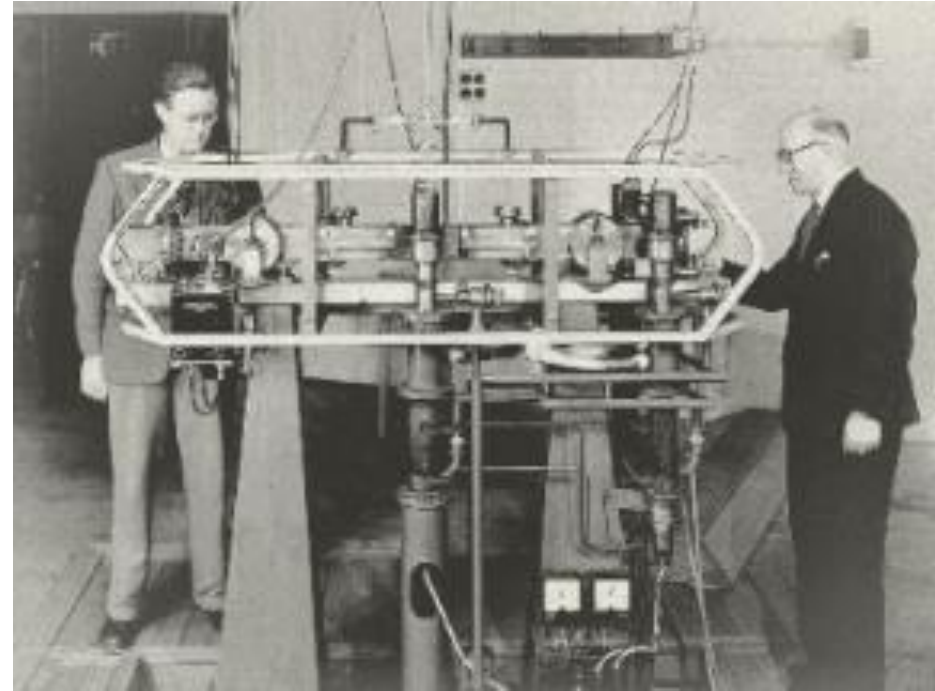
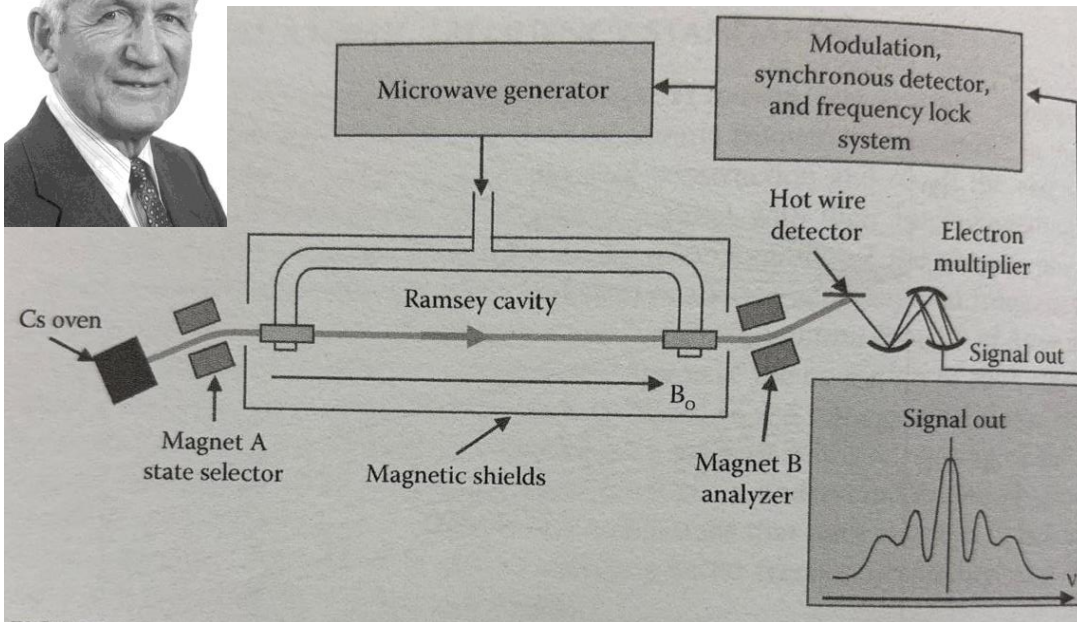


Cesium atoms in a
superposition of
 $F=3$ and $F=4$



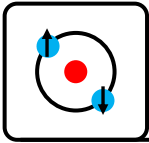


Cs BEAM Frequency Standards



Proposal: Norman F. Ramsey, A molecular beam resonance method with separated oscillating fields, [Phys. Rev. 78, 695 \(1950\)](#).

Experiment: L. ESSEN & J. V. L. PARRY, An Atomic Standard of Frequency and Time Interval: A Cæsium Resonator, *Nature* **176**, 280–282 (1955)



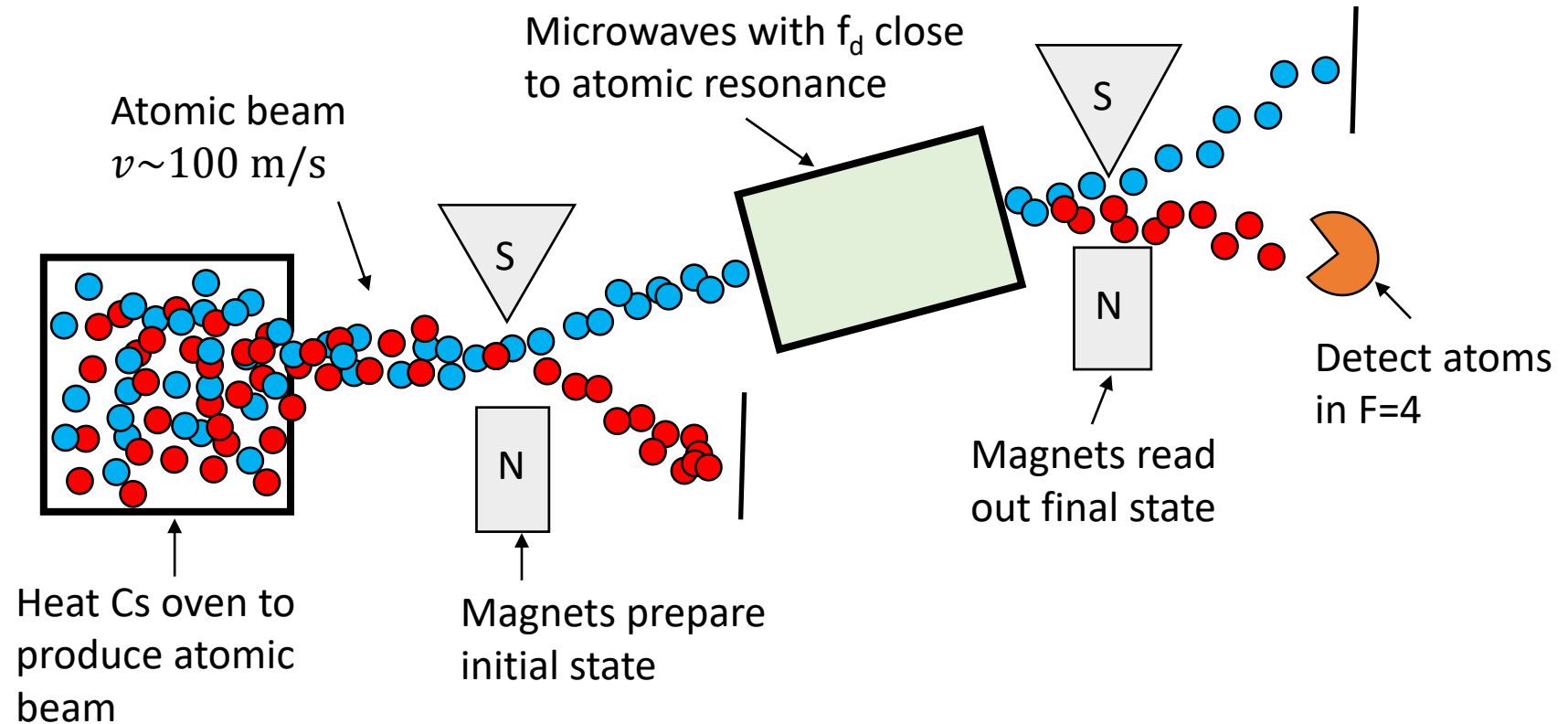
Cesium beam clocks

- Developed in the 1950s¹, Cs beam clocks are still manufactured and used today.
- For a 1 m long interaction region, $t_{\text{interact}} \approx 10$ ms.
- Linewidth ≈ 100 Hz, $\frac{\Delta f}{f} \approx 10^{-8}$
- To achieve an accuracy of 10^{-14} , the line center must be measured to 1 part per million.

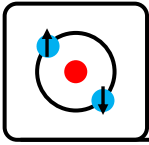
Prepare atoms
in a known state

Atoms interact
with field

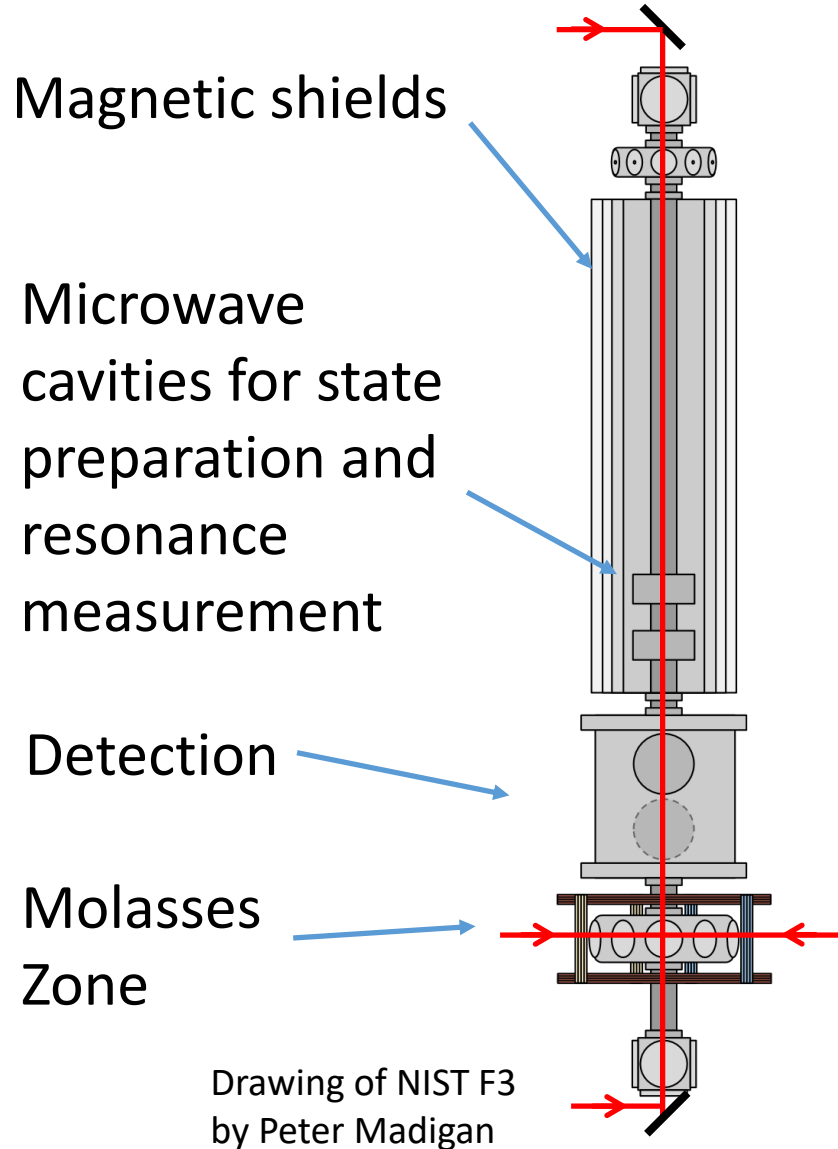
Measure
final state

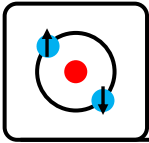


¹ N. F. Ramsey, "History of Early atomic clocks", Metrologia, **42** (2005).

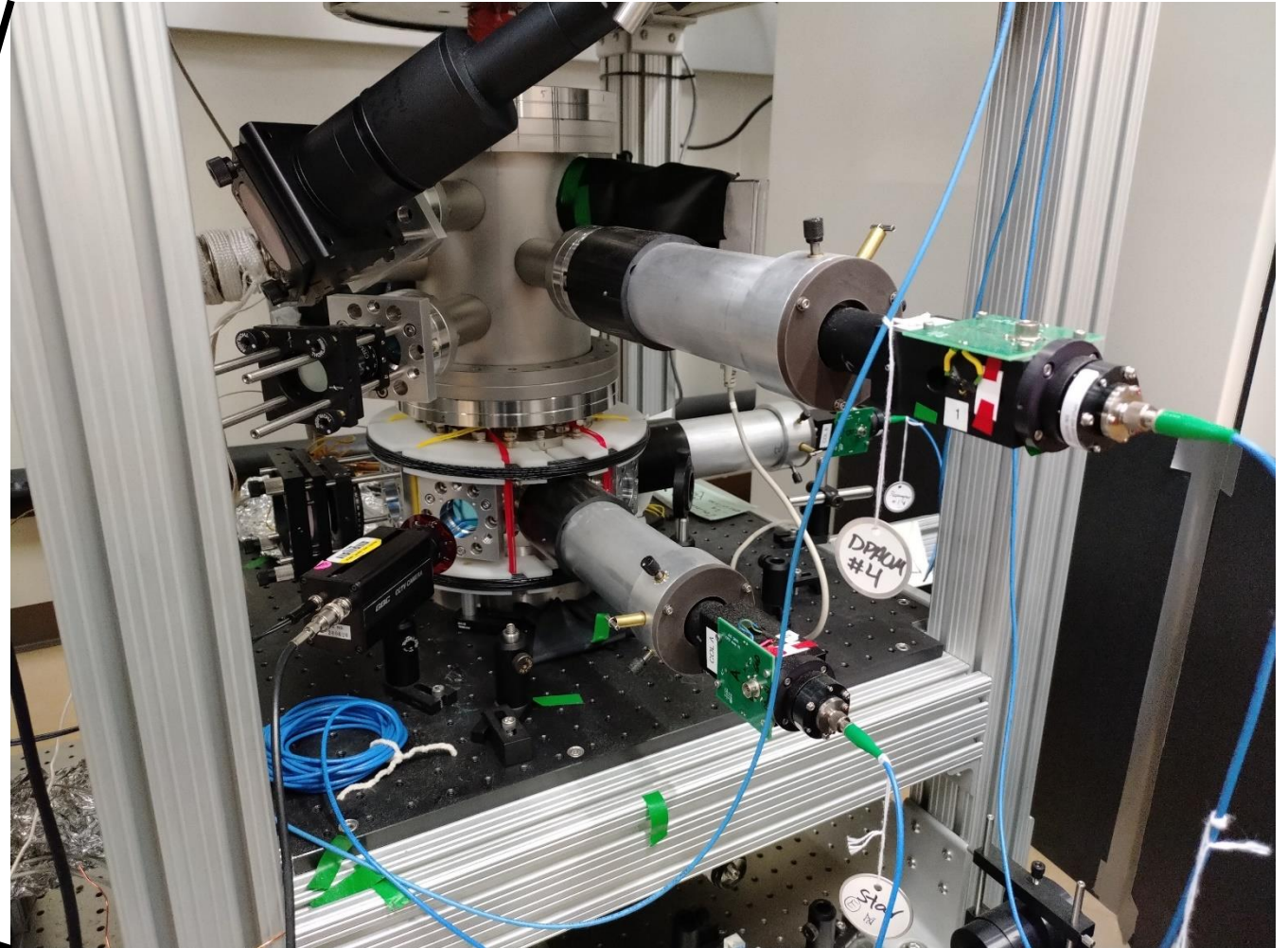
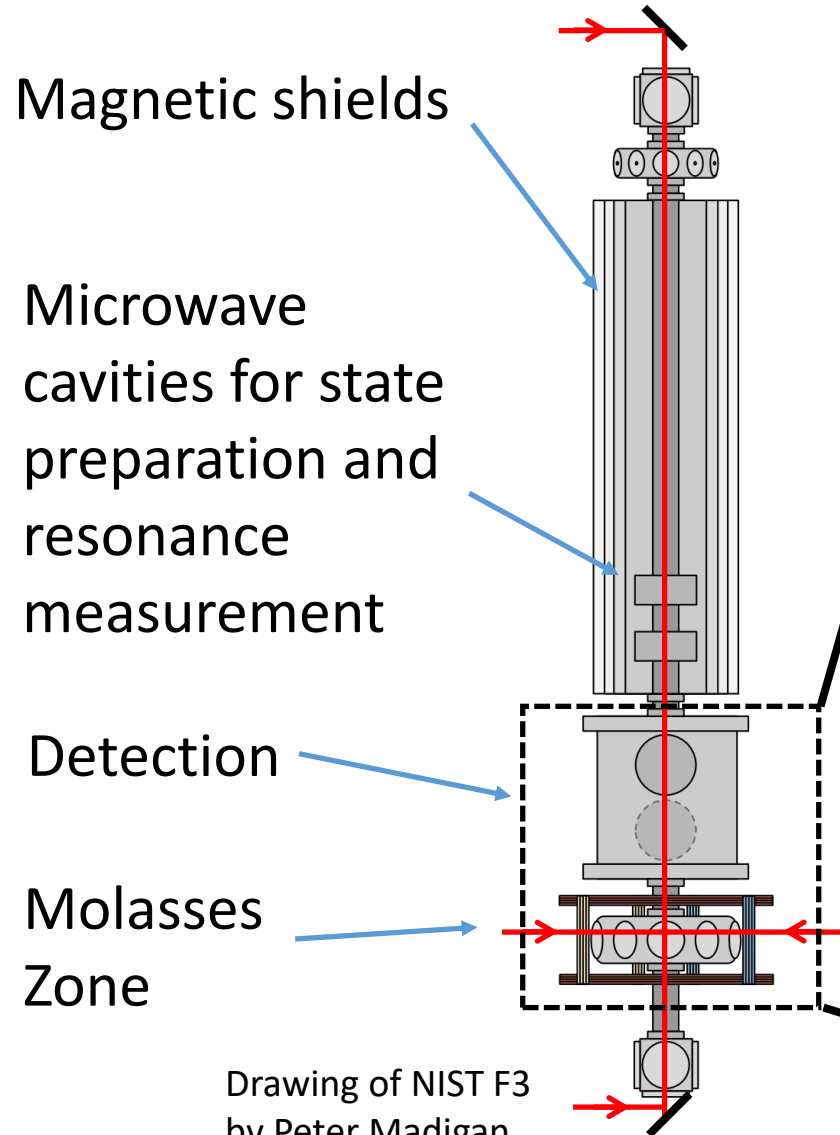


Anatomy of a cesium fountain

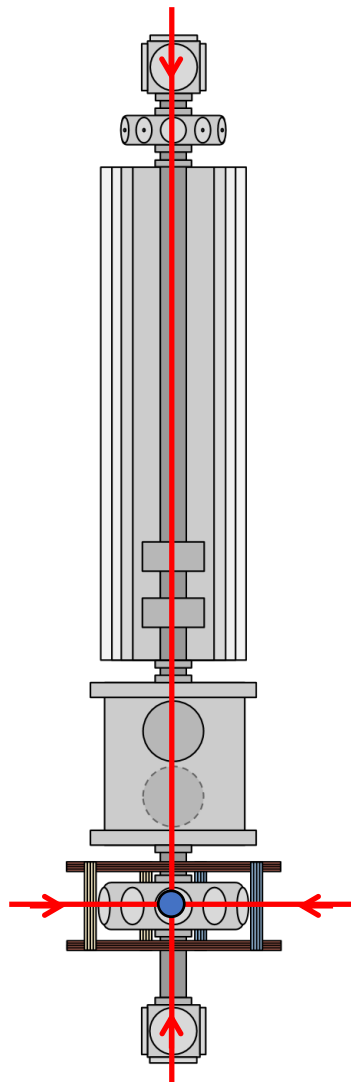




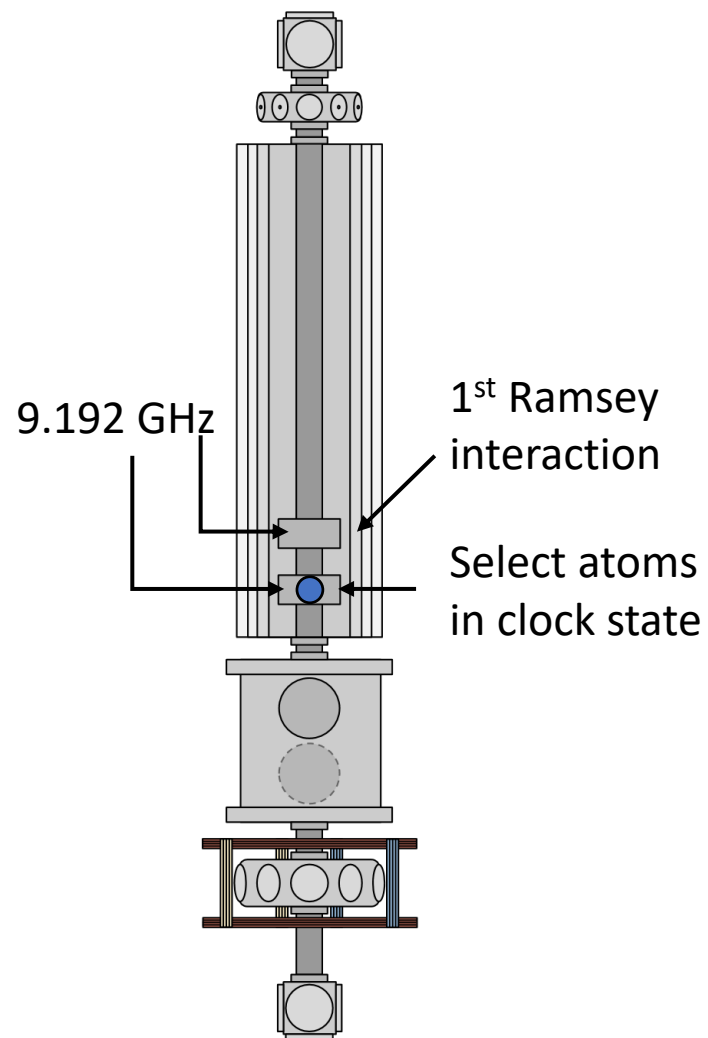
Anatomy of a cesium fountain



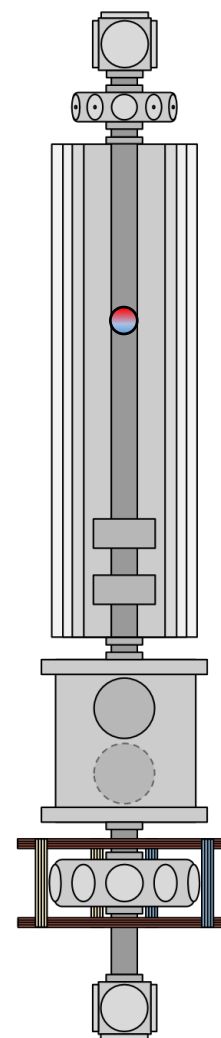
$t = 0$



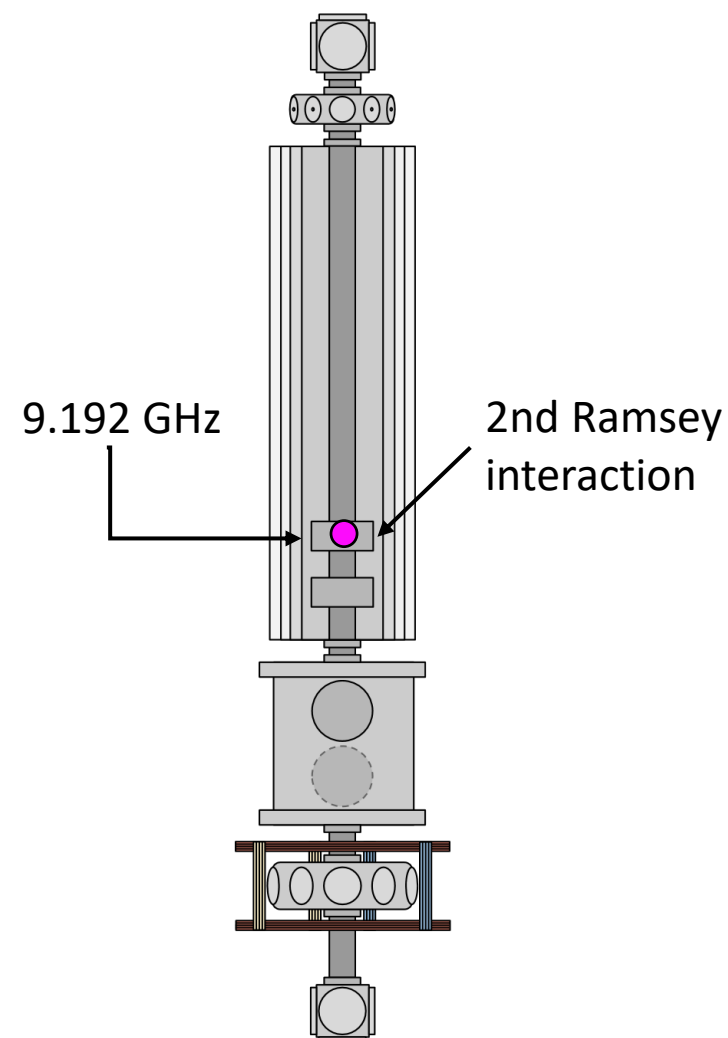
$t \approx 150 \text{ ms}$



$t \approx 400 \text{ ms}$



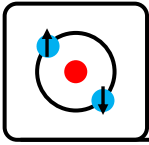
$t \approx 600 \text{ ms}$



Load $\sim 10^6$ cold atoms and launch
the cloud upwards ($v \approx 4 \text{ m/s}$)

Atoms reach apogee, $h \approx 0.78 \text{ m}$
above the molasses zone

Drawing of NIST F3
by Peter Madigan

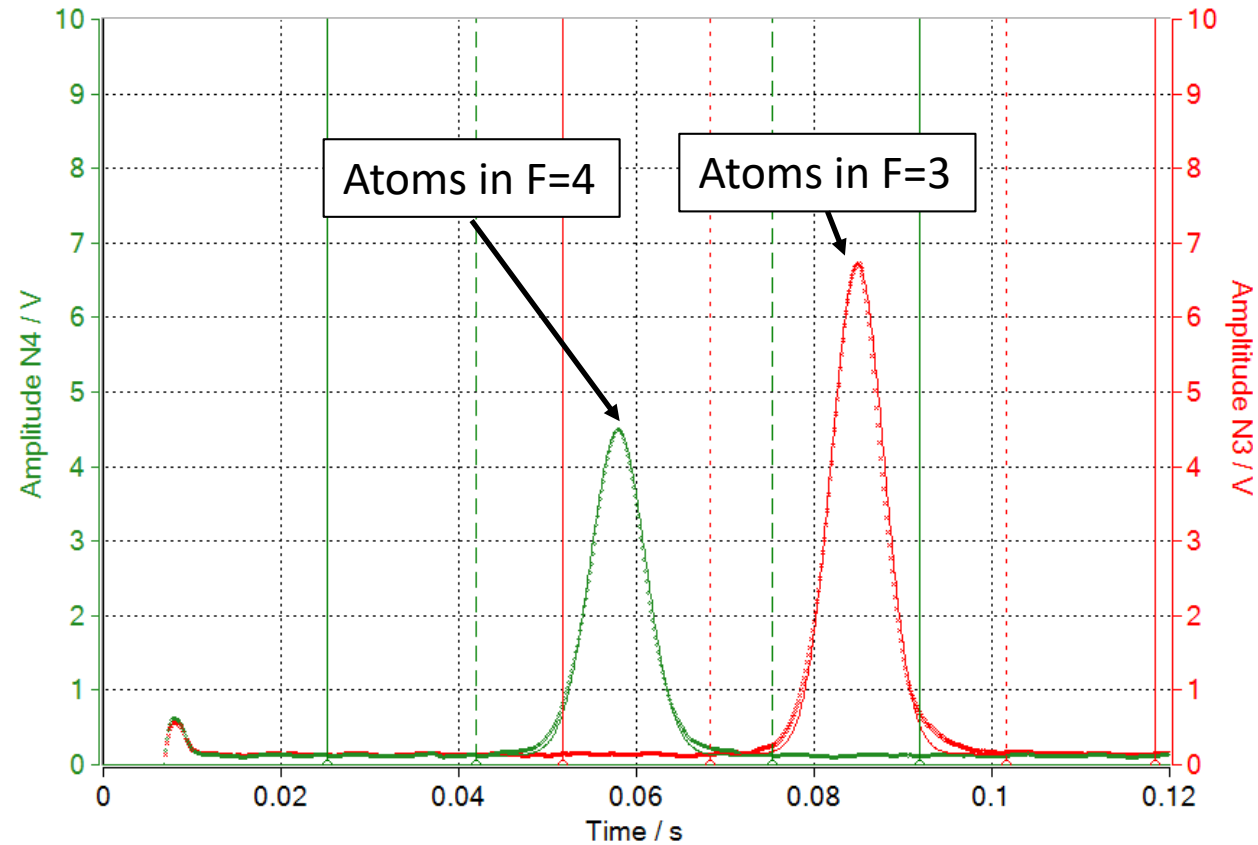
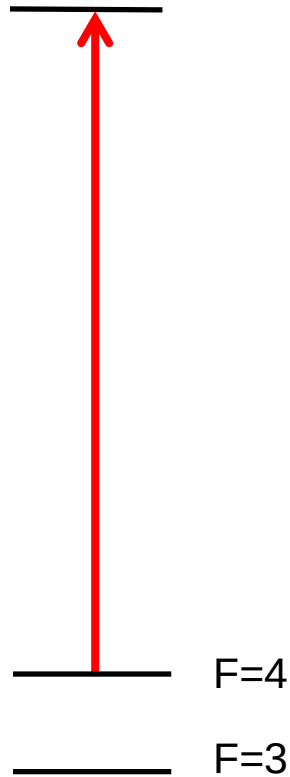
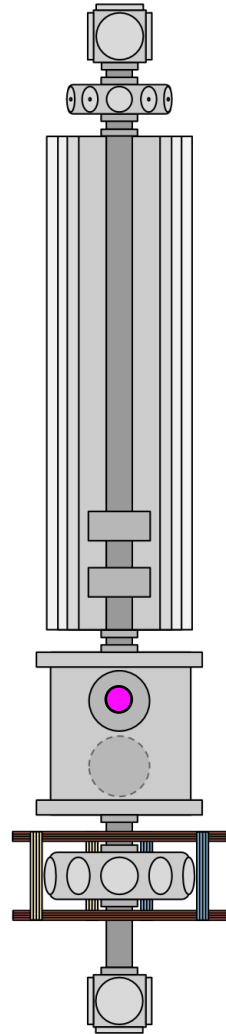


Detection

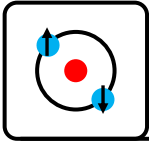
As the atoms fall down the fountain, they cross several beams of resonant light

Detection has three steps:

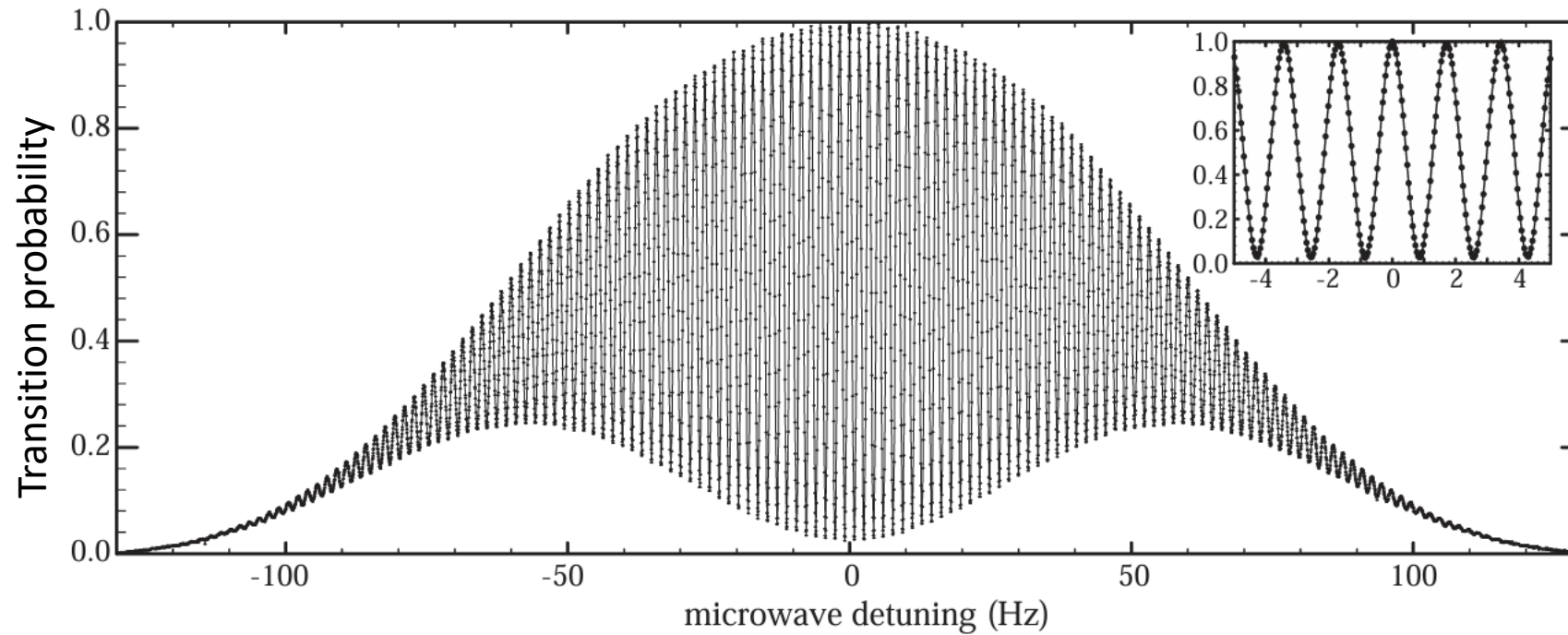
1. Collect fluorescence from atoms in $F=4$.
2. Pump atoms from $F=3$ to $F=4$.
3. Collect fluorescence from atoms that were in $F=3$



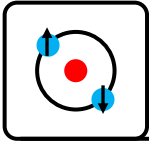
Collected fluorescence as a function of time in NIST-F3



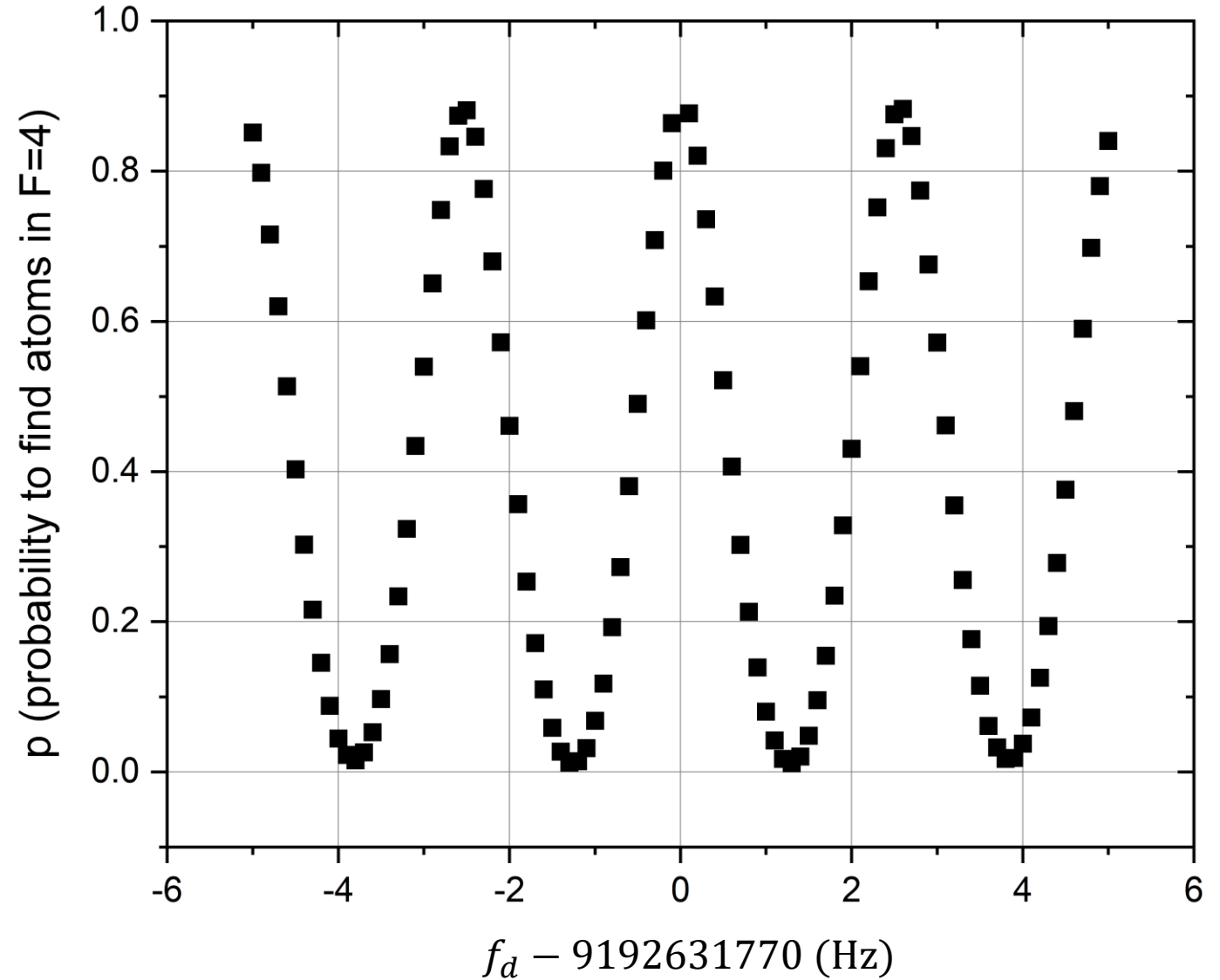
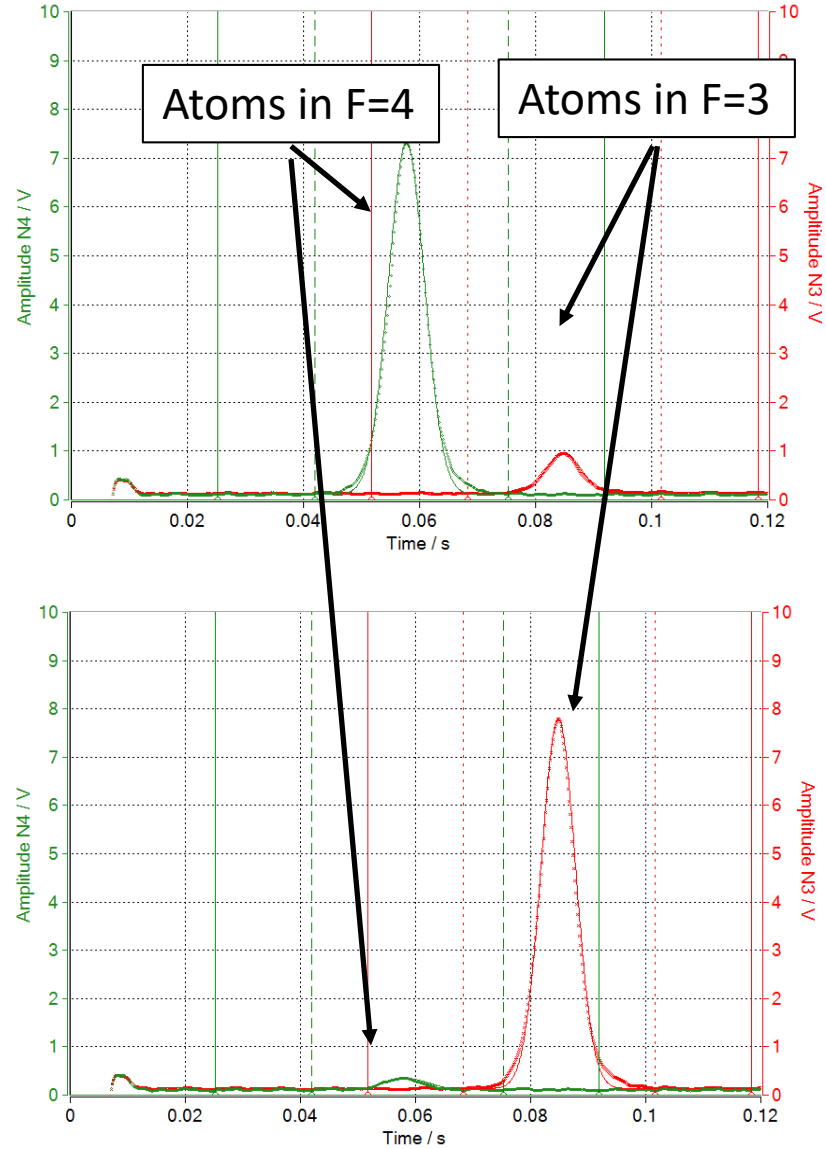
Ramsey spectroscopy

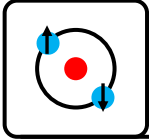


Ref: R Wynands and S Weyers 2005 *Metrologia* **42** S64



Scanning the microwave frequency



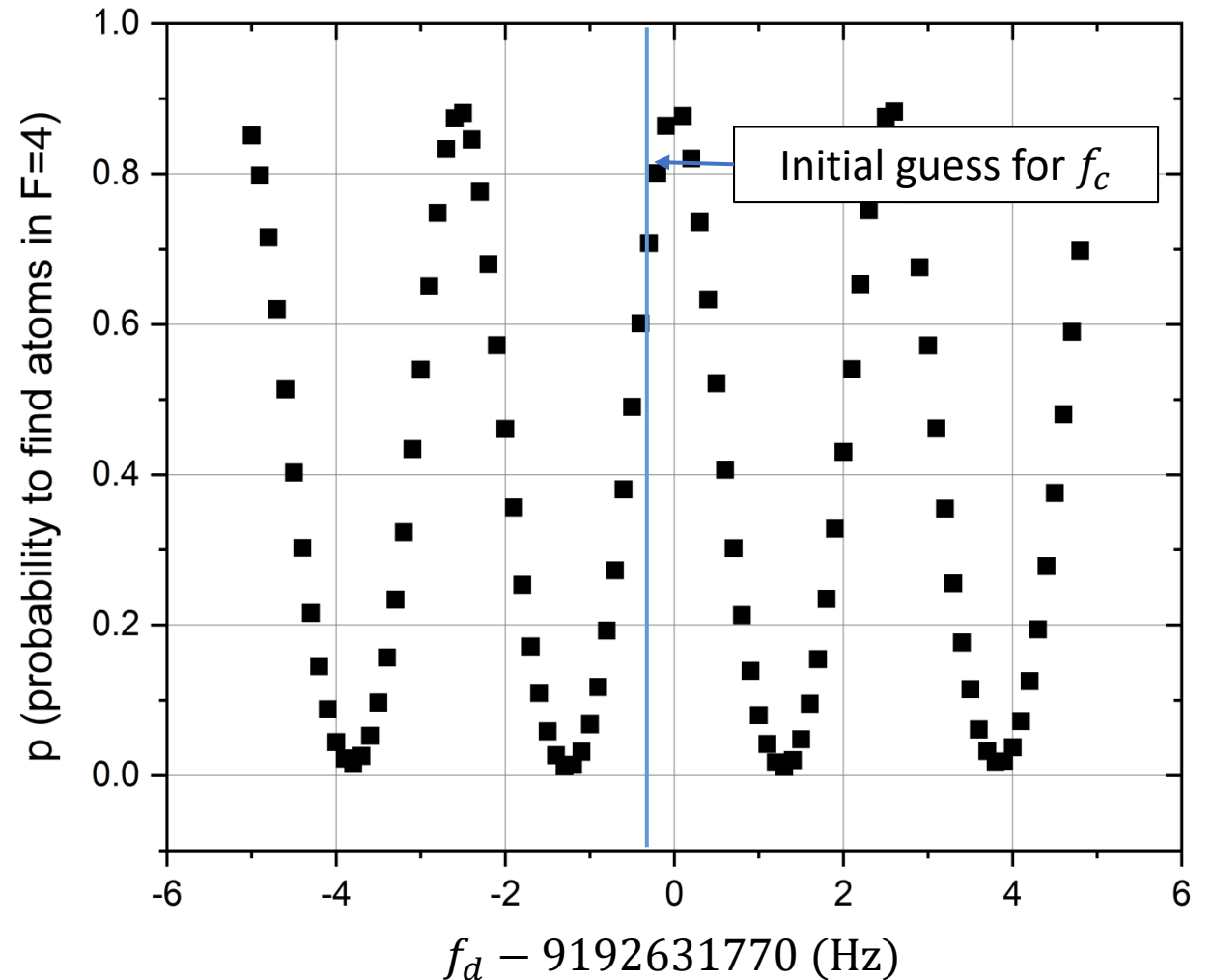


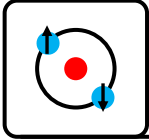
Locking to the central fringe

Fundamental fountain measurement:
determine the frequency offset of the
central fringe.

Three step algorithm:

1. Guess an initial offset frequency.
2. Detune the microwaves by $\pm\Delta f$ to probe on the “left” and “right” side of the fringe. Then,
 - $p_{\text{left}} < p_{\text{right}} \rightarrow \text{increase } f$
 - $p_{\text{left}} > p_{\text{right}} \rightarrow \text{decrease } f$
3. Steer the drive frequency so that $p_{\text{left}} = p_{\text{right}}$



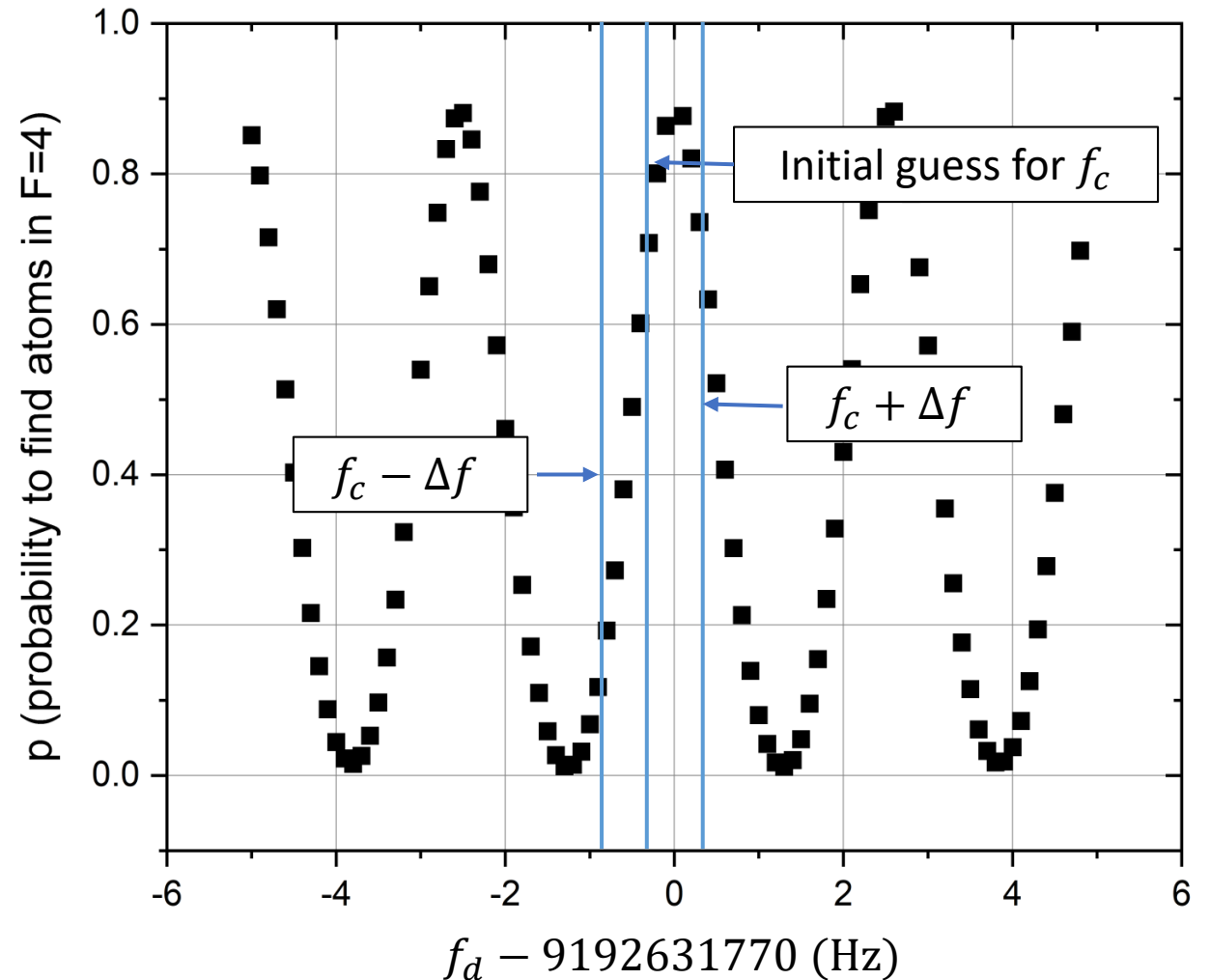


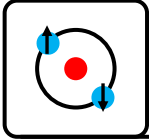
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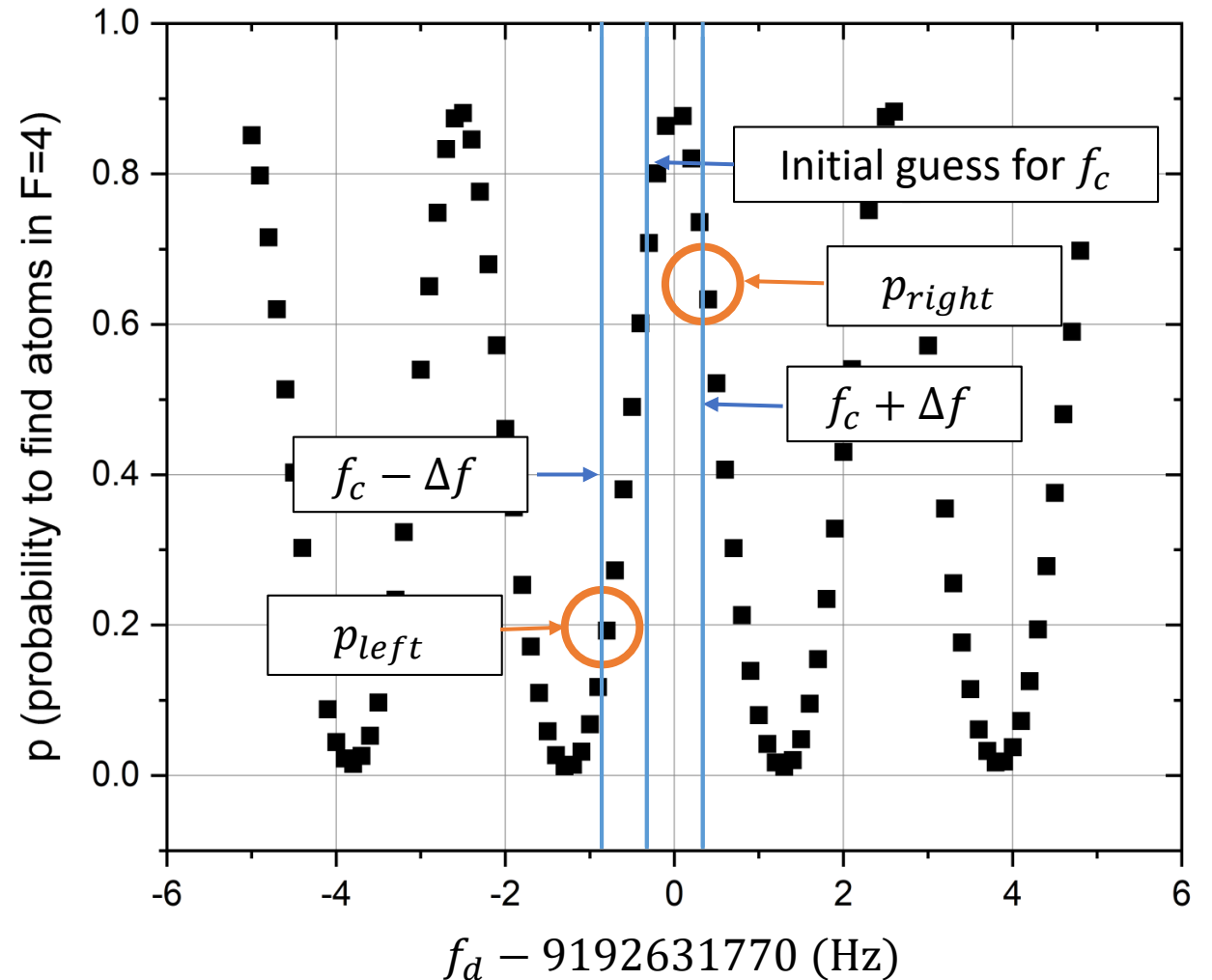


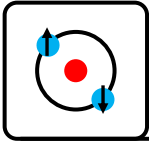
Locking to the central fringe

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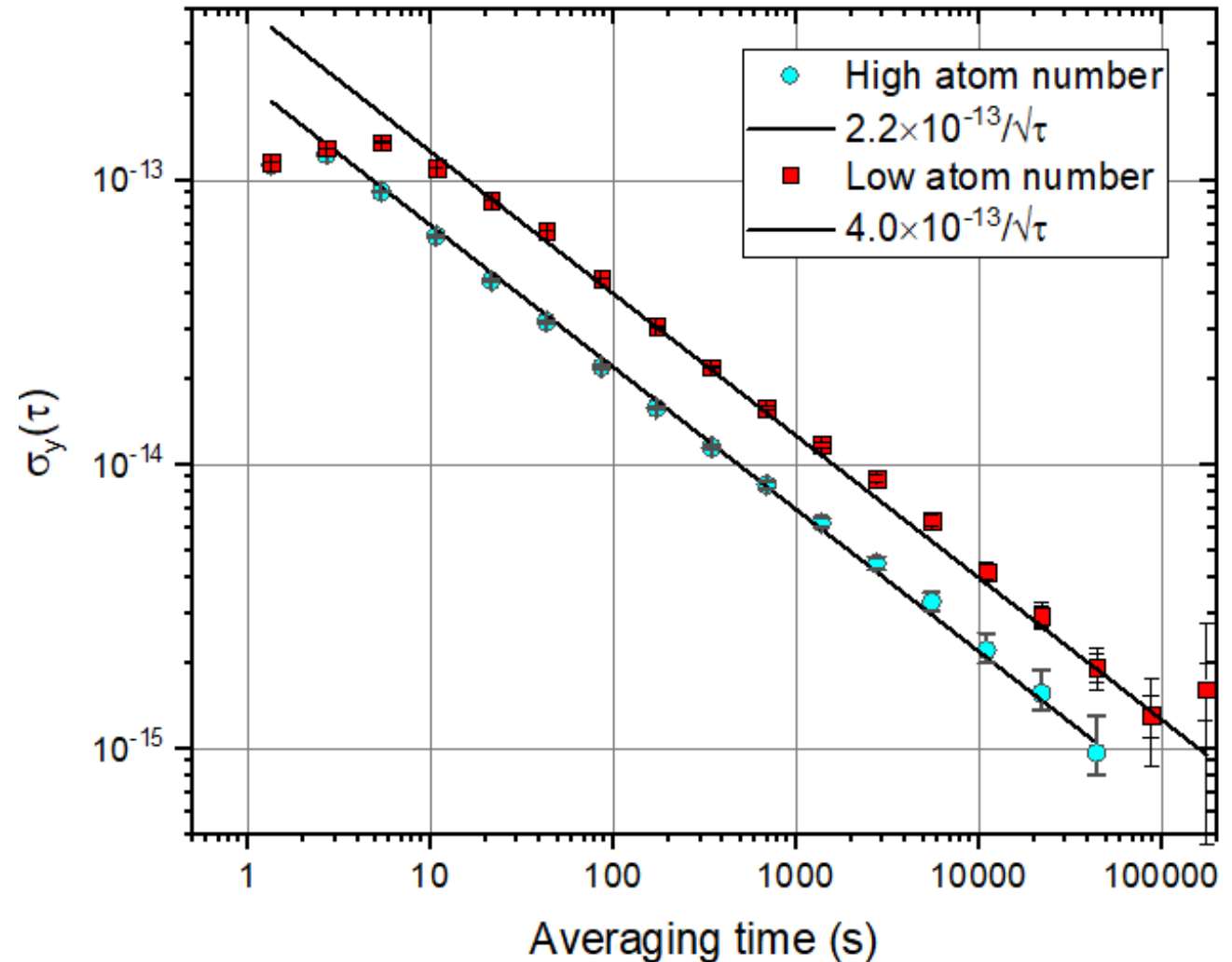
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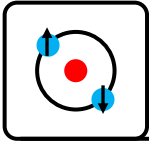




Short term frequency stability

- Dominant noise sources
 - microwave phase noise
 - quantum projection noise (QPN).
- For an intuitive picture of QPN, think of flipping a coin.
- $P(\text{heads}) = 0.5$ on average, but each trial is random.
- Quantum mechanics imposes similar noise on detecting the state of the atoms.
- If you detect N atoms, the signal to noise on the average probability improves like $\sqrt{N}$.





Dehmelt envisioned optical clock in 1982

26 October 1984, Volume 226, Number 4673

SCIENCE

Trapped Ions, Laser Cooling, and Better Clocks

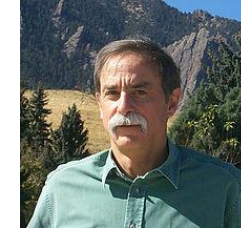
D. J. Wineland

In a recent experiment (1) at the National Bureau of Standards (NBS) in Boulder, Colorado, the frequency of a particular hyperfine transition in the ground state of beryllium atomic ions was measured with an inaccuracy of only about one part in 10^{13} . In this experiment, the ions were confined or "trapped" in a small region of space by

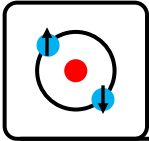
tenths of a percent. On the experimental side, about 8 years ago a measurement (3) of beryllium ion hyperfine structure to an accuracy of 3 ppm (parts per million) was made. Since the accuracy of this previous measurement is presumably good enough to satisfy the theorists for quite some time in the future, one can logically ask why anyone would want to

cally counted; when 9,192,631,770 cycles have occurred, 1 second has passed. Several laboratories are trying to apply this same idea to the internal energy levels of ions; there is reason to believe that the inaccuracy of a time standard based on stored ions can eventually be much smaller than that of the cesium clock, which can have an accuracy of about one part in 10^{13} or less (4).

2) With the extreme accuracy attained with stored ion techniques, it may be possible to measure various small effects which would otherwise be masked by measurement imprecision. As an example, it should be possible to measure nuclear magnetic susceptibility as a small perturbation to atomic hyperfine structure (5). Nuclear magnetic susceptibility, which has not been measured previously, could give a new kind of information about nuclear structure. Trapped ions may provide a way to make the required precise measurements.

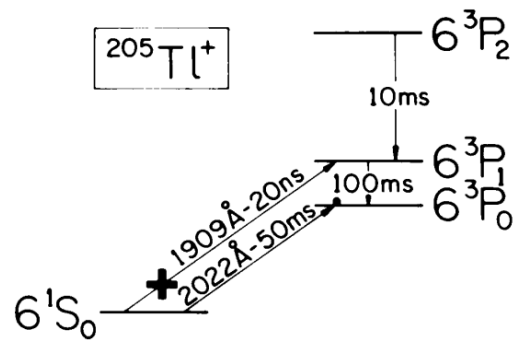


A logical extension of this idea is to go to much higher frequency—for example, to use a narrow optical transition. The anticipated Q in this case can be extremely high, 10^{15} or more. A number of transitions in various ions have been proposed (7); Dehmelt (36) was the first to suggest that such extremely high resolution spectroscopy could be carried out by using single-photon transitions in, for example, single group IIIA ions. For instance (36), the $^1S_0 \leftrightarrow ^3P_0$ transition in Tl^+ ($\lambda = 202$ nm) has a $Q \approx 5 \times 10^{14}$. For such single-photon optical transitions, it is desirable to approximately satisfy the Lamb-Dicke criterion; this is most easily accomplished with single



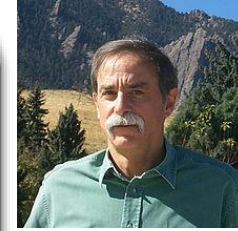
Dehmelt envisioned optical clock in 1982

26 Oct

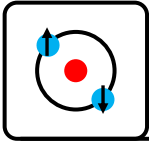


In a recent e
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about one part
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"trapped" in a

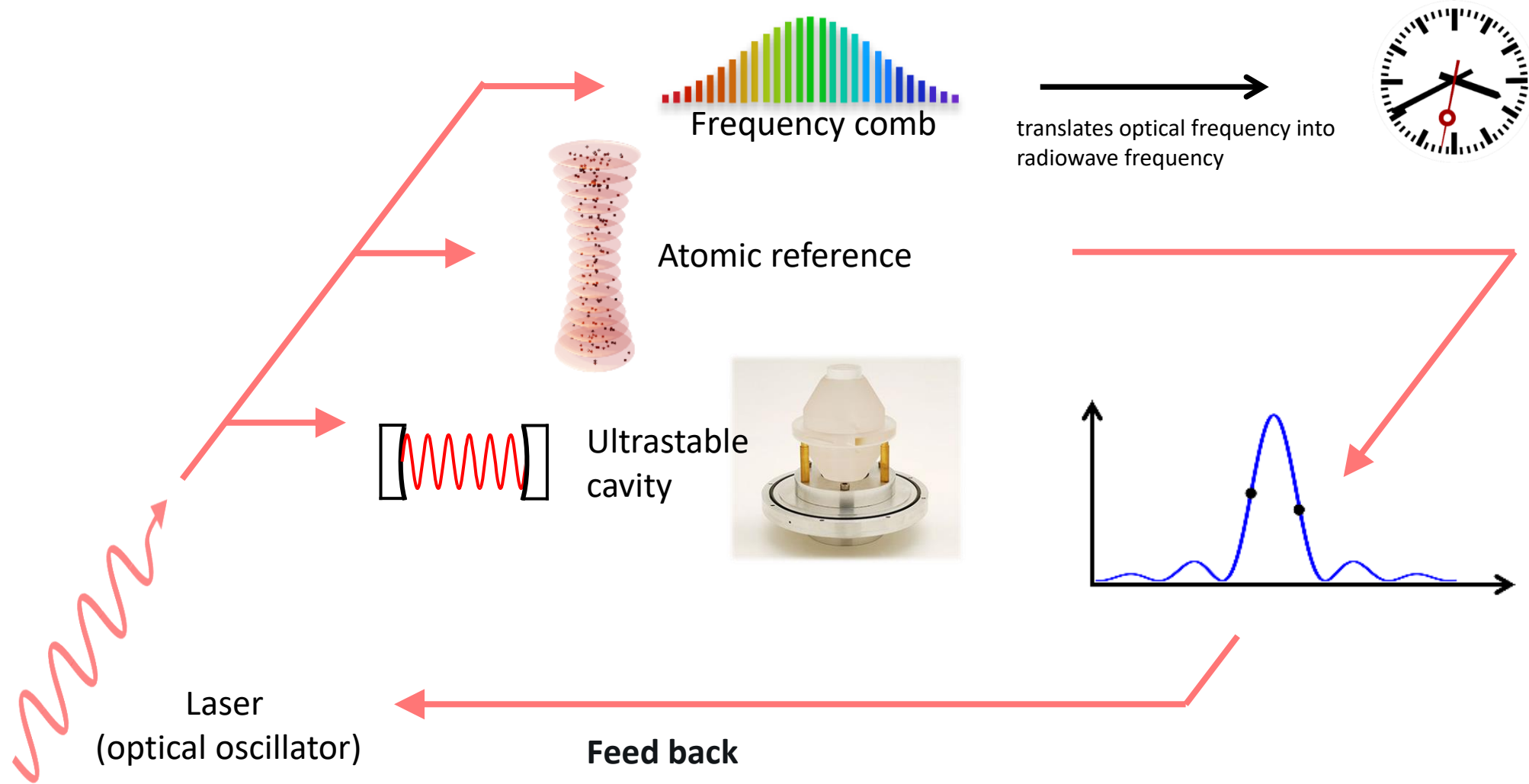
- **Group IIIA** elements (**trapped-ion**), namely Tl^+ , In^+ , Ga^+ , Al^+ , B^+ are excellent candidates for such work.
- 10^{-18} fractional uncertainty in not-too-distant future.
- The benefits to physics, geology, cosmology.
- **Group-II** elements (**neutral atom**) also applies (later suggested by Hitetoshi Katori)



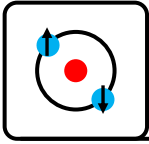
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optical transition. The
this case can be ex-
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py could be carried out
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group IIIA ions. For
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-photon optical transi-
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-Dicke criterion; this is
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Principle of Optical Atomic Clocks

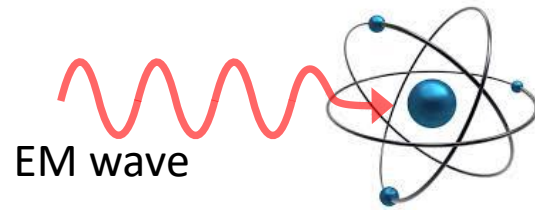


- Scott A. Diddams et al. ,Optical frequency combs: Coherently uniting the electromagnetic spectrum. Science 369, 3676 (2020).
- A. D. Ludlow, M. M. Boyd, J. Ye, E. Peik, and P. O. Schmidt, Optical atomic clocks, Rev. Mod. Phys. 87, 637 (2015).

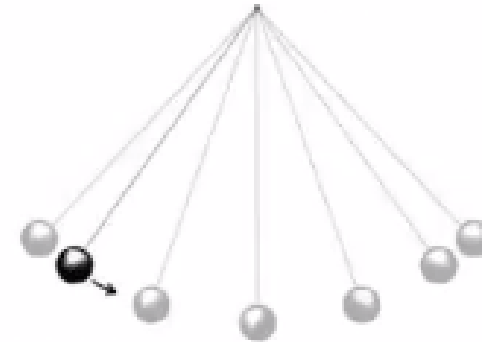
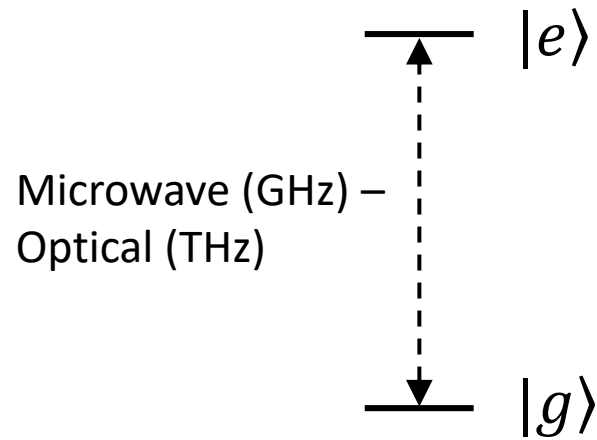


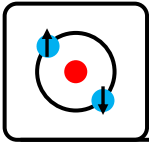
Atomic frequency standards

Atom



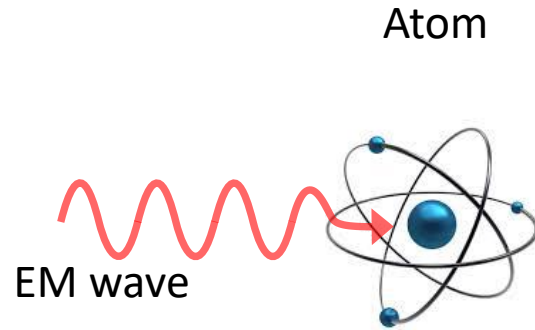
Clock



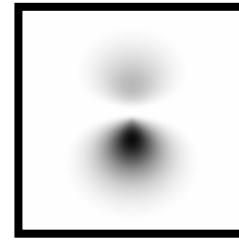


Atomic frequency standards

Atomic energy state **superposition** act like pendulum clock



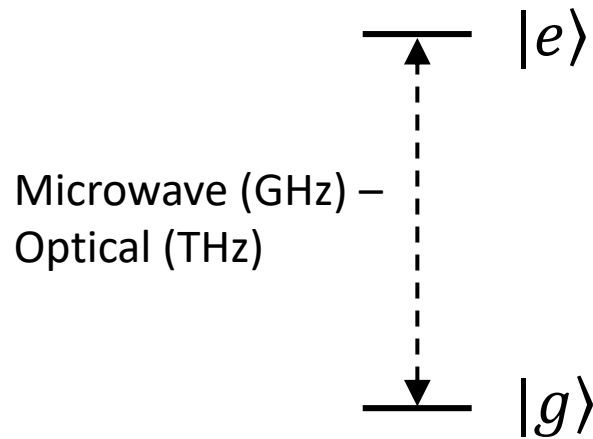
— $|2\rangle$



$$\psi = |1\rangle + e^{-i2\pi f_0 t} |2\rangle$$

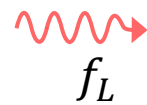
— $|1\rangle$

$$f_0 = \left(\frac{E_2 - E_1}{h} \right)$$



Practical mode of operation

— $|2\rangle$



f_L

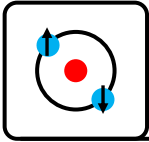
— $|1\rangle$

1. Start in $|1\rangle$

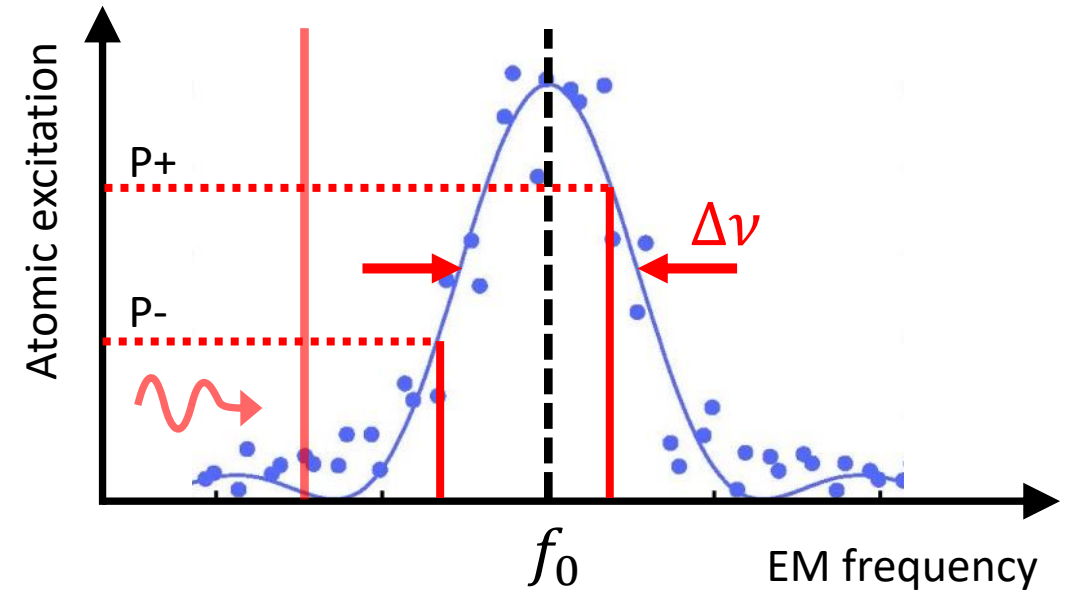
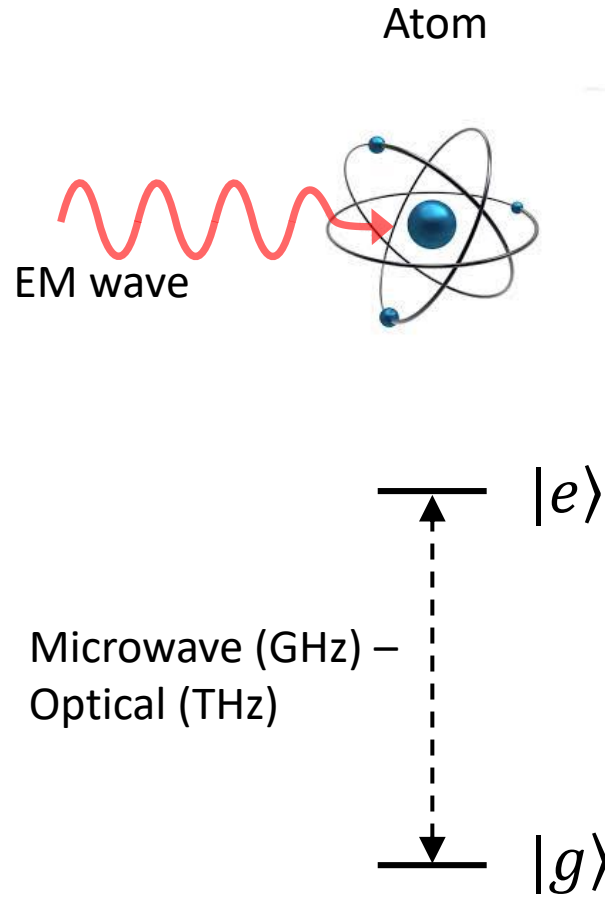
2. Apply radiation at frequency f_L for short time

3. Measure probability of $|2\rangle$

maximum $|1\rangle \rightarrow |2\rangle$ transition probability when $f_L = f_0$

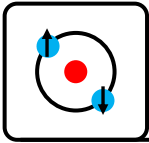


Atomic frequency standards

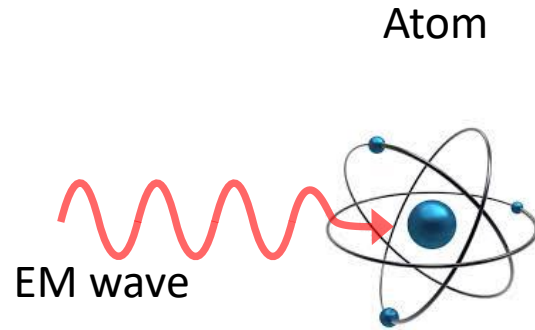


Excitation probabilities used to produce error signal

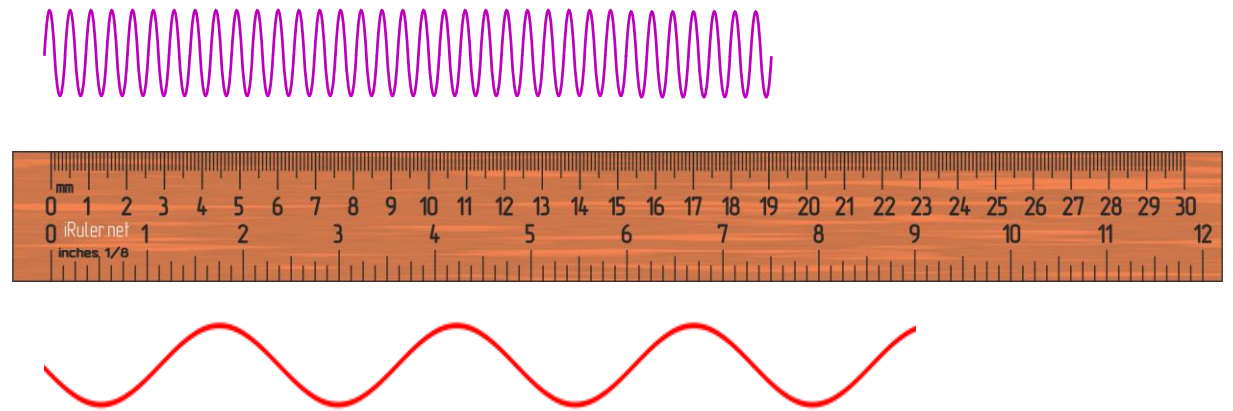
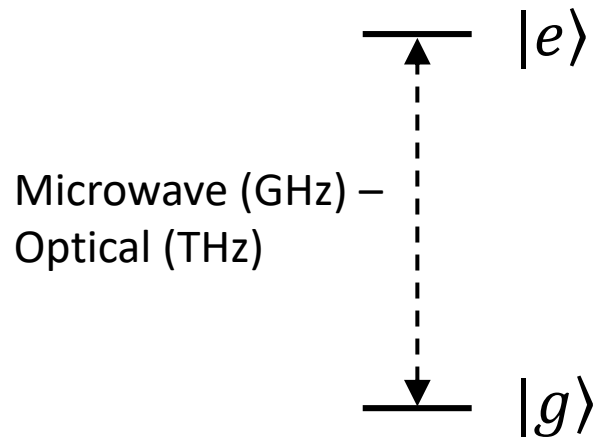
$$\text{Error} = \frac{p_+ - p_-}{p_+ + p_-}$$

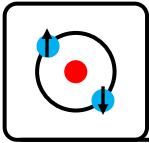


Atomic frequency standards: **Optical** Frequencies

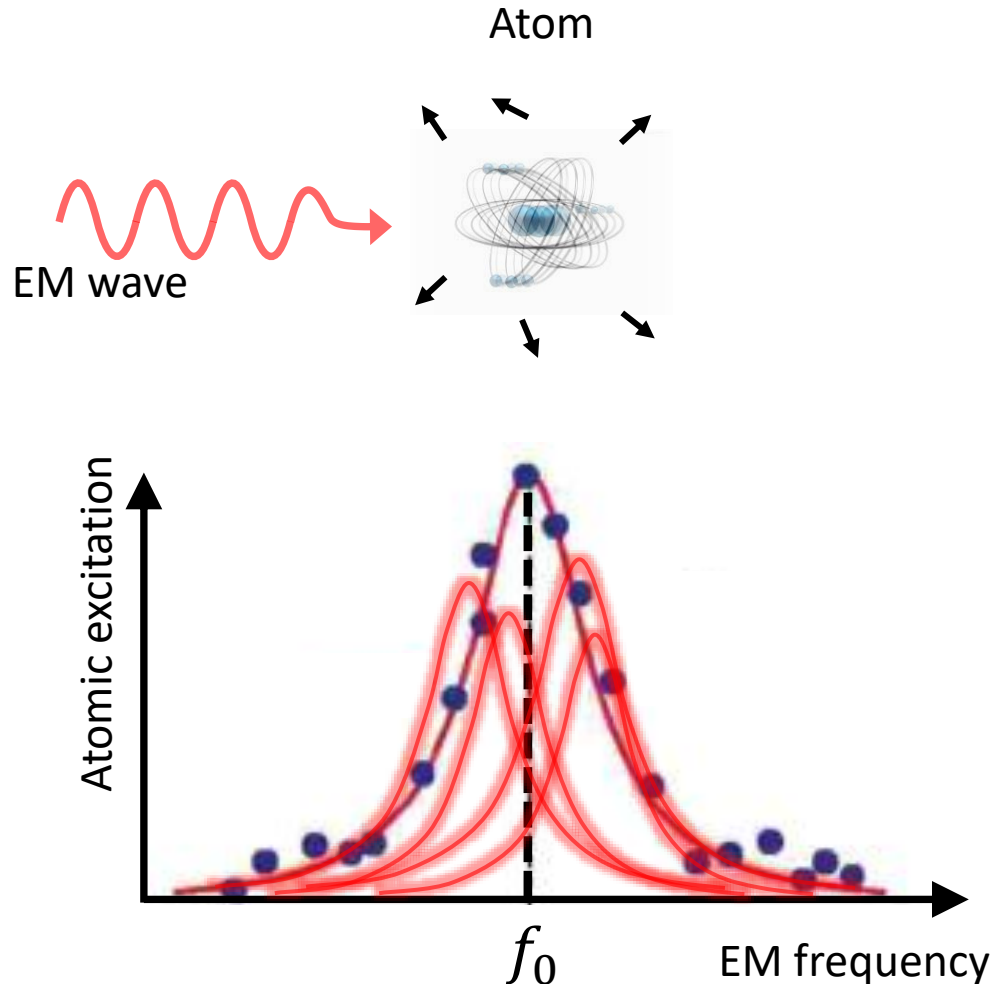


- High operating frequency : **optical** frequency
- Well-defined transition frequency
- States that have **low sensitivity** to and/or can be isolated from the environment





Atomic frequency standards



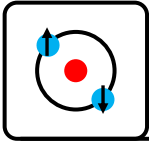
$$\omega = \omega_0 + \mathbf{k} \cdot \mathbf{v} + \frac{\hbar k^2}{2m} - \frac{1}{2} \frac{\omega_0 v^2}{c^2}$$

- Doppler shift
- Recoil shift
- 2nd order Doppler effect (Time dilation shift)
 - Treat energy conservation relativistically

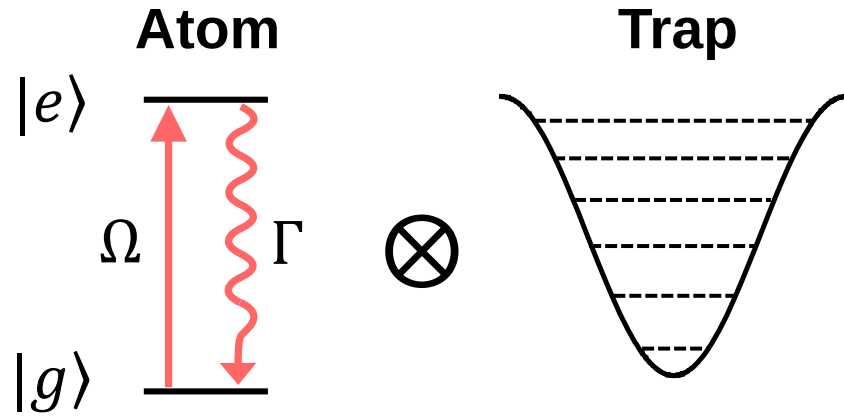
$$E = \sqrt{p^2 c^2 + m_0^2 c^4}$$



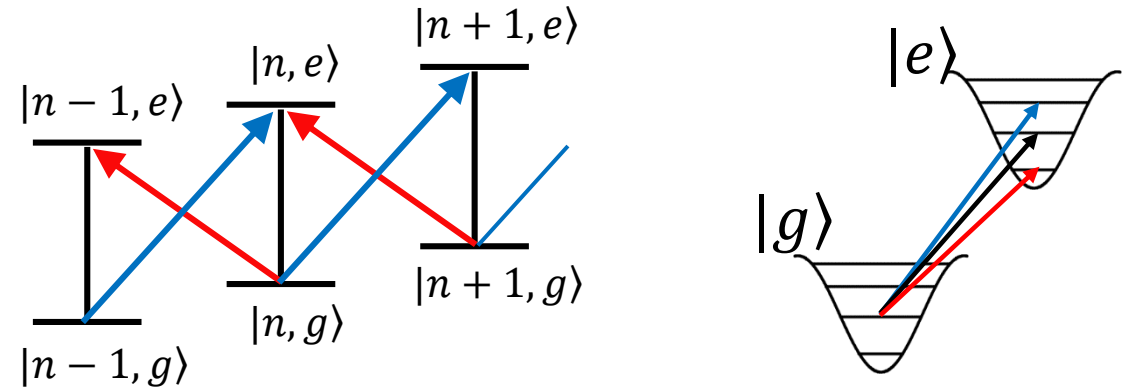
Spectroscopy in the **Lamb Dicke** regime
(freeze the atom)



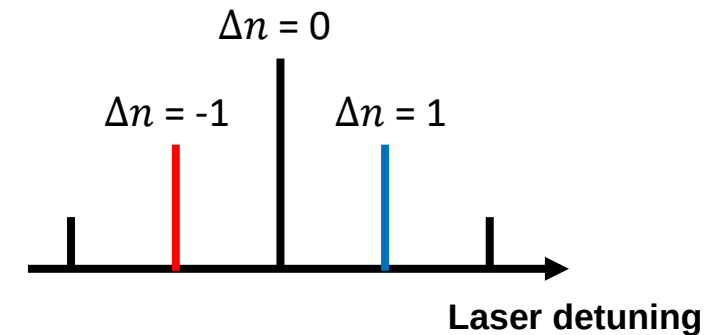
Electronic and Motional excitation

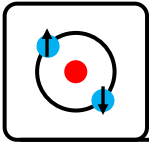


Coupled system:

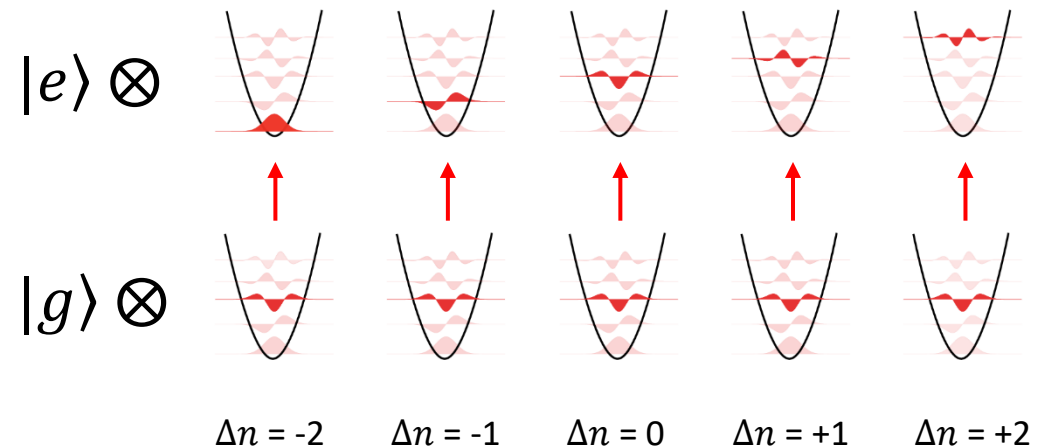
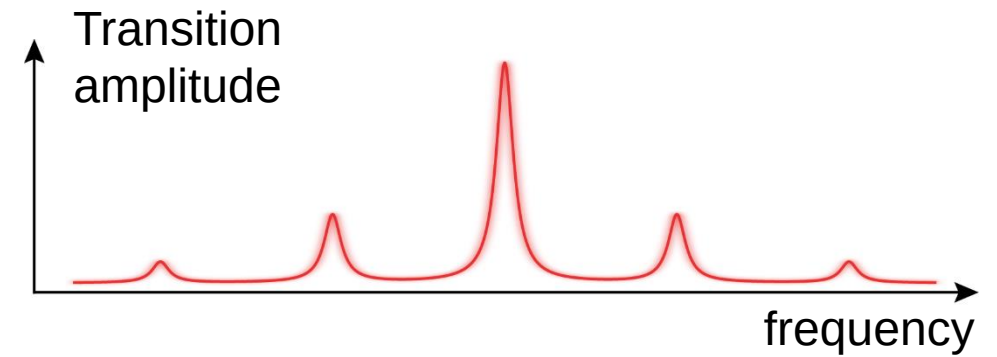
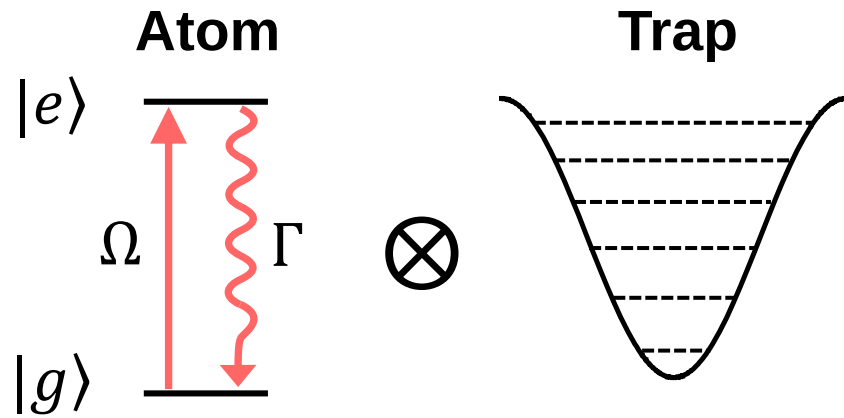


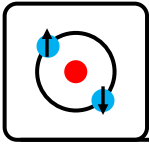
Spectroscopy: carrier and sidebands



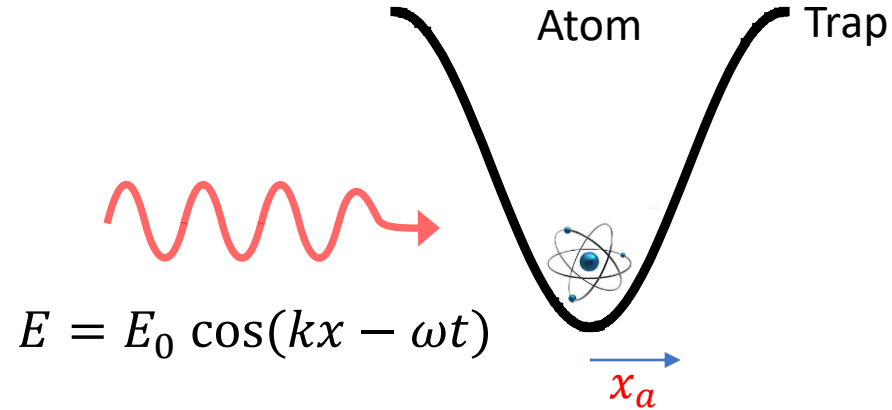


Electronic and Motional excitation





Lamb-Dicke confinement - I



- Light field seen by an atom:

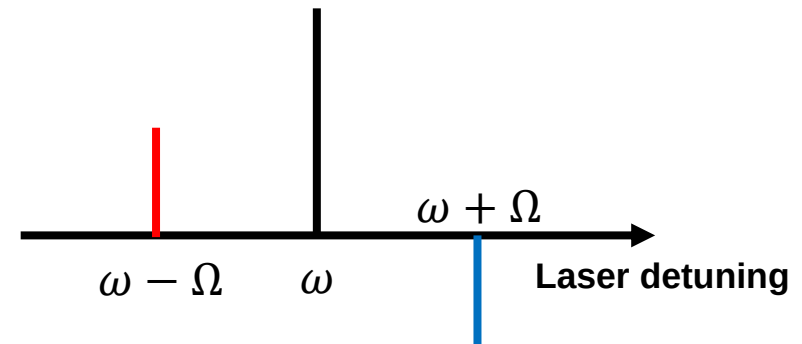
$$E = E_0 \cos(kx_a \sin \Omega t - \omega t)$$

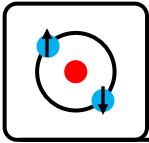
$$\approx E_0 \left[\cos \omega t + kx_a \frac{\cos(\omega - \Omega)t - \cos(\omega + \Omega)t}{2} \right]$$

Lamb-Dicke condition

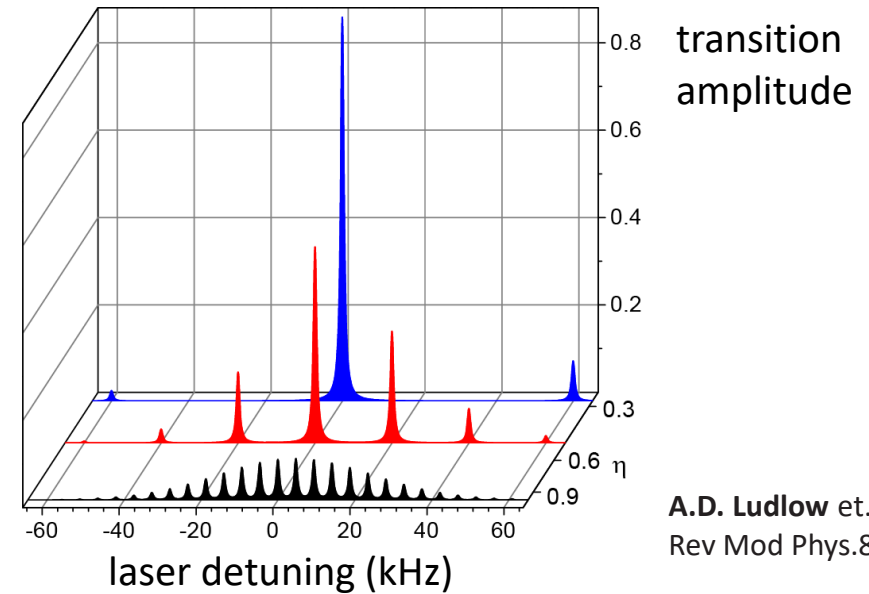
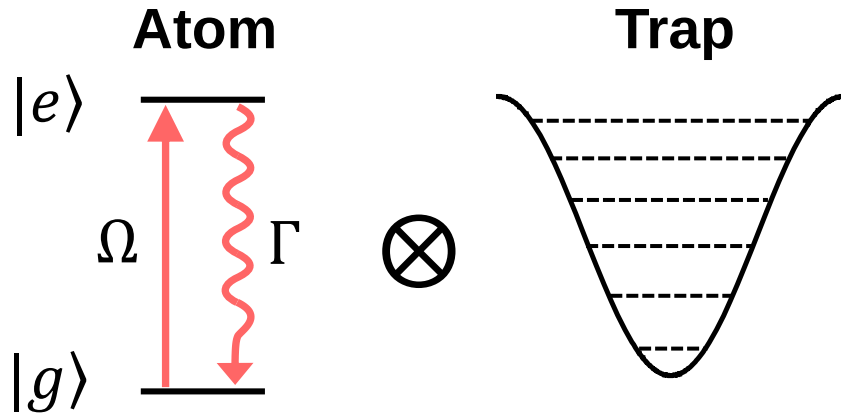
$$kx_a \approx x_a/\lambda \ll 1$$

Spatial confinement $\ll \lambda$





Lamb-Dicke confinement - II



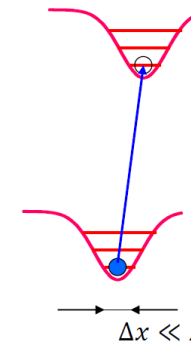
A.D. Ludlow et.al,
Rev Mod Phys.87.637(2015)

- Frozen the external degree of freedom to attain, Doppler-free, Recoil free spectroscopy.
- Lamb Dicke regime: ($\eta \ll 1$)

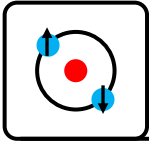
$$\eta = k \frac{x_0}{\sqrt{2}} = k \sqrt{\frac{\hbar}{2m\omega_{ho}}} = \sqrt{\frac{\omega_{rec}}{\omega_{ho}}}$$

Length scale:

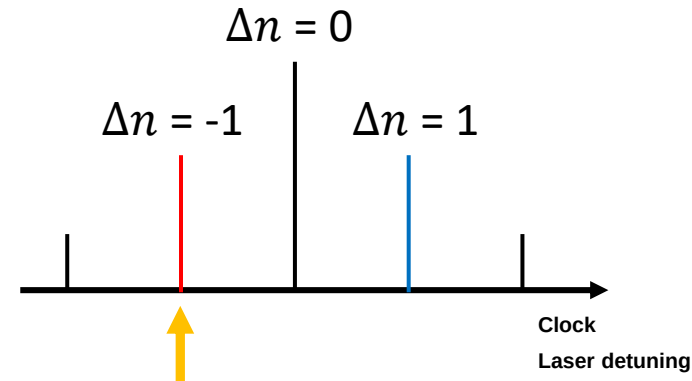
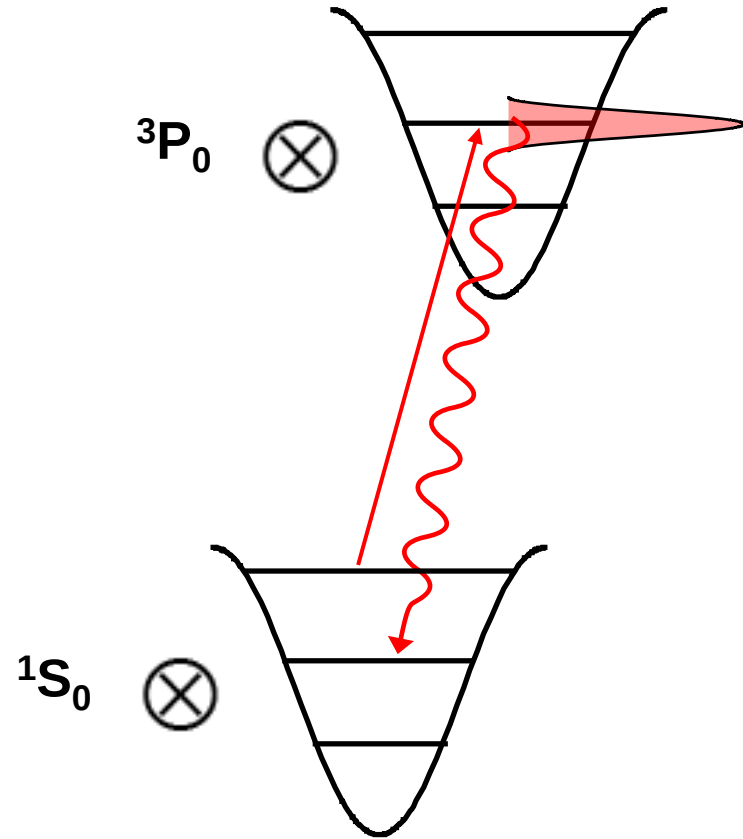
Energy scale:



$$E_R = \frac{\hbar^2 k^2}{2m} \ll \hbar \omega_{ho}$$



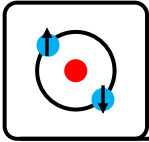
Motional sideband $\rightarrow$ red sideband cooling



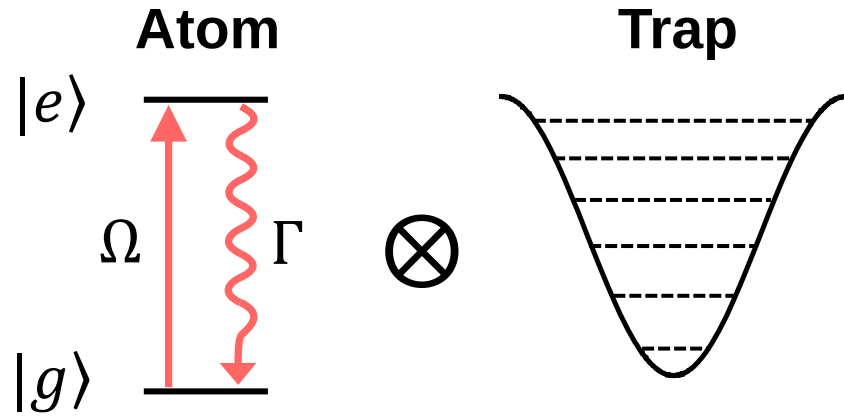
Red sideband cooling: $\Delta n = -1$

Repetitive red sideband cooling

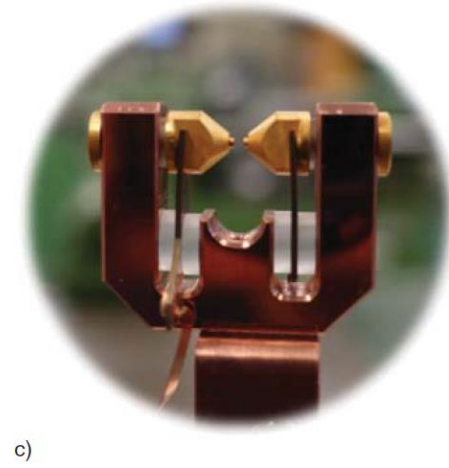
$\rightarrow$ Atoms accumulate in the motional ground state



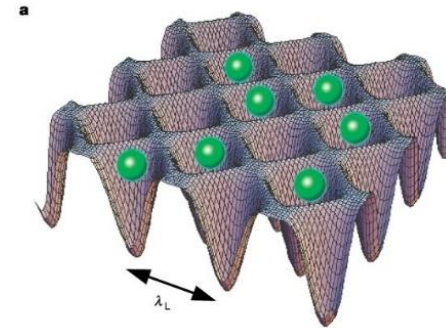
Electronic and Motional excitation

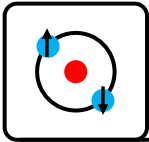


Trapped ion



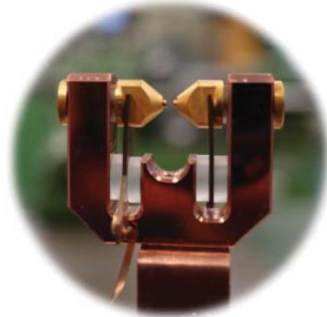
Optical lattice clock





Internal energy perturbation-free trap

Trapped ion



c)

Electric field strongly controls ion motion

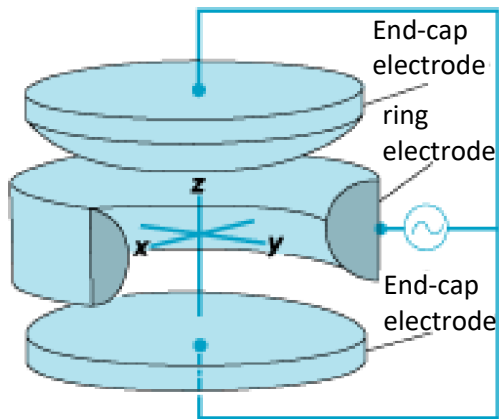
$$m\mathbf{a} = q\mathbf{E}$$

without strongly perturbing the internal energy of an ion,

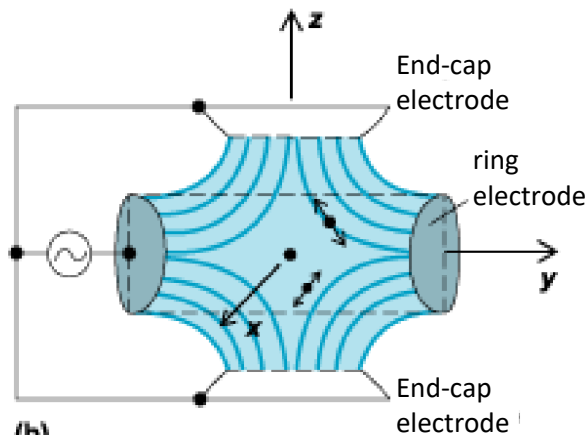
ac Stark effect :

$$U = -\frac{1}{2} \alpha |\mathbf{E}|^2$$

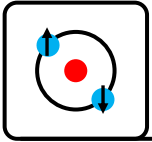
Earnshaw's theorem: three-dimensional trapping is not possible with static fields,
Use oscillating field.



(a)

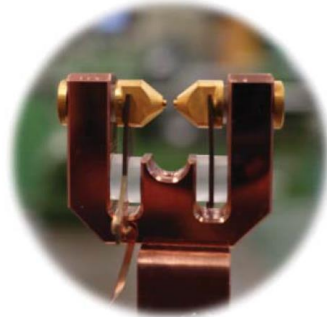


(b)

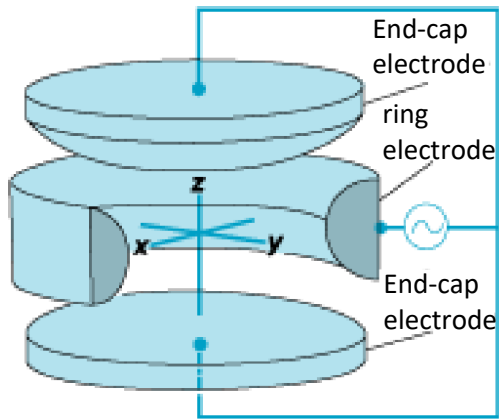


Internal energy perturbation-free trap

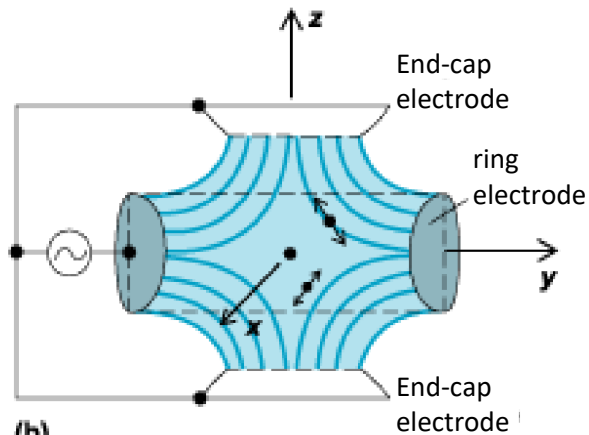
Trapped ion



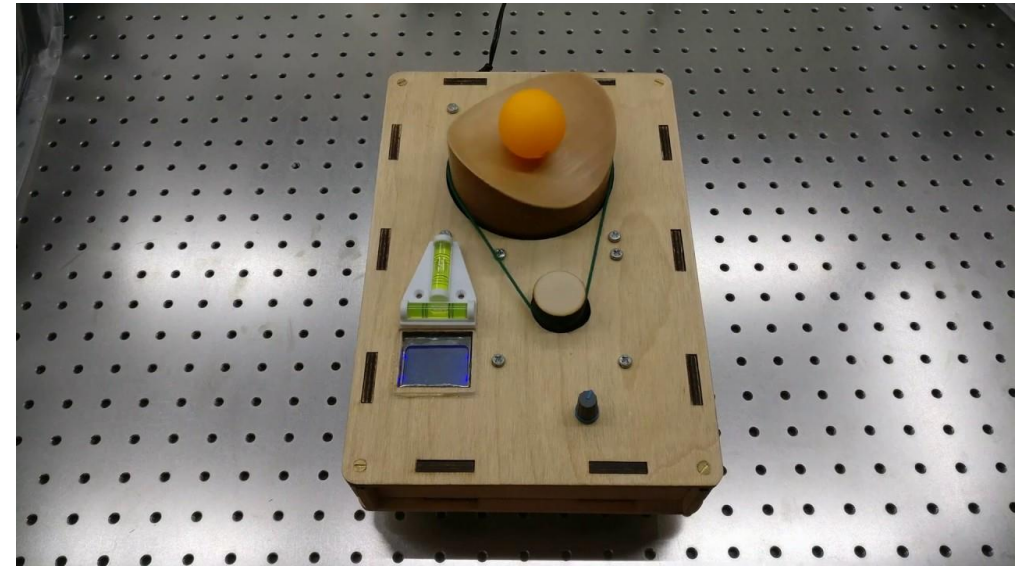
c)



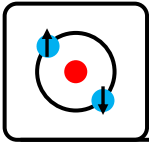
(a)



(b)

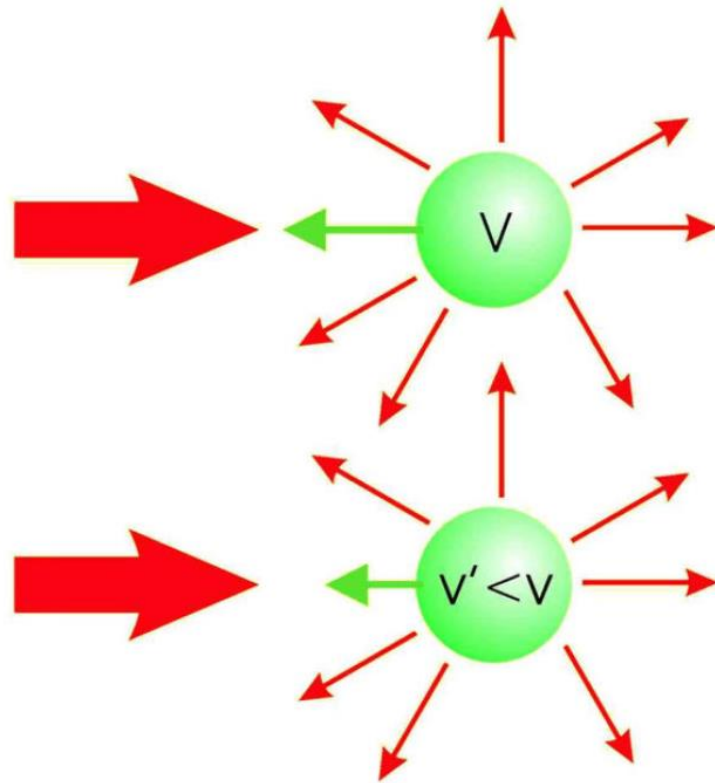


[Ion Trap toy \(youtube.com\)](https://www.youtube.com/watch?v=...)



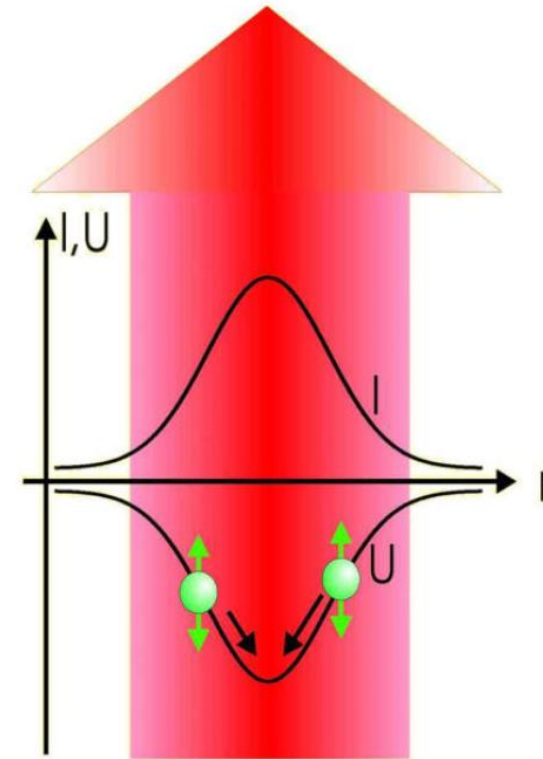
Trapping neutral atoms ?

Dissipative: Spontaneous Scattering

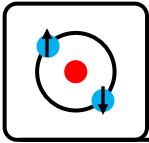


Laser Cooling, Preparation, and Readout

Conservative: Dipole Force



Dipole Trapping

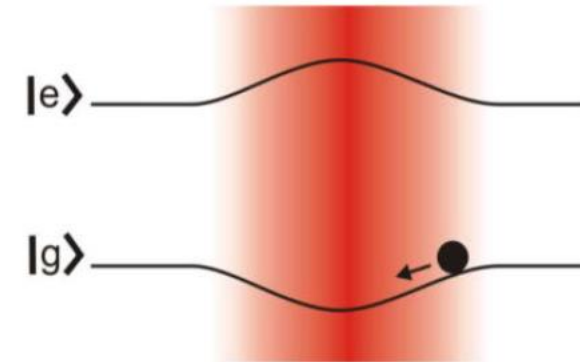


Trapping neutral atoms ?

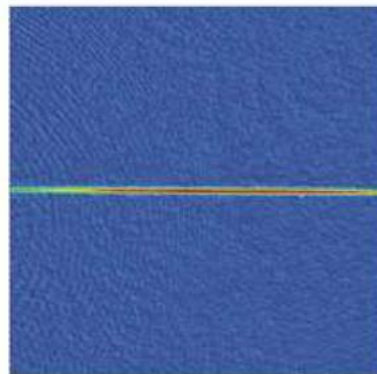
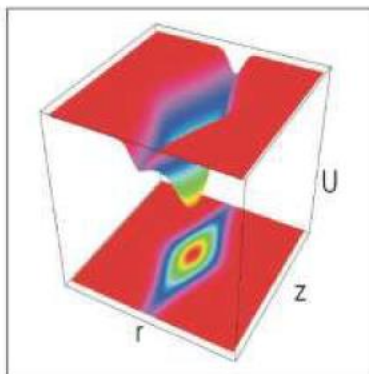
- Energy Shift Proportional to Intensity

$$U(r) \sim -\frac{I(r)}{\Delta}$$

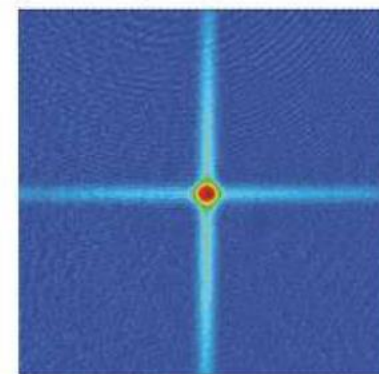
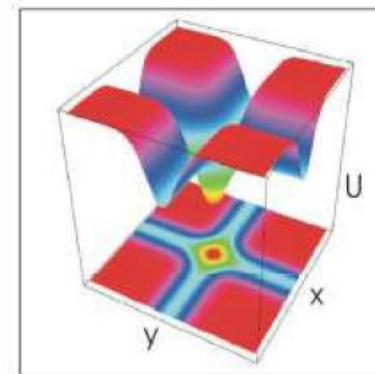
- Red detuning: $\omega_0 > \omega_L$
→ Atoms attracted to intensity maximum
- Large Detuning → Conservative Potential

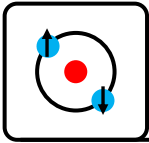


Single Beam Dipole Trap

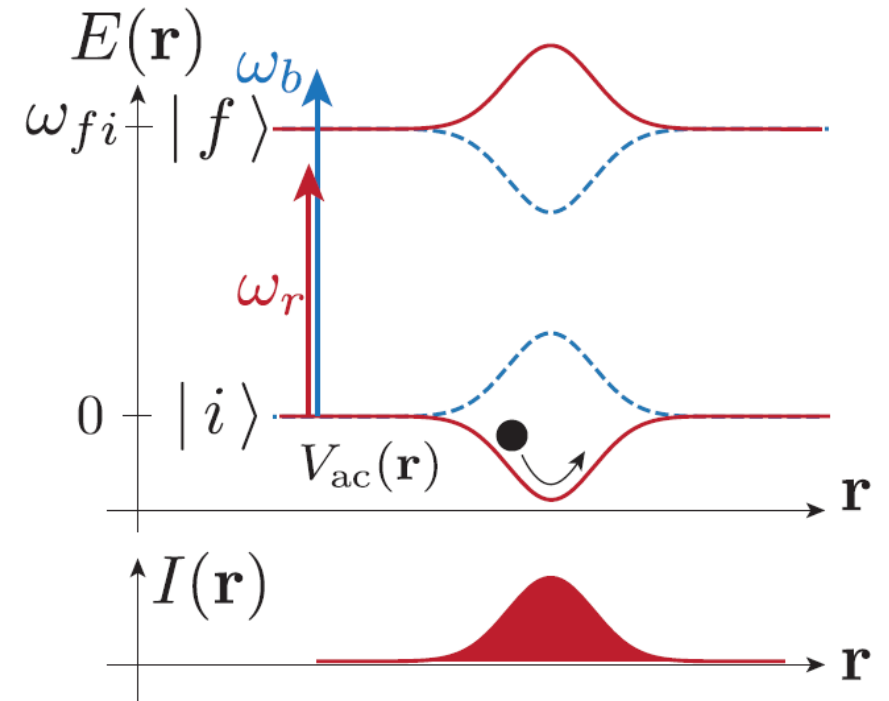
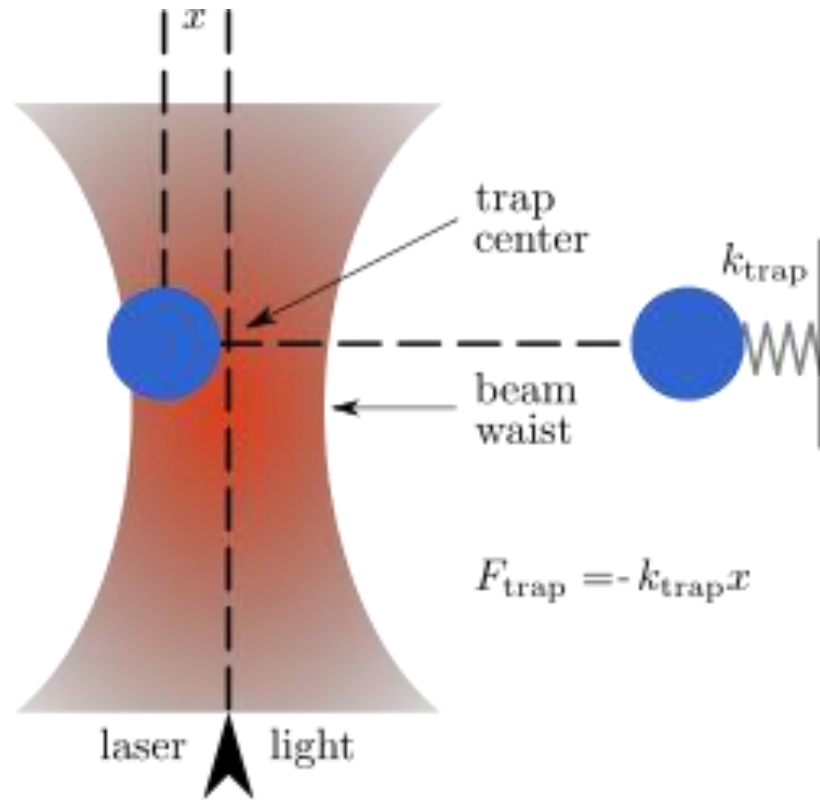


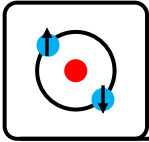
Crossed Beam Dipole Trap



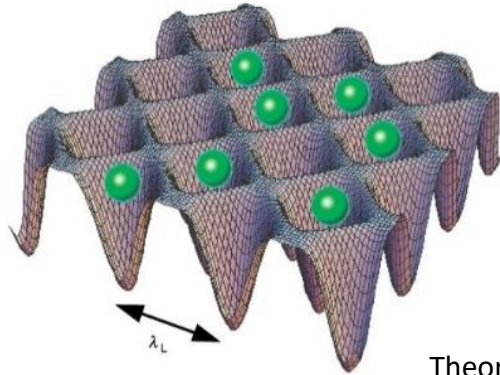


Trapping neutral atoms ?

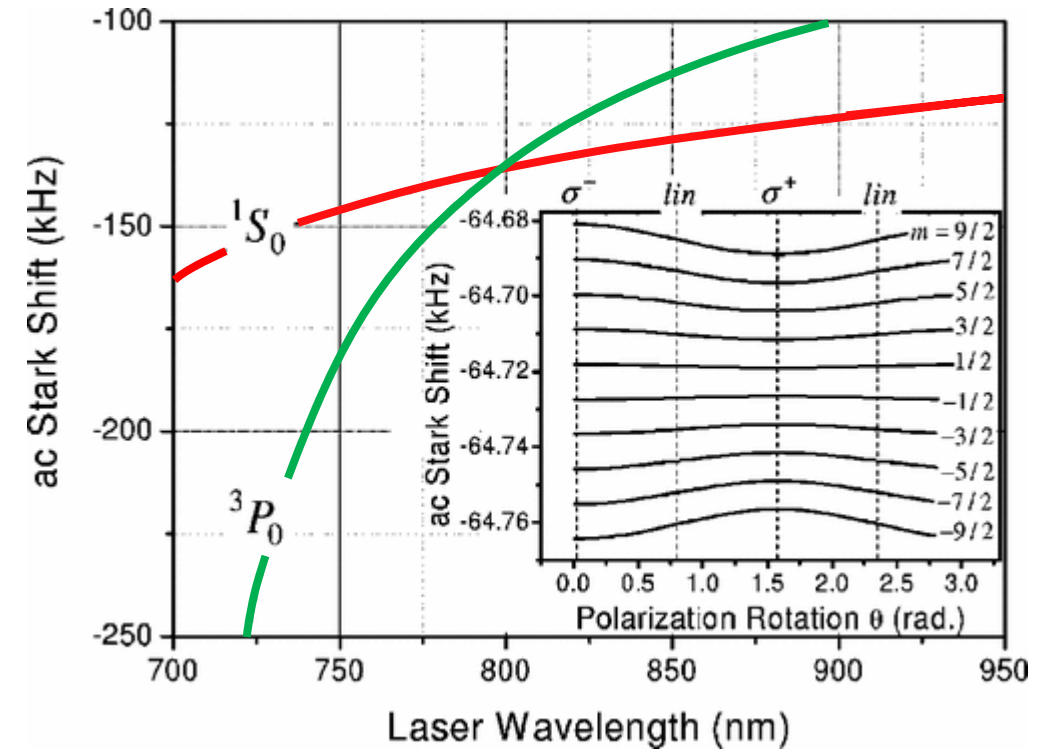
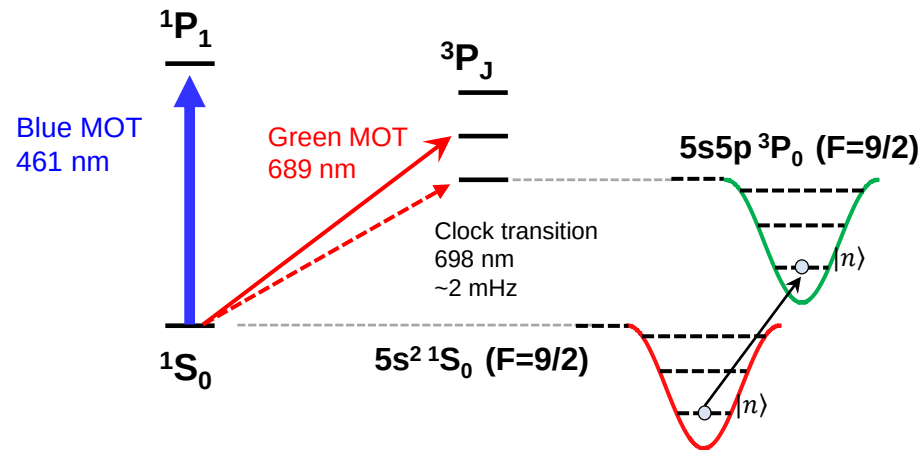




Engineered light shift trap (magic wavelength lattice)



Theory : Katori et.al, PRL (2003)
Experiment : Katori, Nature (2005)

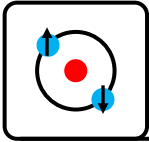


Hidetoshi Katori, Masao Takamoto, V. G. Pal'chikov, and V. D. Ovsiannikov, PRL **91**, 173005 (2003)

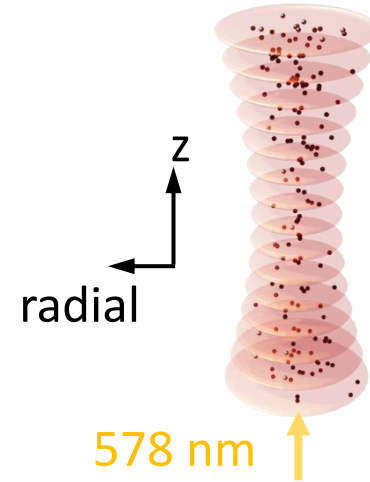
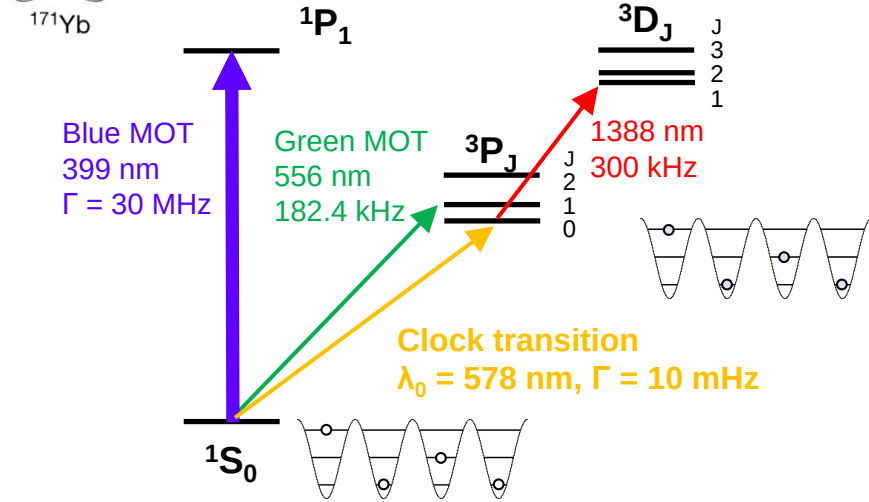


Clock Figures of Merit 101

- N. D. Lemke
PhD thesis, 2012, Optical Lattice Clock with Spin-1/2 Ytterbium Atoms
- N. Hinkley
PhD thesis, 2016, An atomic clock for 10^{-18} timekeeping
- F. McGrew
PhD thesis, 2020, An Ytterbium Optical Lattice Clock with Eighteen Digits of Uncertainty, Instability, and Reproducibility
- McGrew, ... A.D. Ludlow,
Atomic clock performance enabling geodesy below the centimetre level.
Nature **564**, 87–90 (2018)

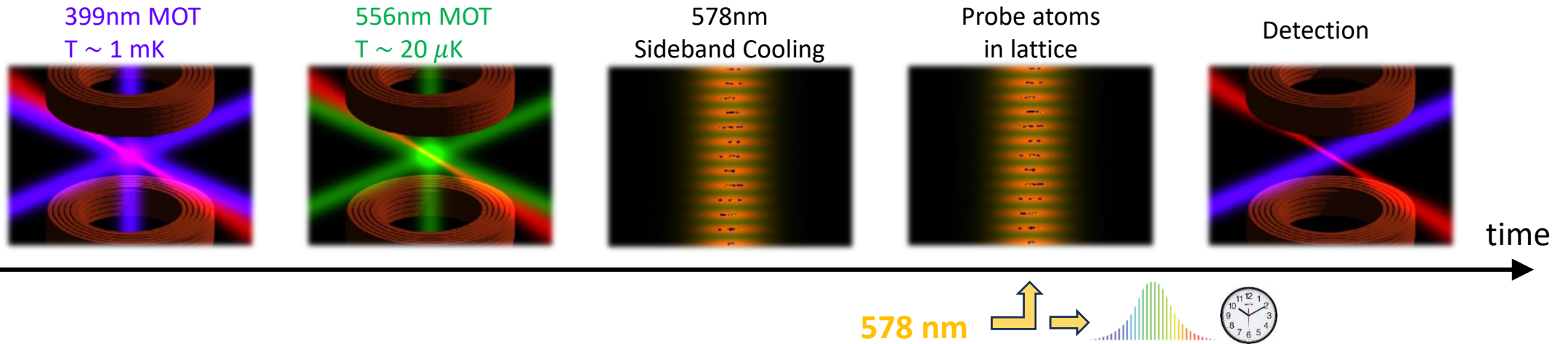


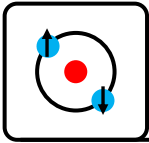
Optical lattice clock



	($\times 10^{-18}$)
blackbody radiation shift	0.9
lattice light shift	0.9
all others	0.6
TOTAL	1.4

McGrew, ... A.D. Ludlow, Nature (2018)





Clock Figures of Merit 101



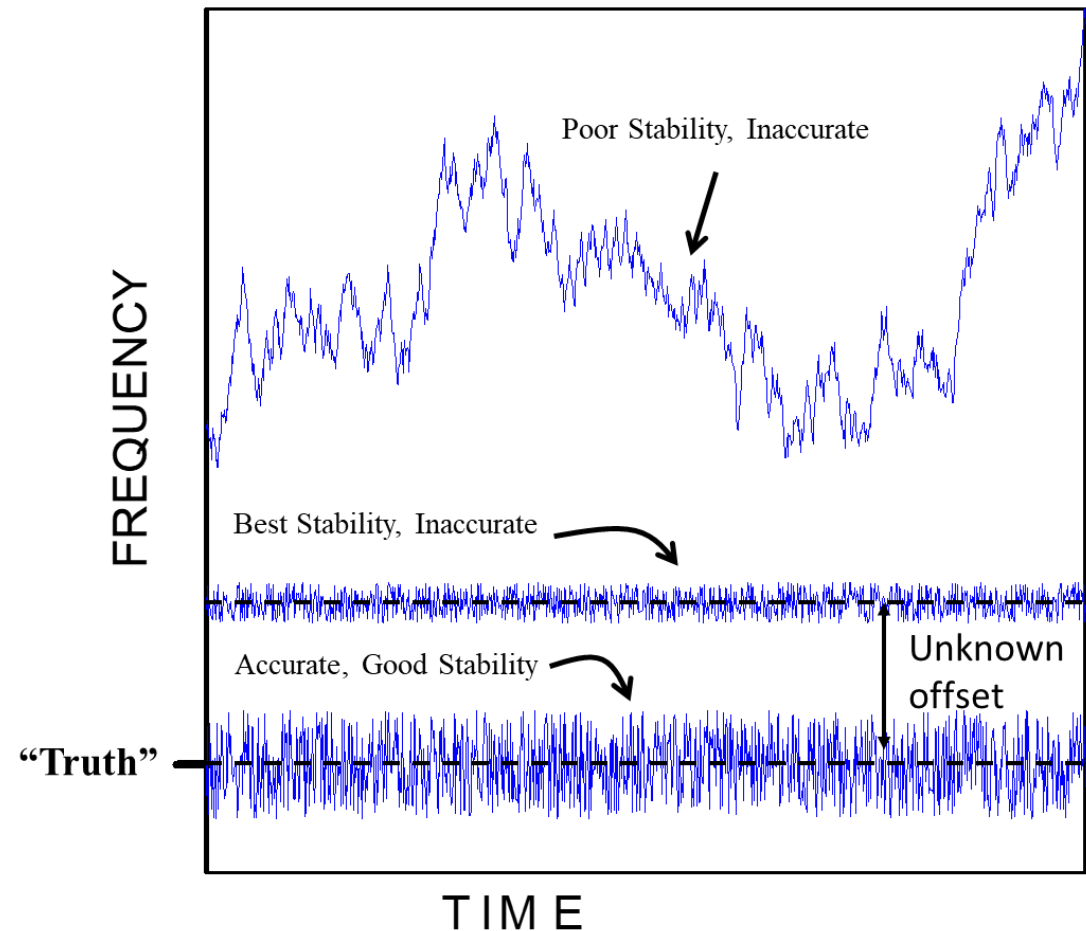
$$f(t)/f_0 = 1 + \epsilon_1 + y_1(t)$$

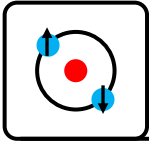
$$f(t)/f_0 = 1 + \epsilon_2 + y_2(t)$$

- Accuracy
- Stability
- Reproducibility

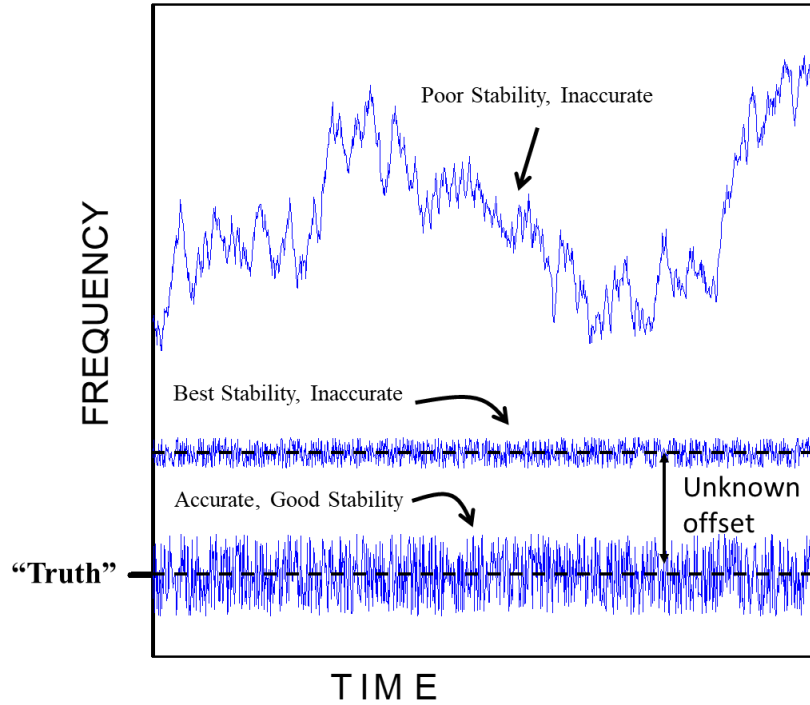
On stability and accuracy

- To realize an accurate frequency standard, frequency stability is essential, but that's not the end of the story.
- We also need to understand (and ideally measure) all the effects that can shift the standard's output frequency.
- After all frequency shifts have been quantified, we should have an accurate standard.
- How do we know? Better build another one to check!

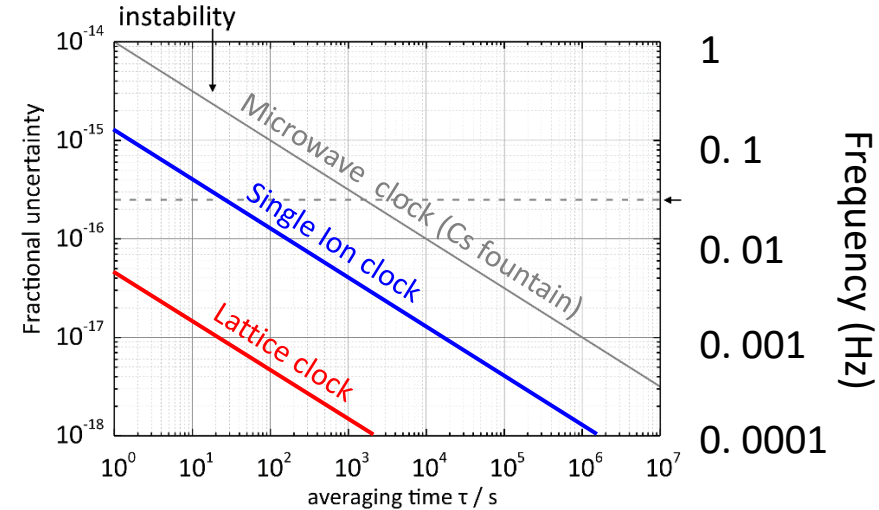




Clock Figures of Merit : **Instability**

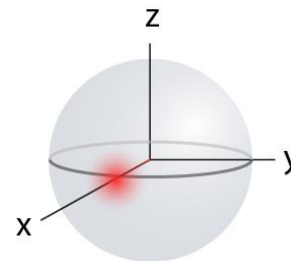


Allan deviation ($<10^{-18}$ at long times)



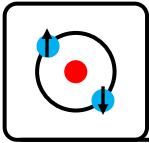
$$\sigma_y(\tau) = \sqrt{\left\langle \frac{(\bar{y}_{i+1} - \bar{y}_i)^2}{2} \right\rangle}$$

Quantum projection noise

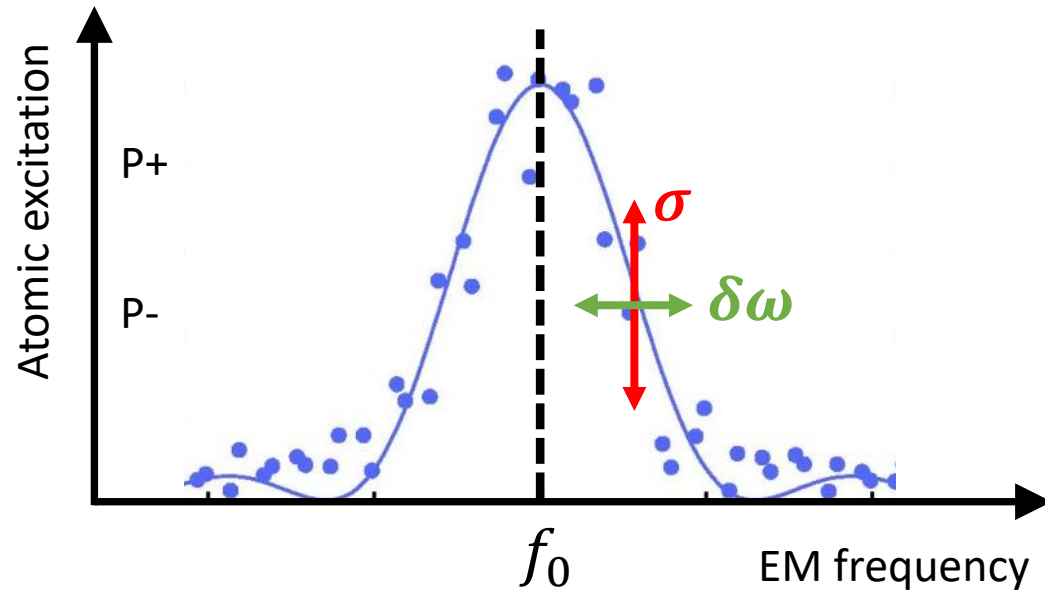


N uncorrelated atoms

Noise limit : $1/\sqrt{N}$



Quantum projection noise limit



How precise can we measure atomic resonance?

Measurement process project $|\phi(t)\rangle$ onto either $|g\rangle$ or $|e\rangle$ state.

Single particle: (Itano, *et.al*, PRA **47**, 3554 (1993))

$$\text{Variance } \sigma^2 = p_e (1 - p_e)$$

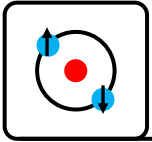
For an ensemble of uncorrelated **N** particles

$$\text{Variance } \langle \Delta N \rangle^2 = N \sigma^2 = N p_e (1 - p_e)$$

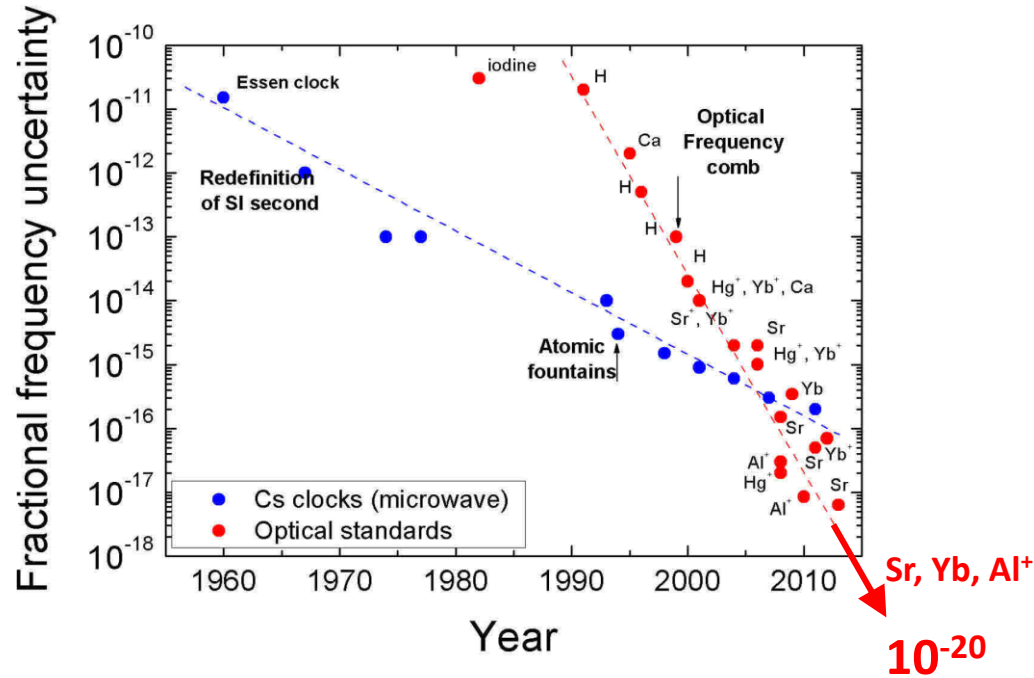
Uncertainty in **estimating frequency** (sensitive at point $p_e = 0.5$)

$$|\delta\omega_0| = \langle \Delta N \rangle \left| \frac{d\omega}{d(Np_e)} \right| \leq \frac{1}{T\sqrt{N}}$$

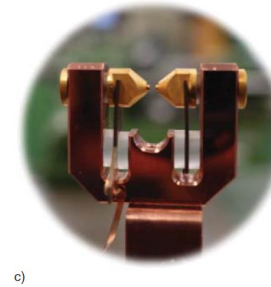
$$\text{Clock stability : } \sigma = \frac{\delta\omega_0}{\omega_0} = \frac{1}{T\omega_0} \frac{1}{\sqrt{N}}$$



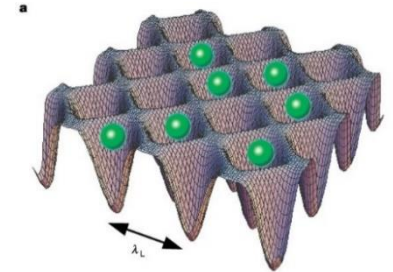
Atomic clock: sensors of Time & Space



Trapped ions



Neutral atoms

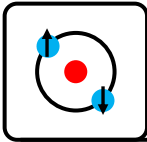


$$\sigma_{QPN}(\tau) \propto \frac{\Delta f}{f_0} \sqrt{\frac{1}{T_m N \tau}}$$

Poli... Tino, Rivista del Nuovo Cimento 36 (2014)

Boulder Atomic Clock Optical Network (BACON) Collaboration*, Nature (2021)

- Current accuracy $\sim 10^{-18}$ = grav. redshift @ 1 cm
- Precision $\sim 1 \times 10^{-19}$ (today)



Clock Figures of Merit : **Instability**

So ... how **good** is your clock?

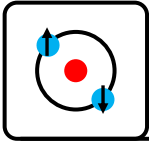


lattice clock numbers

$$10^{-17} \sigma_y(\tau) = \frac{\Delta\nu}{\nu_0} \frac{1}{\sqrt{N}} \frac{\sqrt{T}}{\sqrt{\tau}}$$

Annotations for the equation:

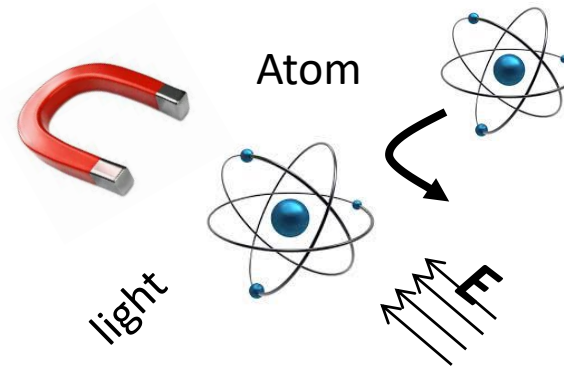
- $\Delta\nu$: Width of feature (<1 Hz)
- ν_0 : Oscillation frequency (500×10^{12} Hz)
- N : Atom number ($10^4 - 10^5$)
- T : Measurement cycle period (~ 1 s)
- τ : Total averaging period (1 s)



Clock Figures of Merit : **Accuracy**

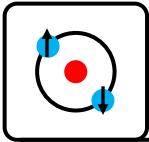
So ... *how good* is your clock?

UNCERTAINTY IN REALIZING UNPERTURBED EIGENFREQUENCY OF ATOM
→ characterized with an **uncertainty evaluation**



	Yb-1 ($\times 10^{-18}$)	
Systematic effects	Shift	Uncertainty
BBR Stark	-2361.2	0.9
Lattice statistical	-1.5	0.8
Lattice model	0	0.3
Lattice traveling wave	0	<0.1
Second-order Zeeman	-118.1	0.2
Spin polarization purity	0	0.3
Cold collision	-0.21	0.07
Background gas collision	-5.5	0.5
DC Stark	0	0.04
Probe AC Stark	0.02	0.01
First-order Doppler	0	0.02
Second-order Doppler	0	<0.1
Tunnelling	0	<0.01
Servo error	0.04	<0.1
Line Pulling	0	<0.1
AO Phase Chirp (OFS)	0	0.1
Total	-2486.5	1.4

Brown et al., PRL **119**, 253001 (2017);
Beloy et al., PRL (2018);
McGrew et al., Nature **564** 87 (2018)

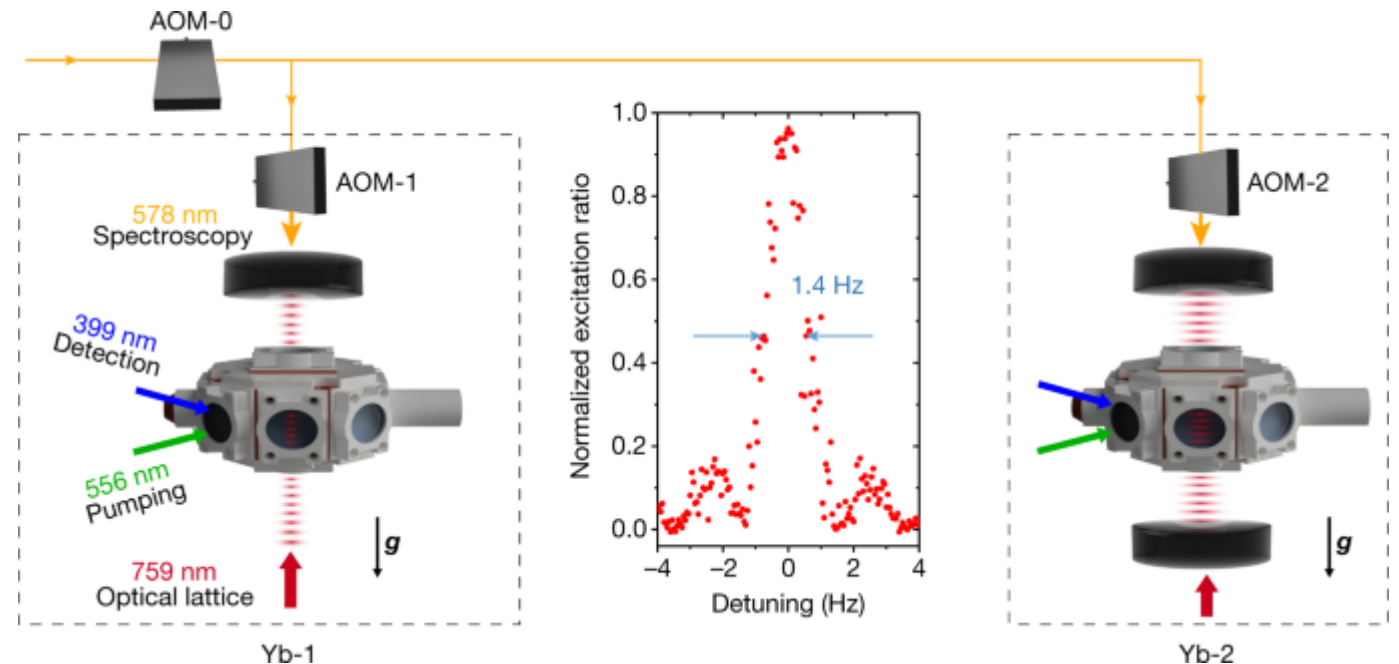


Clock Figures of Merit : **Reproducibility**

So ... *how good* is your clock?



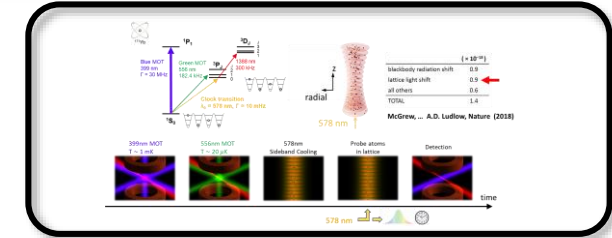
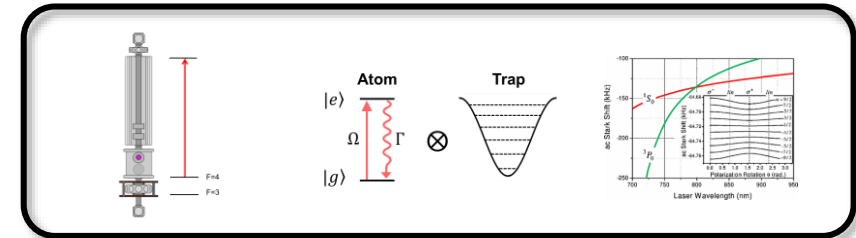
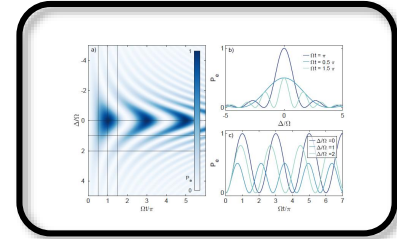
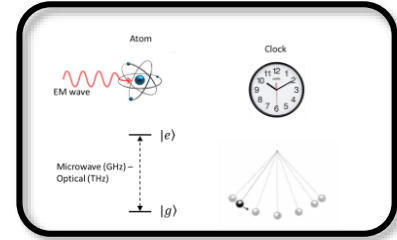
HOW WELL THE TIMEBASES OF TWO SIMILAR STANDARDS AGREE
→ **should be** determined by systematic uncertainty and stability

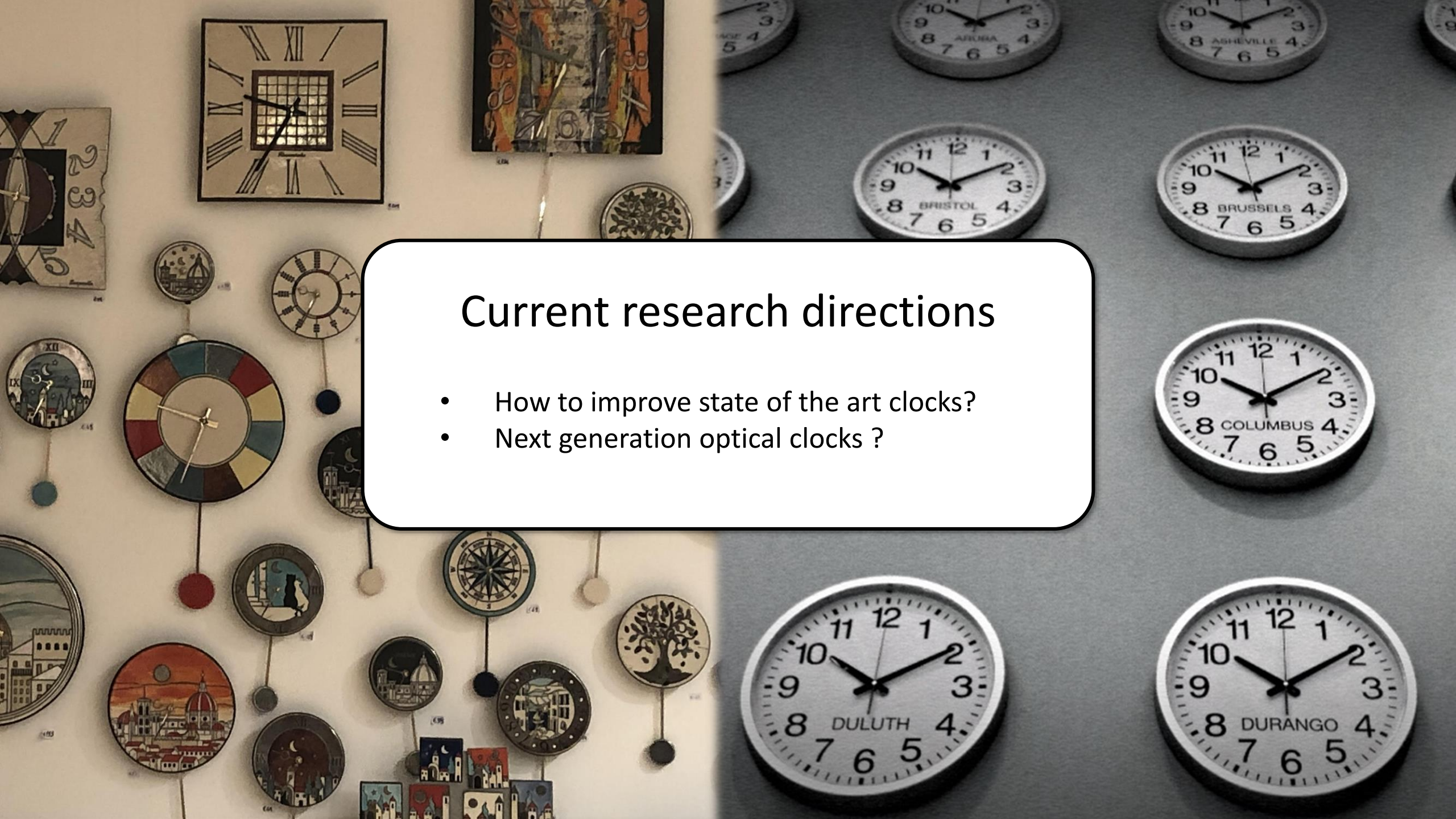


Brown et al., PRL **119**, 253001 (2017);
Beloï et al., PRL (2018);
McGrew et al., Nature **564** 87 (2018)

Outline

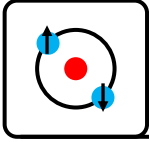
- Introduction
- Two-level system
 - Atom light interaction
 - Highly accurate two level systems – mHz forbidden optical transitions
- How optical atomic clock works:
 - Cs Atomic frequency standards
 - Physics of optical atomic clocks – (optical lattice clock)
- Current research directions
 - How to improve state of the art atomic clocks (optical lattice clock) ?
 - Next generation optical clocks ?





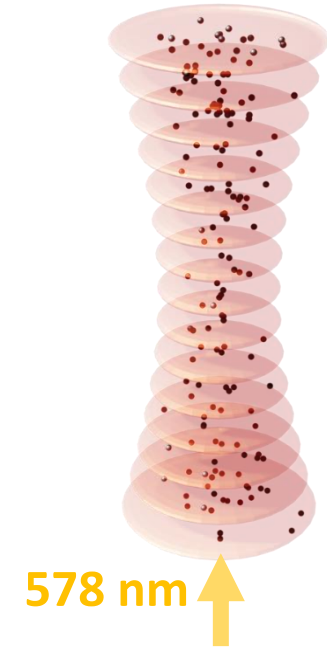
Current research directions

- How to improve state of the art clocks?
- Next generation optical clocks ?

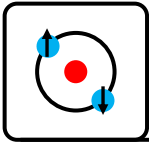


Systematic Uncertainty

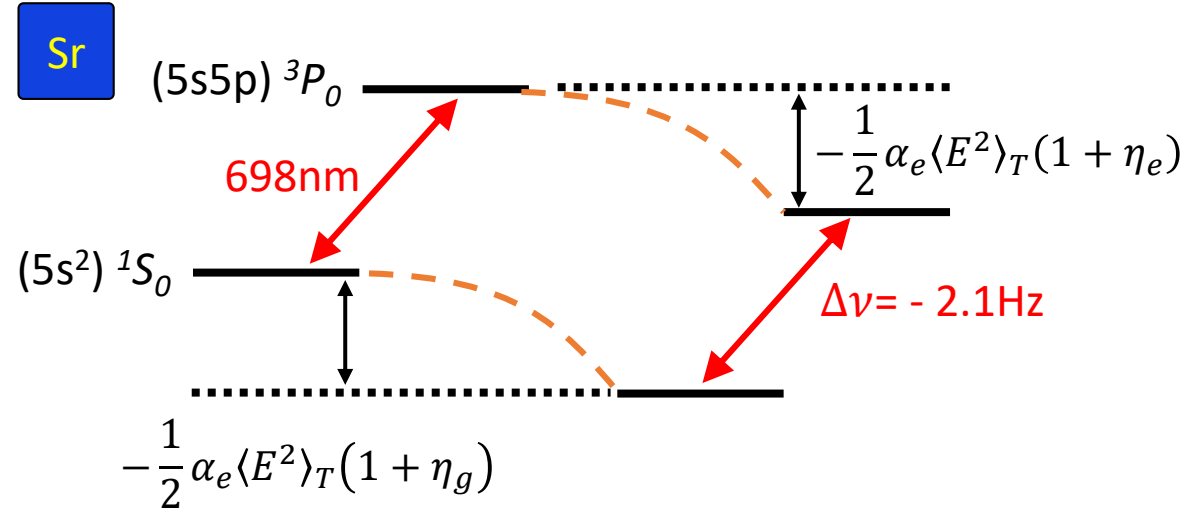
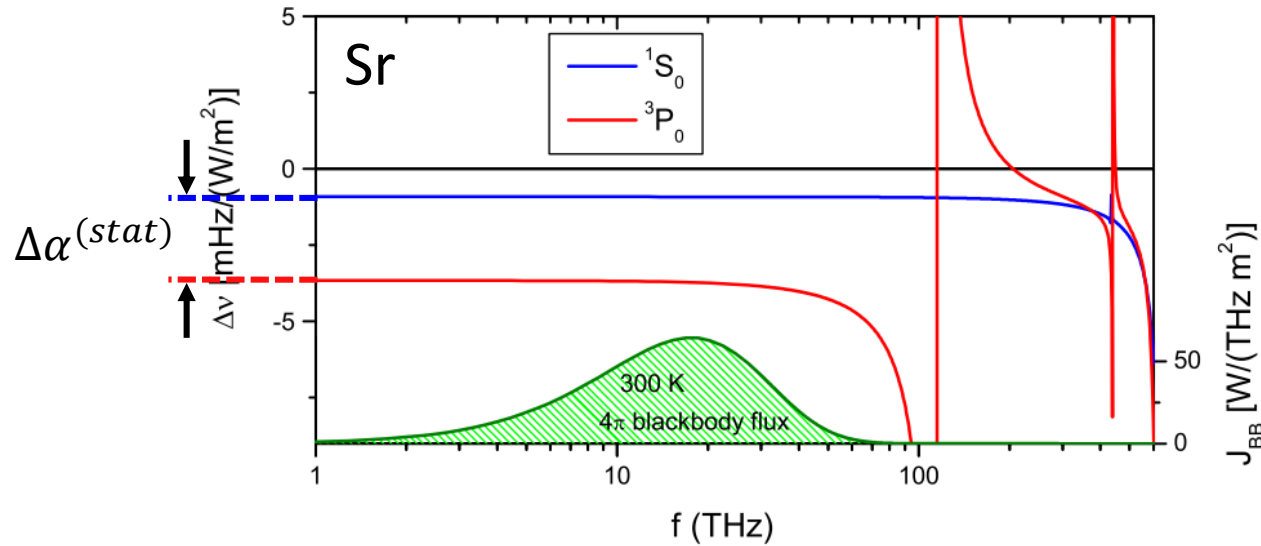
Blackbody radiation Shift



	($\times 10^{-18}$)
blackbody radiation shift	0.9
lattice light shift	0.9
all others	0.6
TOTAL	1.4



Blackbody radiation shift (BBR)



Blackbody radiation spectrum: $u(\nu) = \frac{8 h \nu^3}{c^3} \frac{1}{e^{h\nu/kT} - 1}$

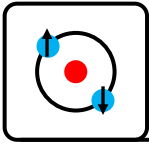
Energy density of BBR: $\langle E^2 \rangle = \int_0^\infty u(\nu) d\nu = \frac{8\pi^5 k_B^4}{15 c^3 \epsilon_0 h^3} T^4 = c_1 T^4$
 $\approx (8.319 \text{ V/cm})^2 \left(\frac{T}{300 \text{ K}}\right)^4$

$$\Delta\nu_{BBR}(T) = -\frac{1}{2h} \Delta\alpha(0) \langle E^2 \rangle_T [1 + \eta_{clock}(T)]$$

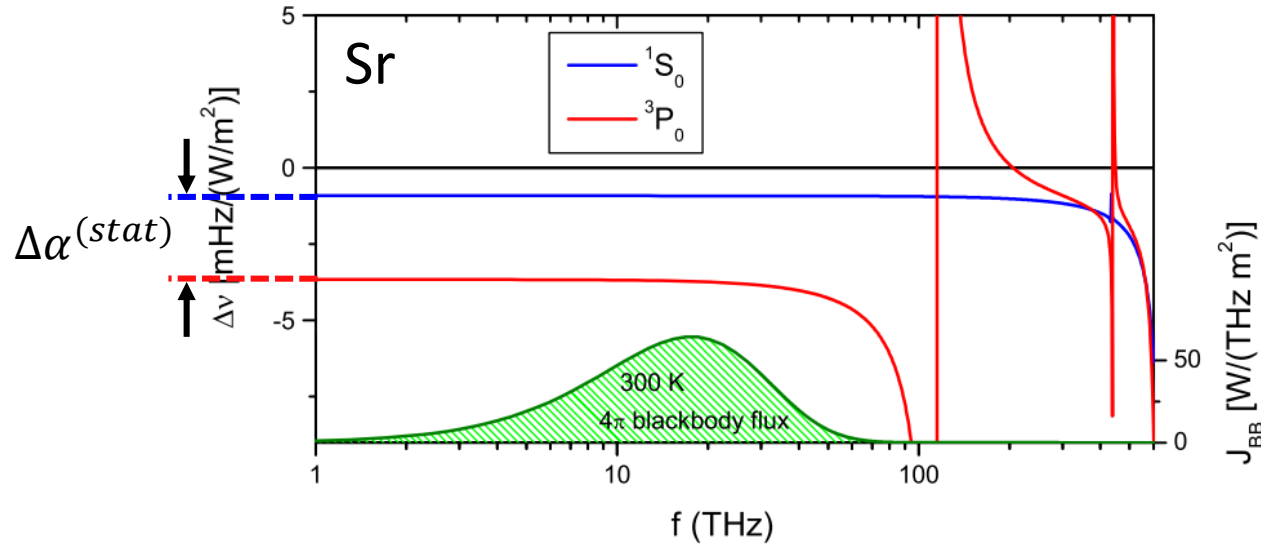
$$= \Delta\nu^{(stat)}(T) + \Delta\nu^{(dyn)}(T)$$

Polarizability: level shift by EM field

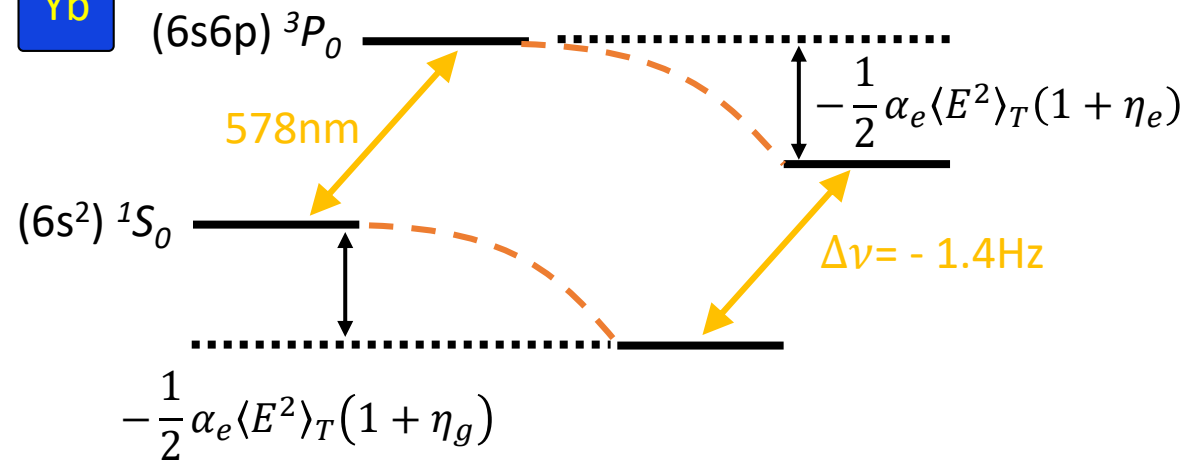
$$\delta E_i(\omega) = \frac{1}{2\hbar} \sum_k \frac{\omega_{ik} |\langle i | e\vec{r} \cdot \vec{\epsilon} | k \rangle|^2}{\omega_{ik}^2 - \omega^2}$$



Blackbody radiation shift (BBR)



Yb



Blackbody radiation spectrum: $u(\nu) = \frac{8 h \nu^3}{c^3} \frac{1}{e^{h\nu/kT} - 1}$

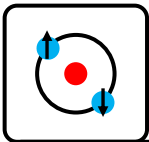
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 $\approx (8.319 \text{ V/cm})^2 \left(\frac{T}{300 \text{ K}}\right)^4$

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$$\delta E_i(\omega) = \frac{1}{2\hbar} \sum_k \frac{\omega_{ik} |\langle i | e\vec{r} \cdot \vec{\epsilon} | k \rangle|^2}{\omega_{ik}^2 - \omega^2}$$



Blackbody radiation shift (BBR)

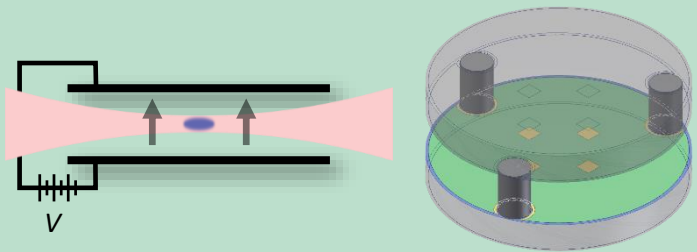
	($\times 10^{-18}$)
blackbody radiation shift	0.9
lattice light shift	0.9
all others	0.6
TOTAL	1.4

McGrew, ... A.D. Ludlow et.al,
Nature **564**, 87–90 (2018)

$$\Delta\nu_{BBR}(T) = -\frac{1}{2h} \Delta\alpha(0) \langle E^2 \rangle_T [1 + \eta_{clock}(T)]$$

now 2×10^{-5} uncertainty

Differential static polarizability



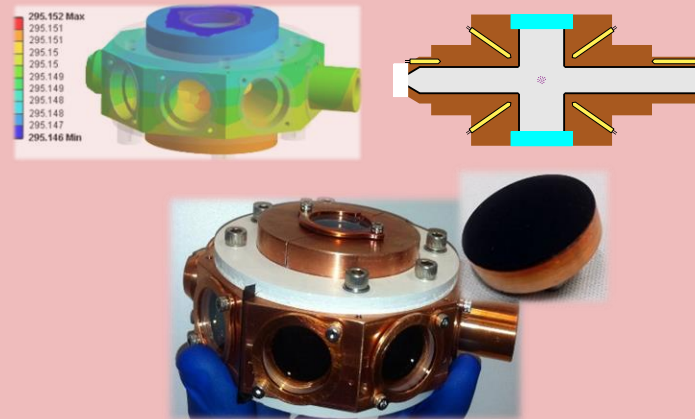
$$\delta\nu = -\frac{1}{2\hbar} \alpha_a \langle E_0^2 \rangle$$

36.2612(7)
kHz (V/cm)⁻²

Sherman et al. PRL **108**, 153002 (2012)

Blackbody

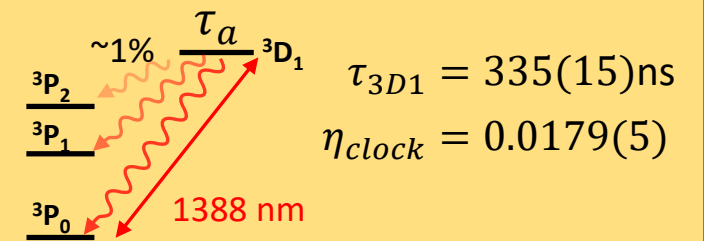
Uniform room temperature blackbody



Beloy et al. PRL **113**, 260801 (2014)

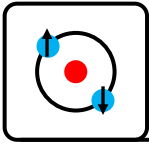
now 5×10^{-4} uncertainty

Dynamic correction



$$\langle 6s5d \ ^3D_1 \| D \| 6s6p \ ^3P_0 \rangle = 2.75(7) e a_0$$

Beloy et al. PRA **86**, 051404(R) (2012)

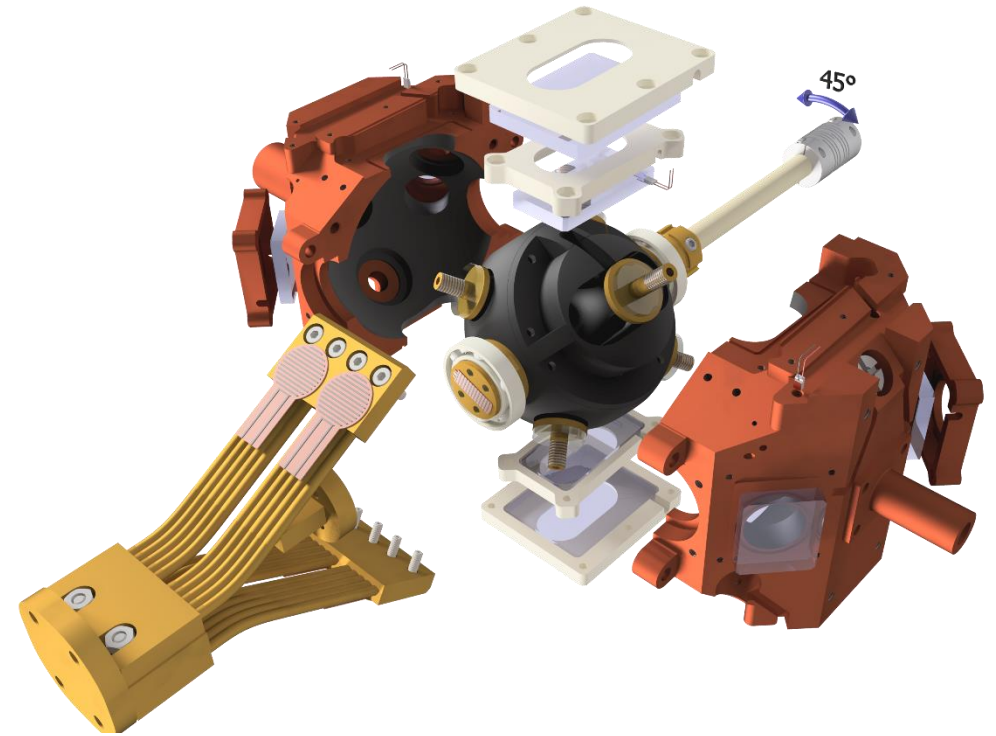


Next generation optical lattice clock – Cryogenic shielding

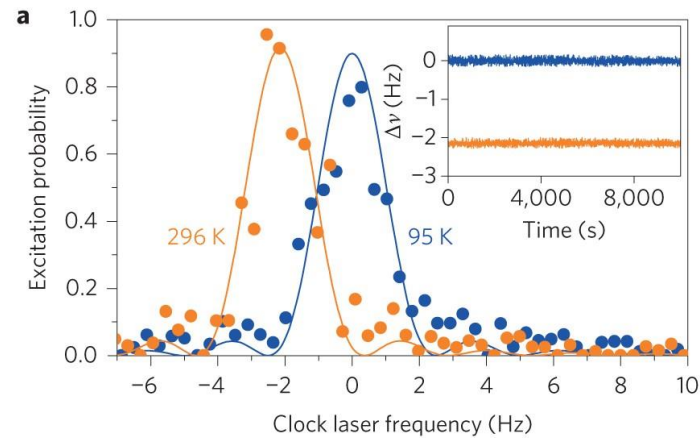
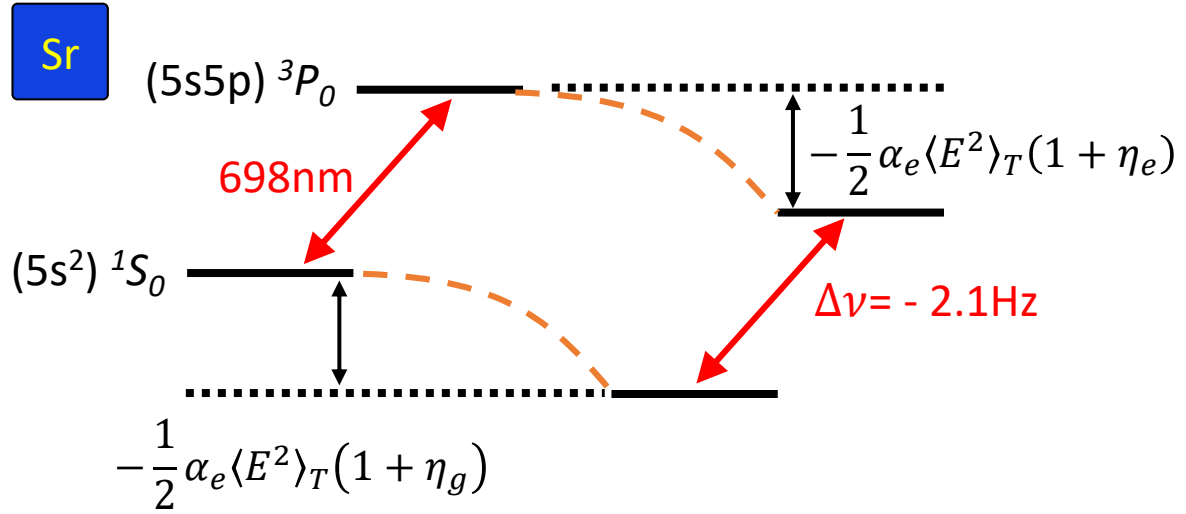
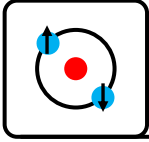
$$\Delta\nu_{BBR}(T) = -\frac{1}{2h} \Delta\alpha(0) \langle E^2 \rangle_T [1 + \eta_{clock}(T)]$$

Energy density of BBR:

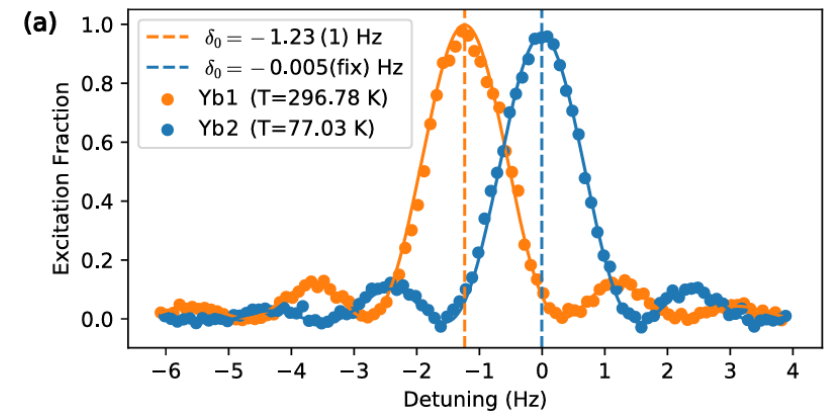
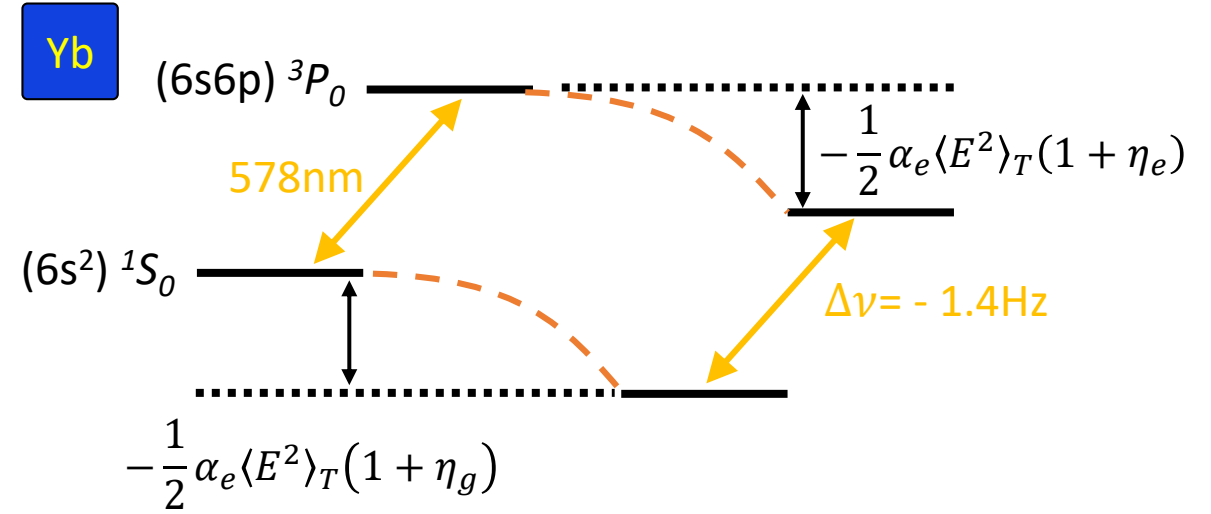
$$\begin{aligned} \langle E^2 \rangle &= \int_0^\infty u(\nu) d\nu = \frac{8\pi^5 k_B^4}{15c^3 \epsilon_0 h^3} T^4 = c_1 T^4 \\ &\approx (8.319 \text{ V/cm})^2 \left(\frac{T}{300 \text{ K}} \right)^4 \end{aligned}$$



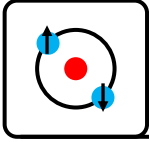
Cold thermal shield ~ 50 K



Ichiro Ushijima, ...Hidetoshi Katori, Nature Photonics 9, 185–189 (2015)

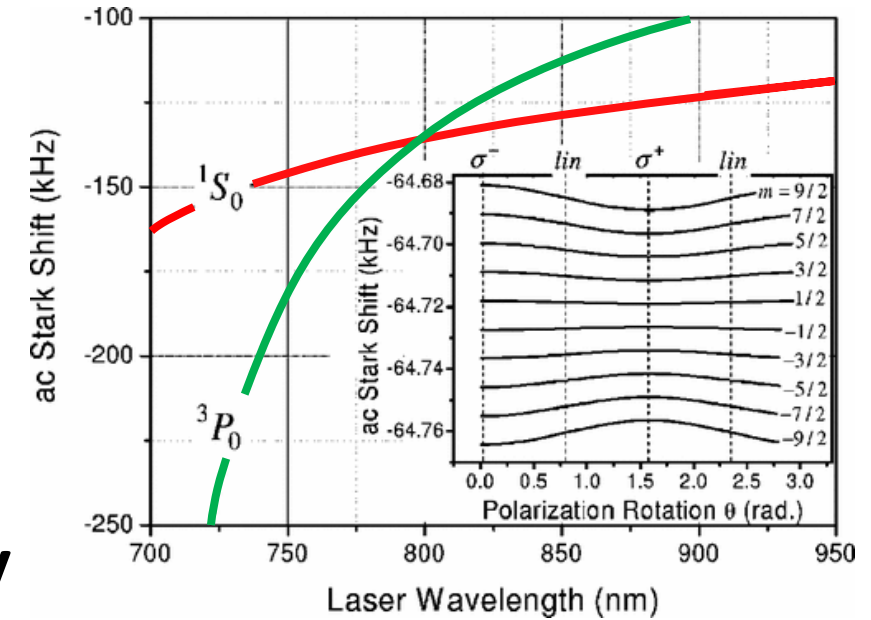


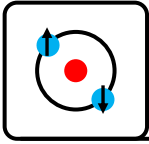
Youssef S. Hassan, ... Andrew D. Ludlow, Phys. Rev. Lett. 135, 063402, 2025



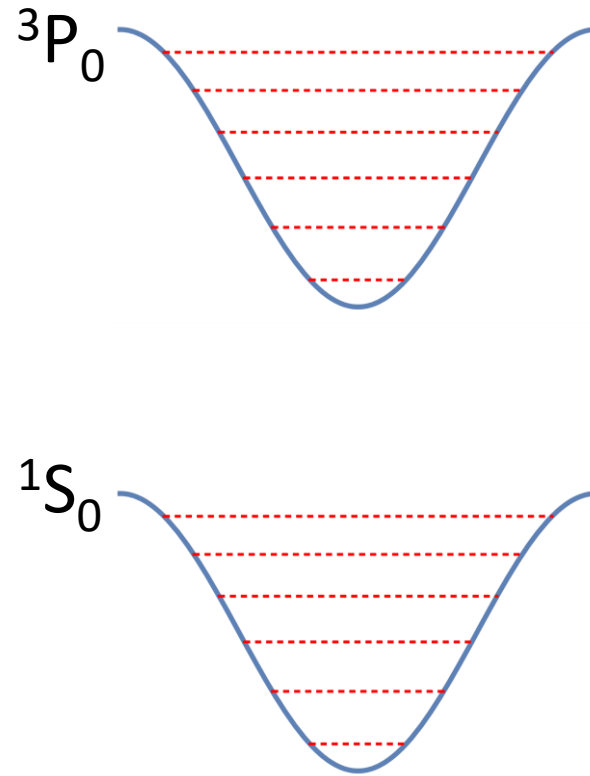
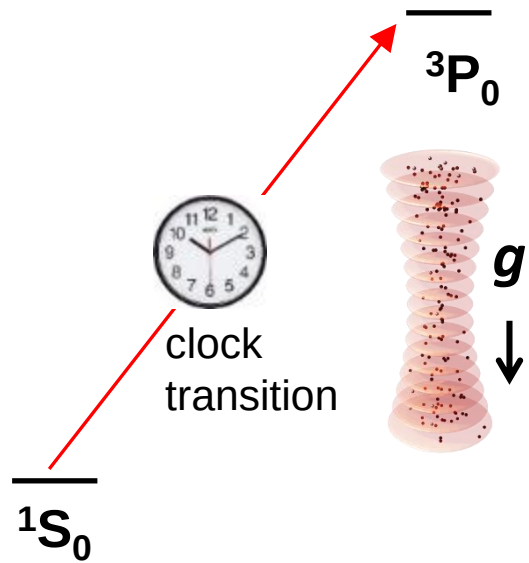
Systematic Uncertainty

Light shift – ac Stark shift





Lattice light shift

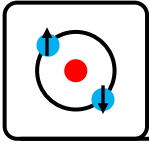


First approximation:

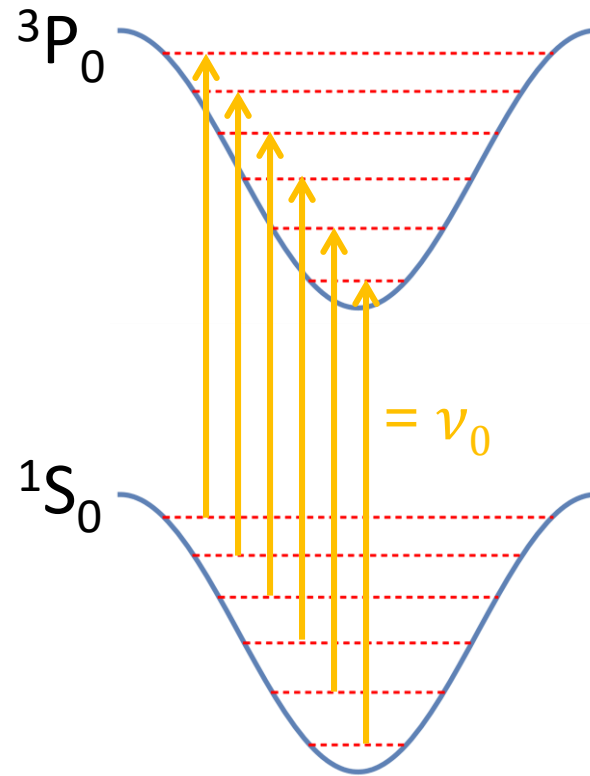
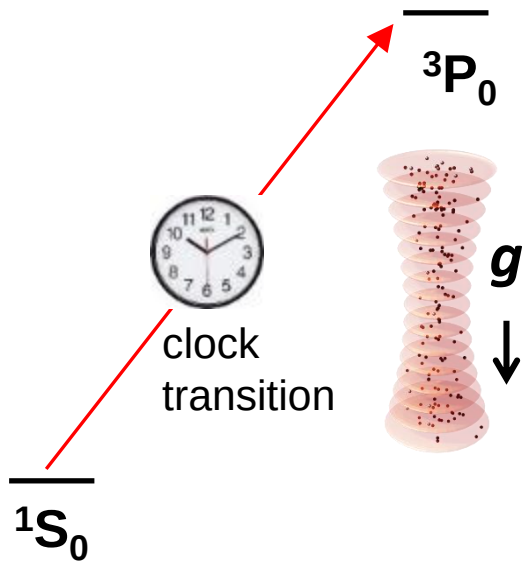
operate lattice at the
magic wavelength



potentials identical for
both clock states



Lattice light shift



First approximation:

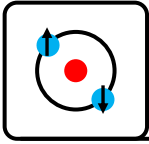
operate lattice at the
magic wavelength



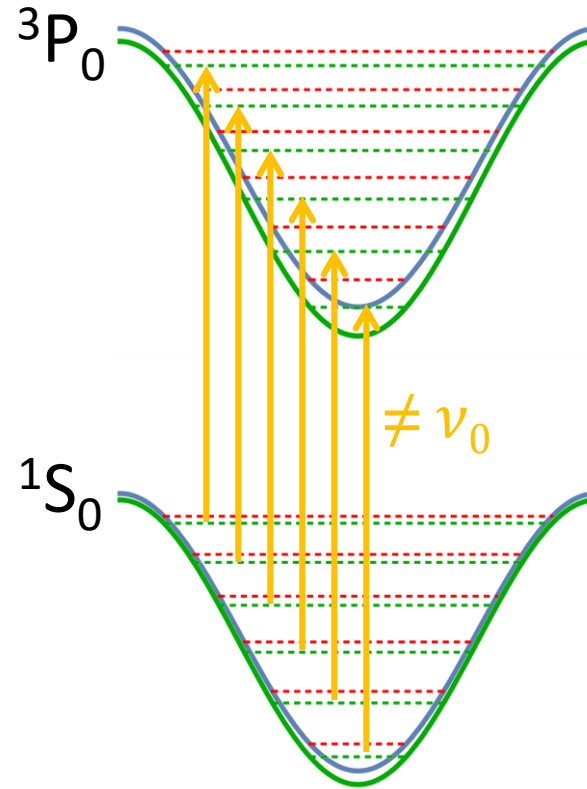
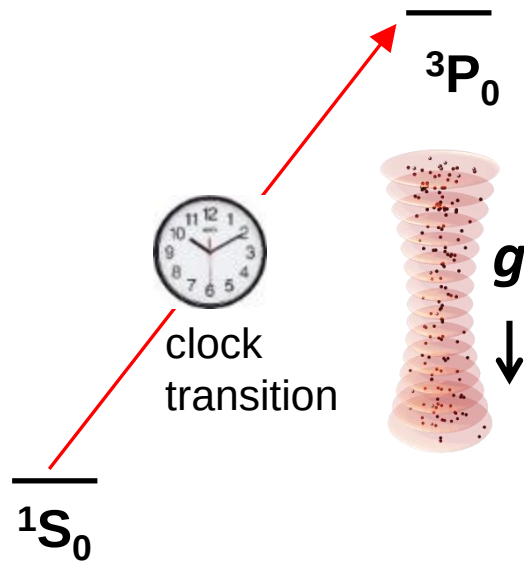
potentials identical for
both clock states



$$\delta\nu = 0$$



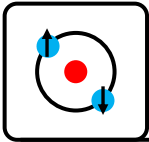
Lattice light shift



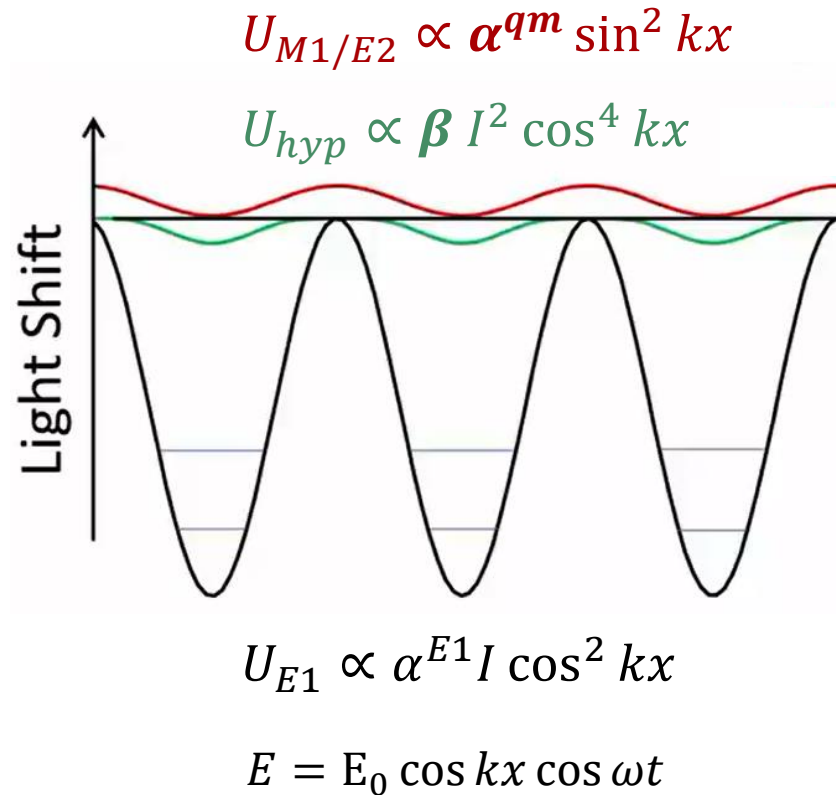
Non-magic effects:

- detuning from magic wavelength (uncancelled 2nd order $E1$)
- higher-multipolar (2nd order $M1$ & $E2$)
- hyperpolarizability (4th order $E1$)

$$\Delta\nu = \Delta\nu^{E1} + \Delta\nu^{M1/E2} + \Delta\nu^{hyper}$$



Light shift – higher order effects



Light shift model

$$U \approx \alpha^{E1} I_0$$

$$u = U/E_r$$

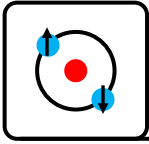
Precise knowledge on higher order polarizabilities
 $(\alpha^{qm}, \beta, \delta_L = \nu^{E1} - \nu)$ is important

$$h\Delta\nu_{LS}(u, \delta_L, n_z) \approx c_{1/2} u^{1/2} + c_1 u + c_{3/2} u^{3/2} + c_2 u^2$$

$$\begin{aligned}
 &\approx \left(\frac{\partial \tilde{\alpha}^{E1}}{\partial \nu} \delta_L - \tilde{\alpha}^{qm} \right) \left(n_z + \frac{1}{2} \right) u^{1/2} \\
 &- \left[\frac{\partial \tilde{\alpha}^{E1}}{\partial \nu} \delta_L + \frac{3}{2} \tilde{\beta} \left(n_z^2 + n_z + \frac{1}{2} \right) \right] u \\
 &\quad + 2 \tilde{\beta} \left(n_z + \frac{1}{2} \right) u^{3/2} \\
 &\quad - \tilde{\beta} u^2
 \end{aligned}$$

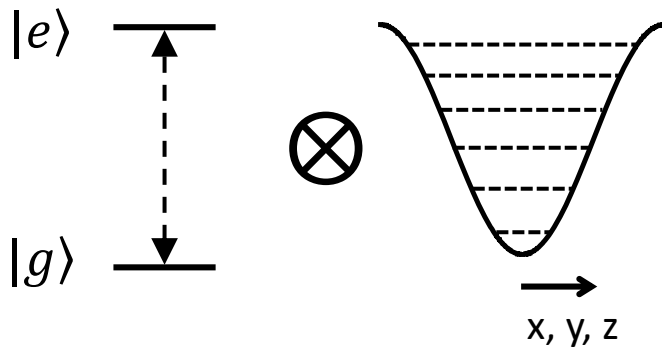
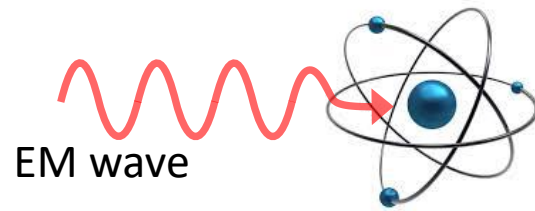
E2/M1 effect
 Hyperpolarizability effect

1. H. Katori et.al., Strategies for reducing the light shift in atomic clocks, PRA **91**, 052503 (2015).
2. R. C. Brown... Ludlow PRL **119**, 253001 (2017)
3. I. Ushijima .. H. Katori, Operational Magic Intensity for Sr Optical Lattice Clocks, PRL **121**, 263202 (2018).



Atomic frequency standards - Precision frontier

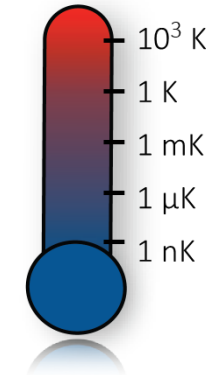
Atom + Trap

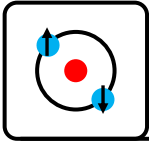


Precision frontier



Temperature



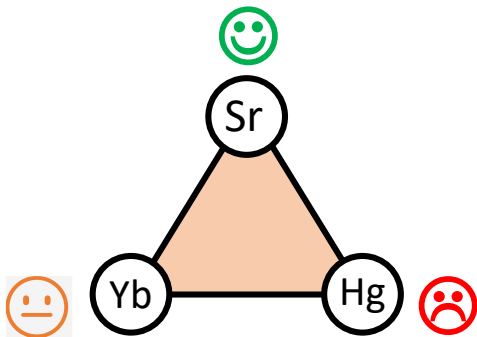
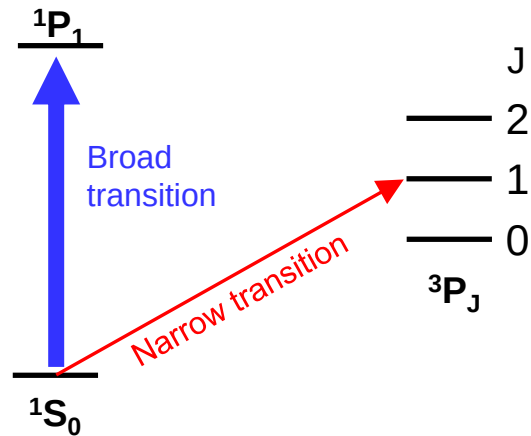
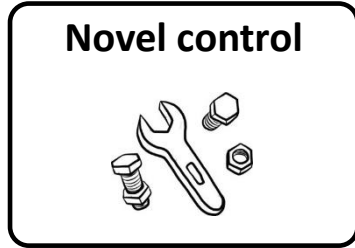


Towards the **optical second** : neutral atom clock

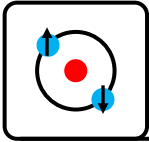


A diagram of a neutral atom. At the top, a circular arrow indicates electron spin. Below it, a central red dot represents the nucleus. Two blue arrows point away from the nucleus, representing the movement of electrons. Below the diagram is a vertical column of periodic table elements: H (1), Li (3), Na (11), K (19), Rb (37), Cs (55), Fr (87), Yb, Hg, and Cd.

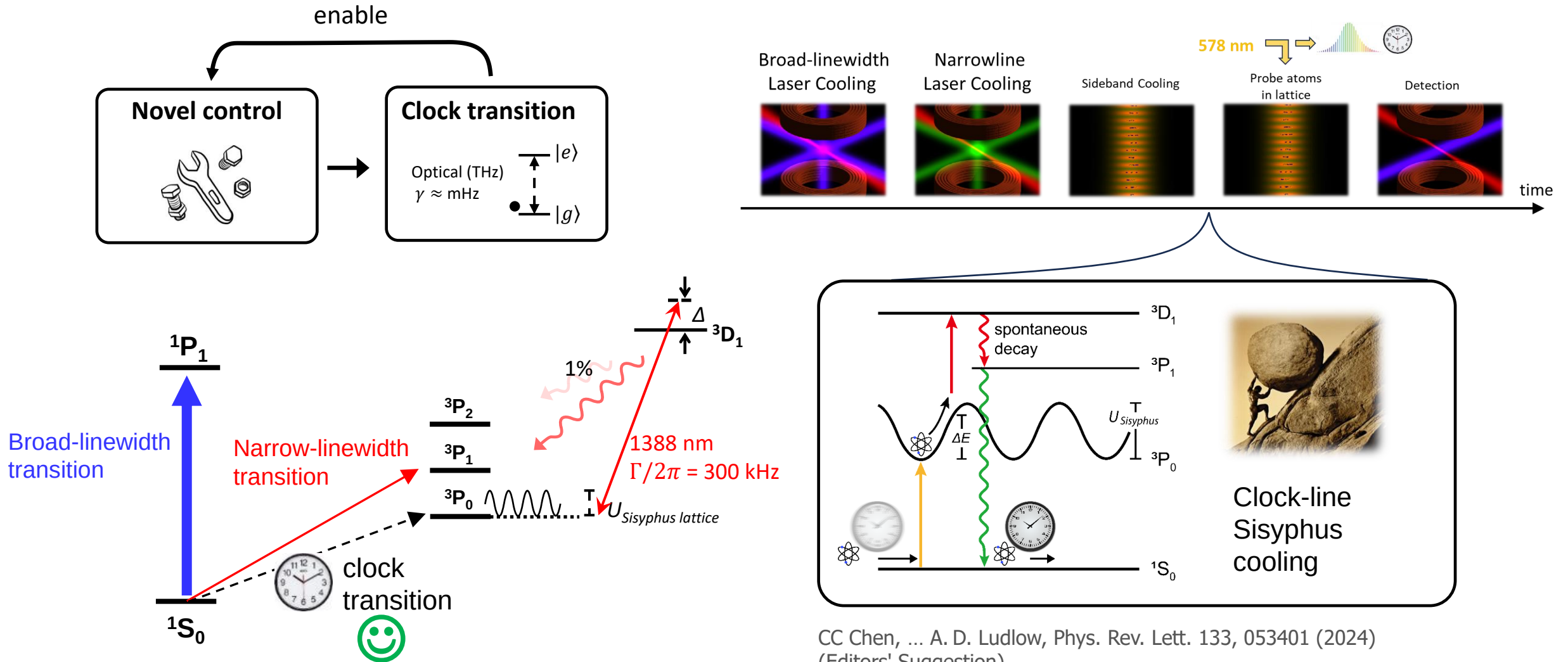
1	H	II	
3	Li	4	Be
11	Na	12	Mg
19	K	20	Ca
37	Rb	38	Sr
55	Cs	56	Ba
87	Fr	88	Ra
Yb			
Hg			
Cd			



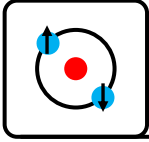
Atomic species	Sr	Yb	Mg	Hg	Cd
Nuclear spin	9/2	1/2	5/2	1/2	1/2
Clock λ (nm)	698	578	458	266	332
Magic λ (nm)	813	759	468	361	420
$1S_0 - 1P_1$ cooling λ (nm)	461	399	285	184 😞	229 😞
$1S_0 - 3P_1$ cooling λ (nm)	689	556	457	253	326
$\Gamma/2\pi$	7.4 kHz	182.3 kHz 😐	40 Hz 😞	1.3 MHz	67 kHz
$T_D/\mu K$	0.18 😊	4.37		31 😞	1.58



Universal cooling method for all optical lattice clocks

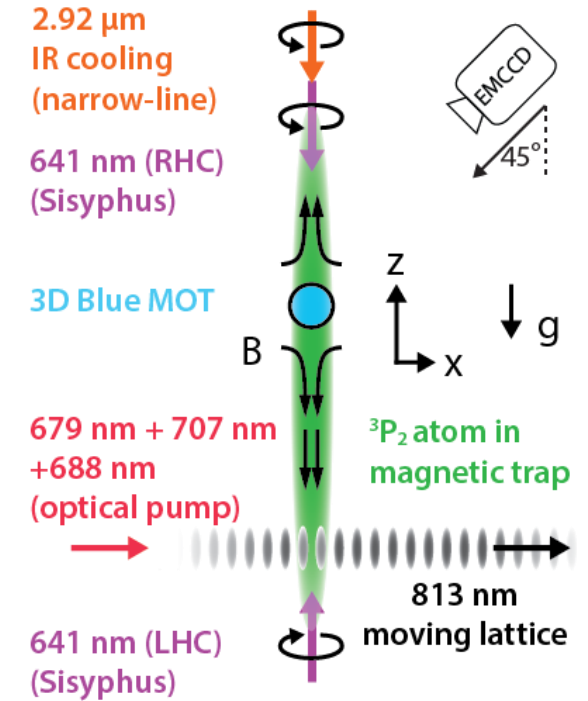


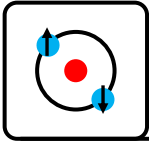
CC Chen, ... A. D. Ludlow, Phys. Rev. Lett. 133, 053401 (2024)
(Editors' Suggestion)
See also Atomic thermometry in optical lattice clocks
Irene Goti, ... Marco Pizzocaro, arXiv:2508.08164 (2025)



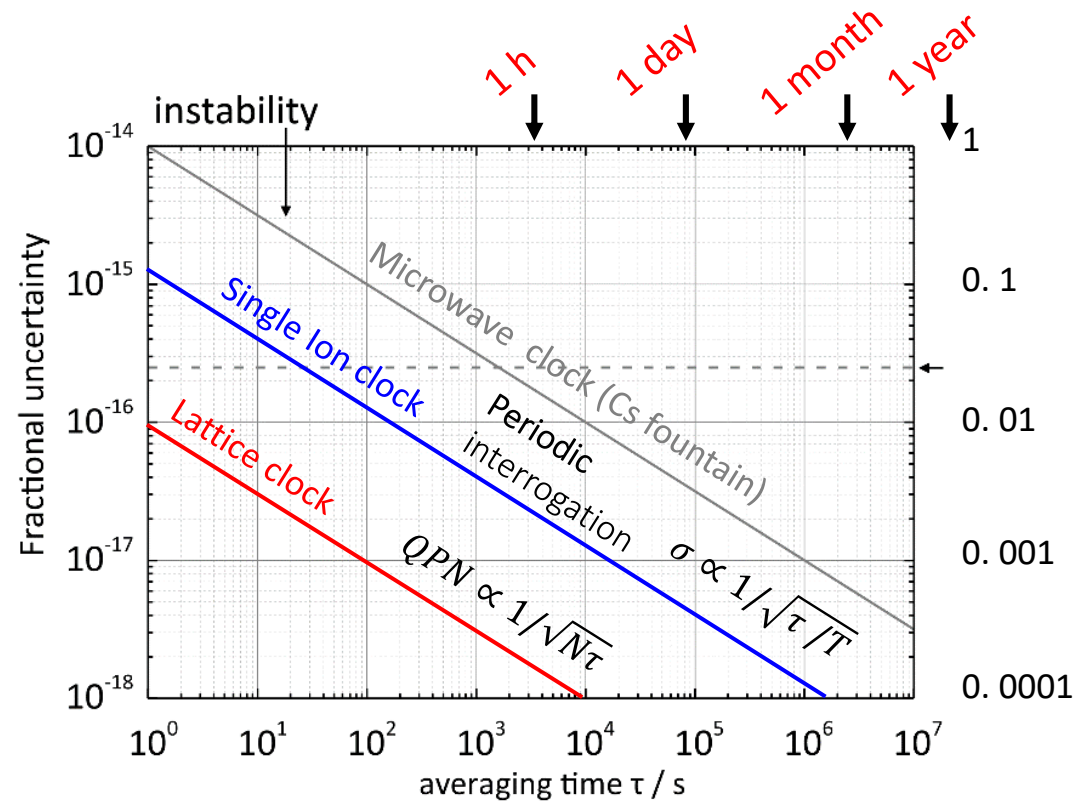
Next generation optical clock

CW- optical clock



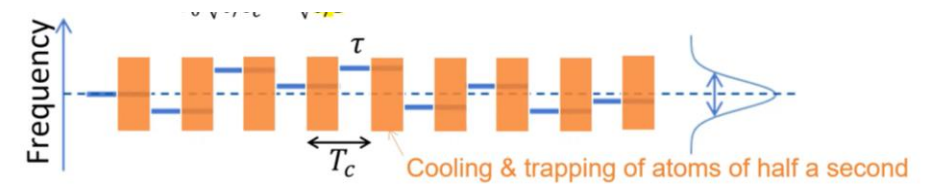


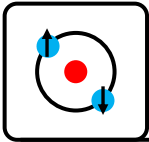
Optical clock: current state of the art



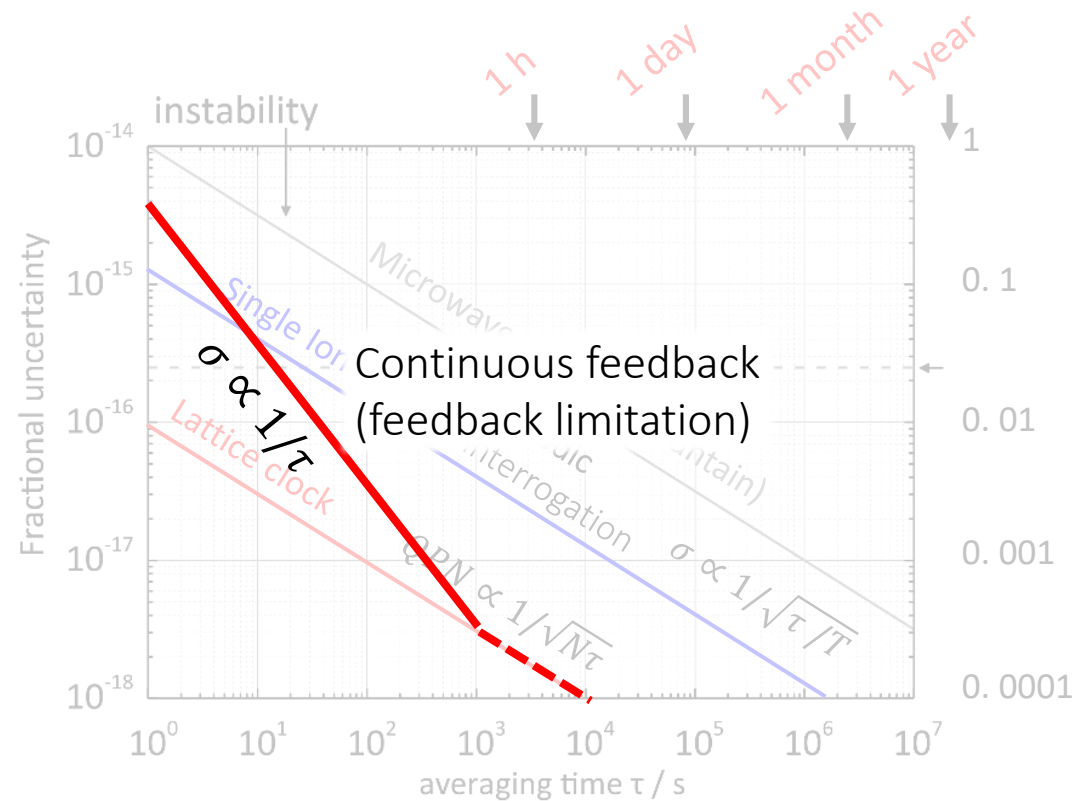
Periodic interrogation :

$$\sigma_{QPN}(\tau) \propto 1/\sqrt{\tau}$$



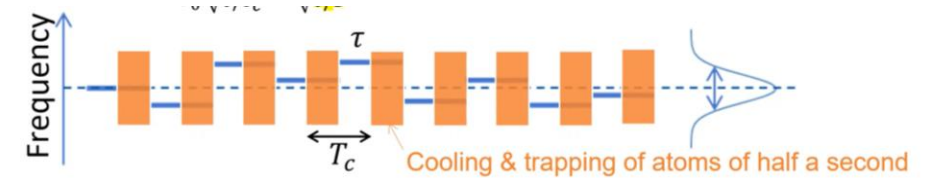


Continuous optical clock



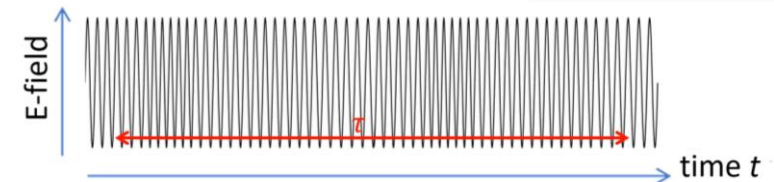
Periodic interrogation :

$$\sigma_{QPN}(\tau) \propto 1/\sqrt{\tau}$$



Continuous interrogation :

$$\sigma = \frac{\delta\nu}{\nu} \propto 1/\tau$$



Continuous measurement and feedback allows $\delta\nu \sim 1/\tau$

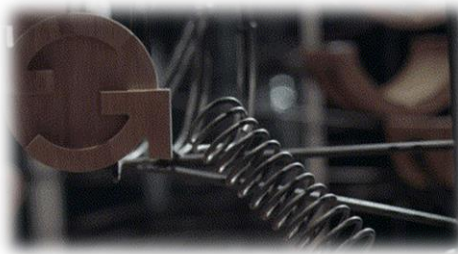
Step 1



Step 2



Step 3



Step 4

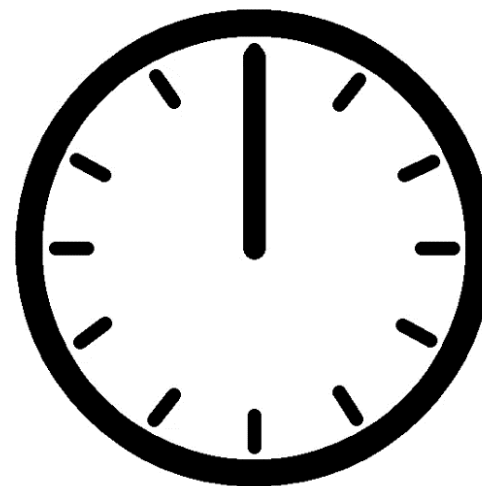
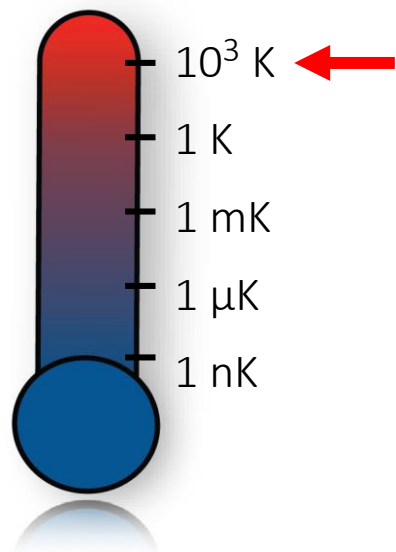


Step 5



time

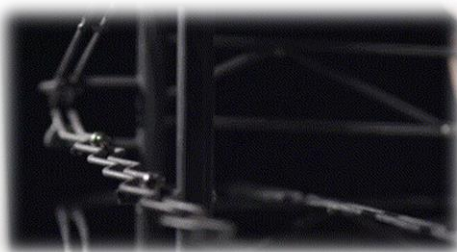
Temperature



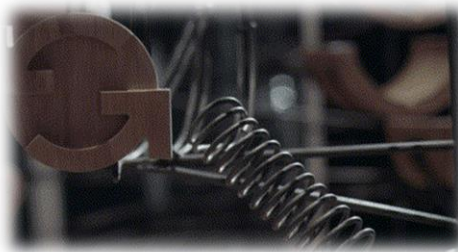
Step 1



Step 2



Step 3



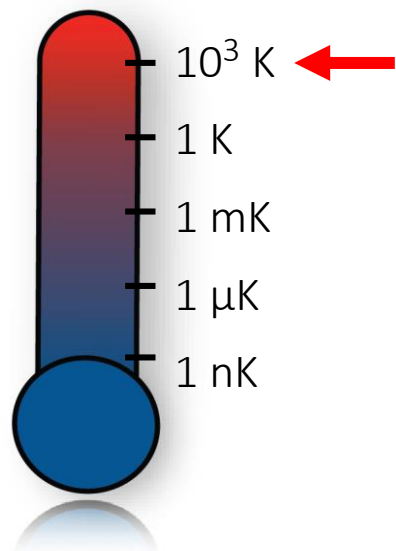
Step 4



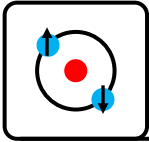
Step 5



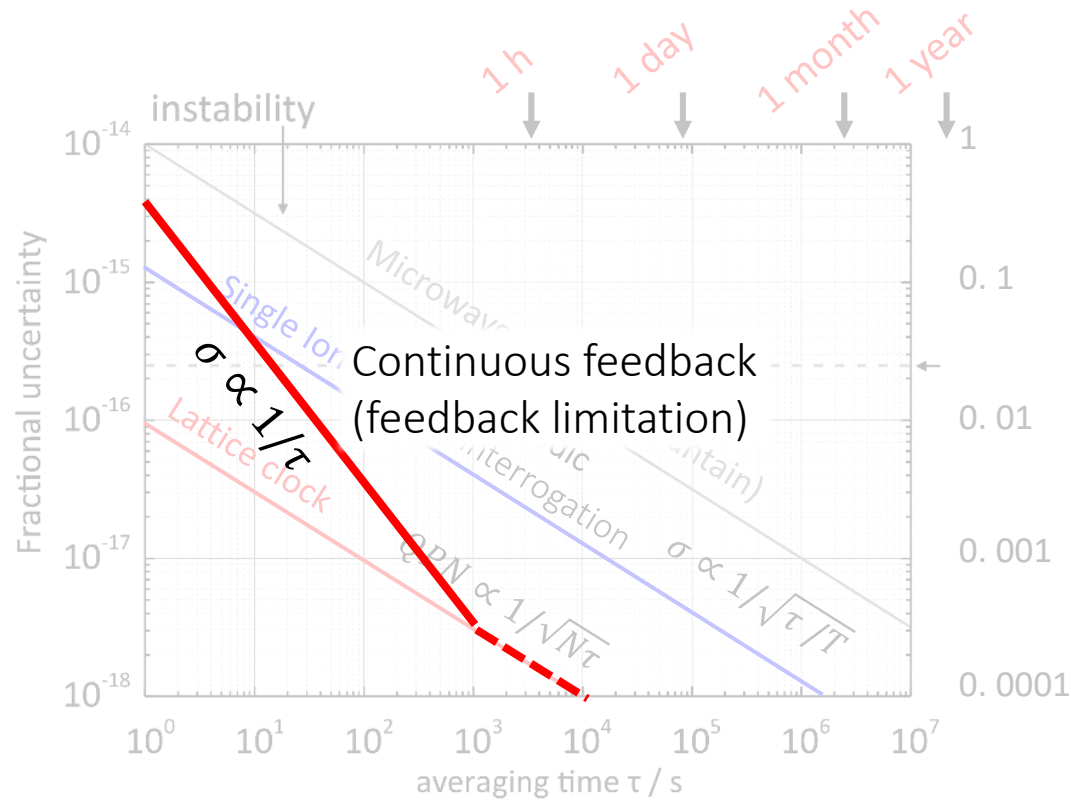
Temperature



time $\nearrow$ space

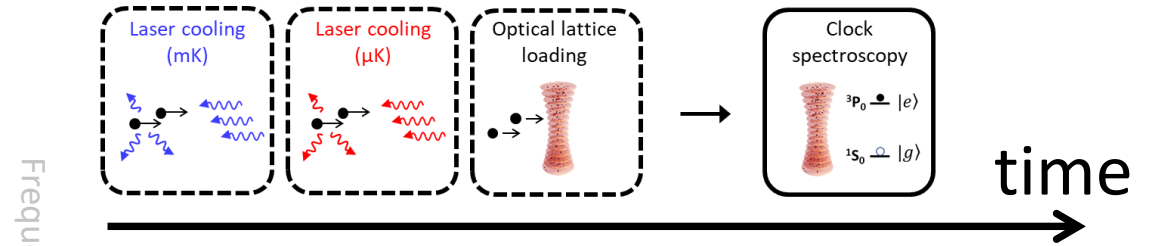


Continuous optical clock



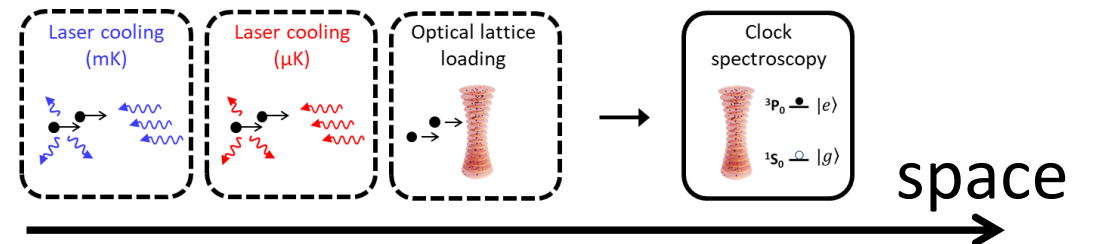
Periodic interrogation :

$$\sigma_{QPN}(\tau) \propto 1/\sqrt{\tau}$$



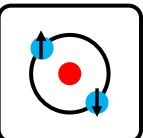
Continuous interrogation :

$$\sigma = \frac{\delta\nu}{\nu} \propto 1/\tau$$



Enabling technology

Continuous ultracold atomic beam loaded in a trap



$^3P_2 - ^3D_3$ Metastable state cooling

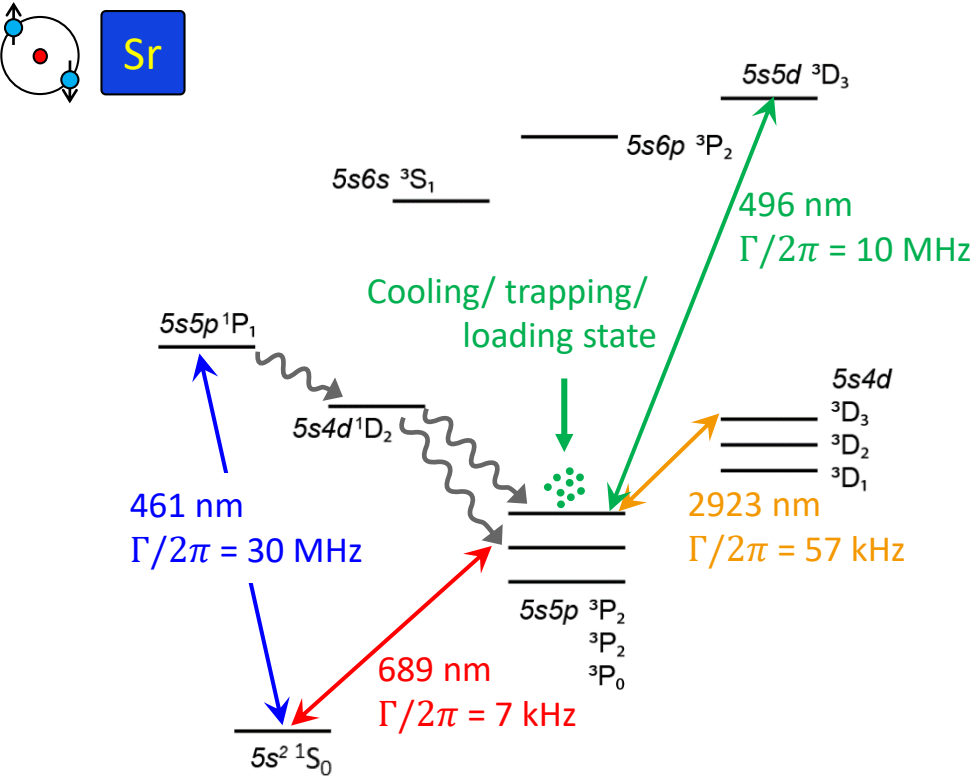


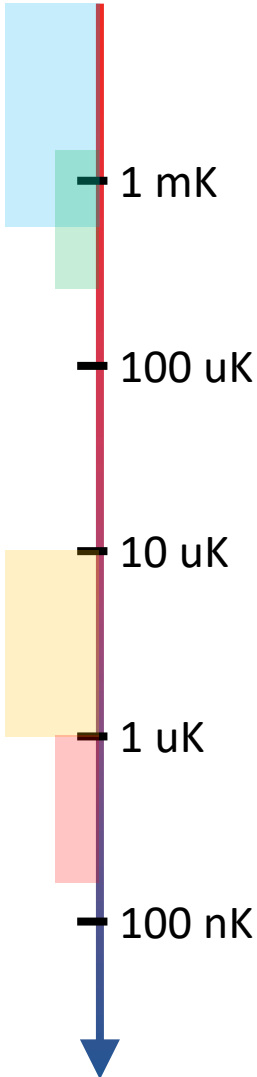
TABLE I. Properties of the cooling transitions of Sr atoms. λ is the transition wavelength, $\gamma_0/(2\pi)$ natural linewidth, I_0 saturation intensity, a_{max} maximum acceleration, T_D Doppler temperature, T_R recoil temperature, and $\epsilon_R = E_R/(\hbar\gamma_0)$ recoil parameter [35].

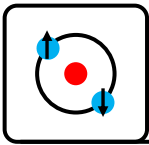
Label	‘Blue’	‘Red’	‘Green’	‘IR’
Lower state	$5s^2\ ^1S_0$	$5s^2\ ^1S_0$	$5s5p\ ^3P_2$	$5s5p\ ^3P_2$
Upper state	$5s5p\ ^1P_1$	$5s5p\ ^3P_1$	$5s5d\ ^3D_3$	$5s4d\ ^3D_3$
λ (nm)	461	689	496	2923
$\gamma_0/(2\pi)$ (Hz)	3.0×10^7	7.5×10^3	9.8×10^6	5.7×10^4
I_0 (mW/cm ²)	40	3.0×10^{-3}	10	3.0×10^{-4}
a_{max} (m/s ²)	9.8×10^5	150	2.8×10^5	270
T_D (μ K)	730	0.18	230	1.4
T_R (μ K)	0.51	0.23	0.44	0.013
ϵ_R	3.5×10^{-4}	0.64	9.4×10^{-4}	4.7×10^{-3}

Sub-Doppler : Green MOT, $\sim 200\ \mu\text{K}$
 IR cooling, $\sim 200\ \text{nK}$

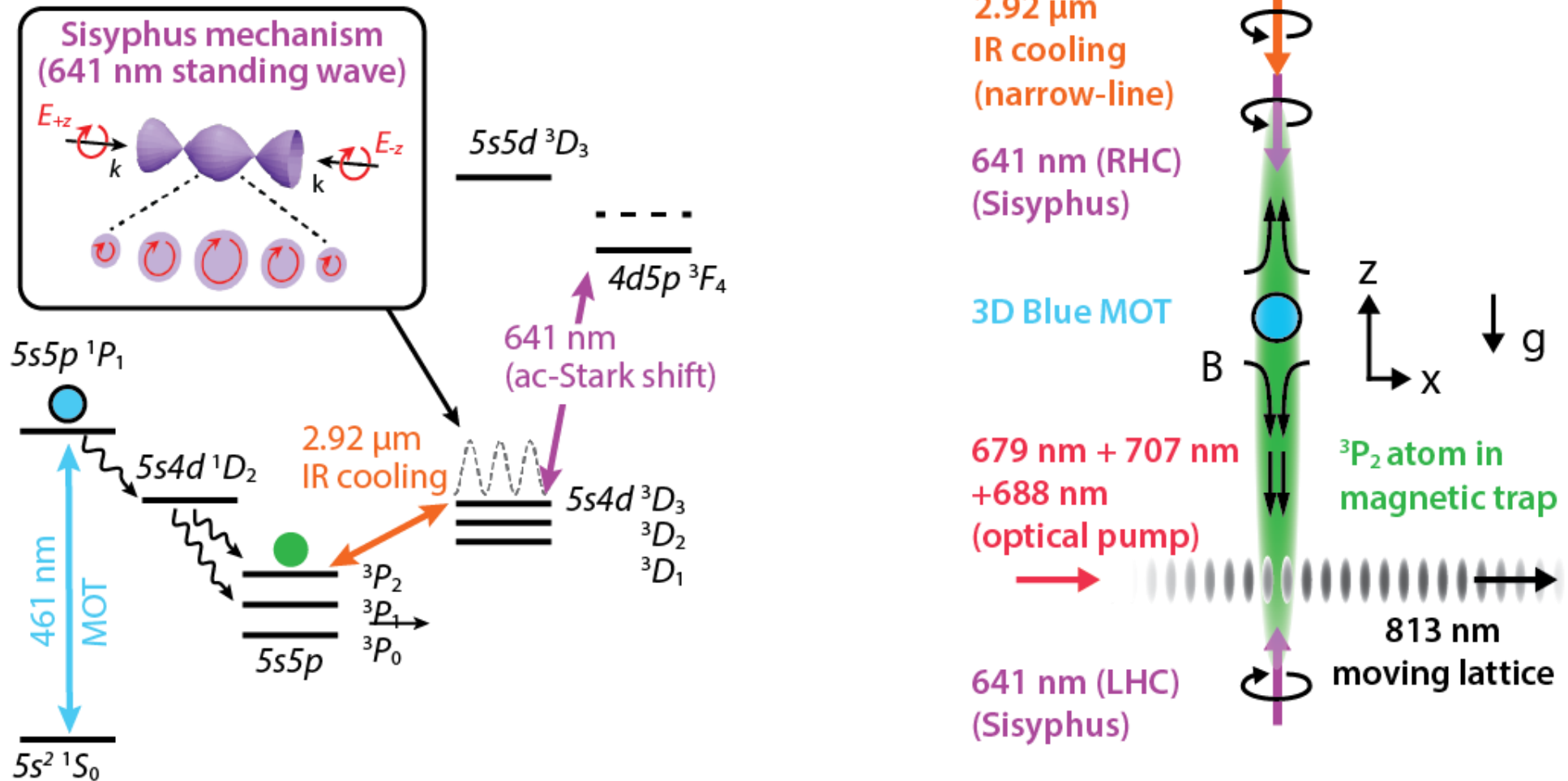
1. Katori et.al, AIP Conference Proceedings 551, 382–396 (2001)
2. Hashiguchi Katori et.al, PRA **100**, 042513 (2019)
3. R. Hobson ... P. Gill, PRA **101**, 013420 (2020)
4. Akatsuka Katori et.al, PRA **103**, 023331 (2021)

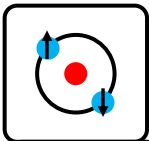
Temperature





Sisyphus cooling in 3P_2 metastable state





Sisyphus cooling in 3P_2 metastable state

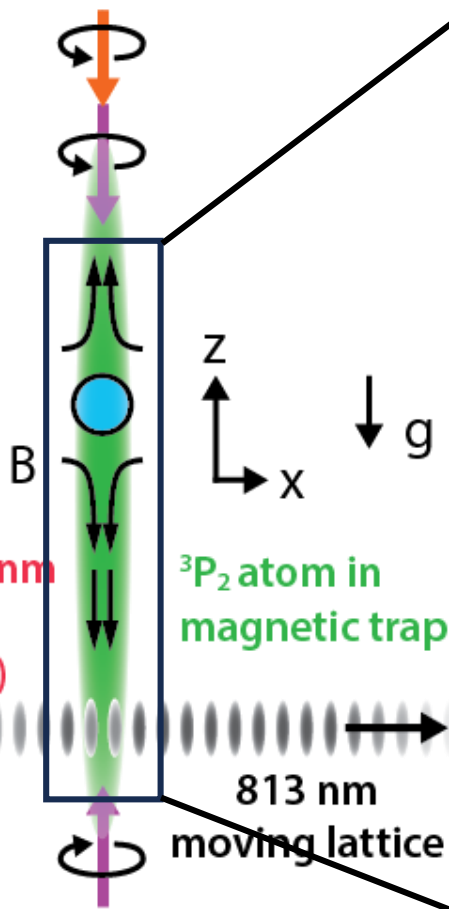
2.92 μm
IR cooling
(narrow-line)

641 nm (RHC)
(Sisyphus)

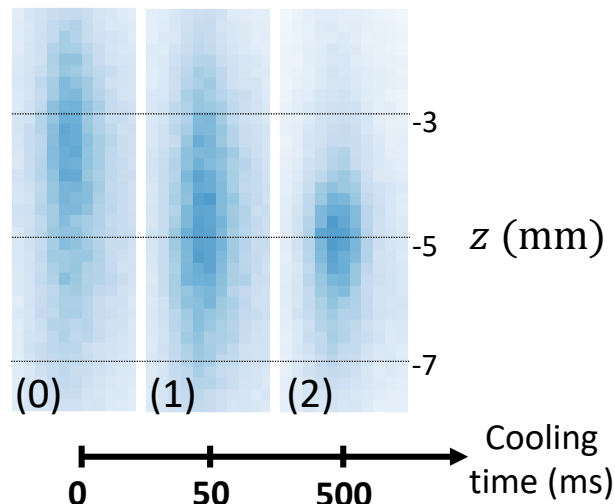
3D Blue MOT

679 nm + 707 nm
+ 688 nm
(optical pump)

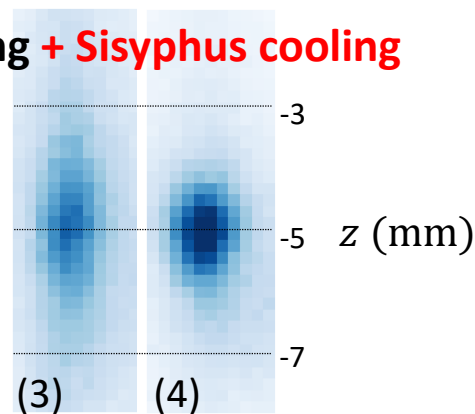
641 nm (LHC)
(Sisyphus)

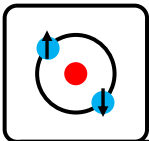


IR cooling

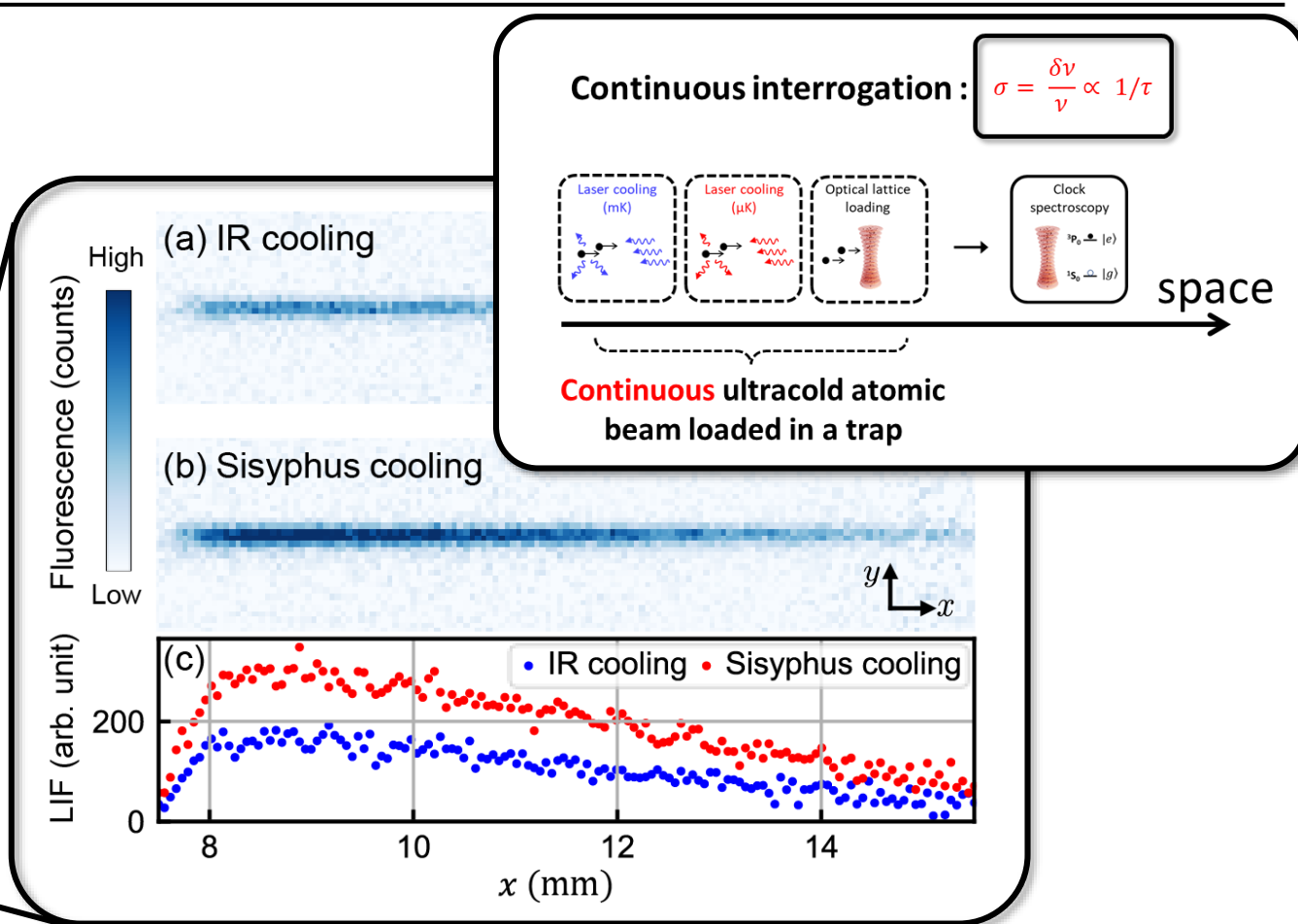
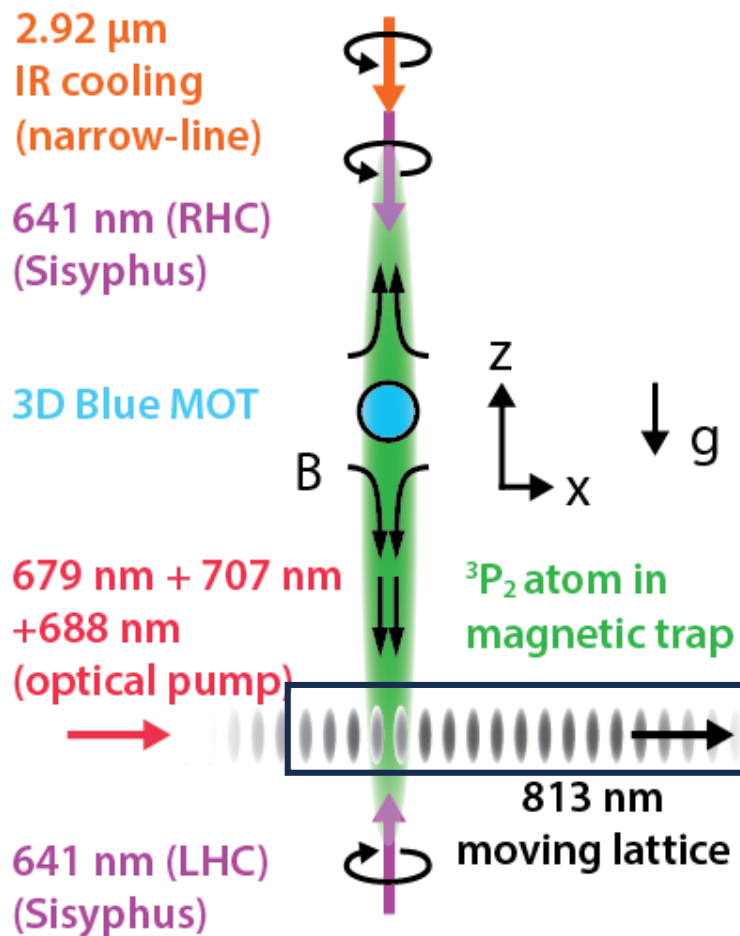


IR cooling + Sisyphus cooling



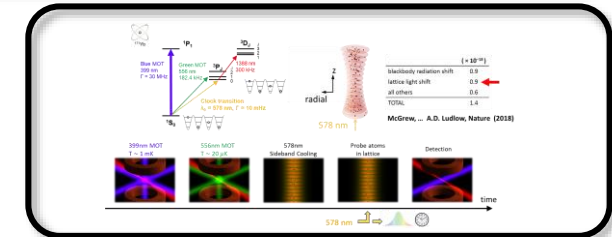
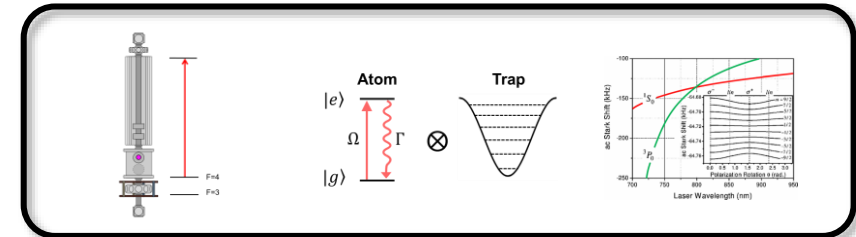
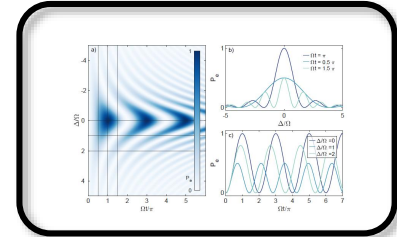
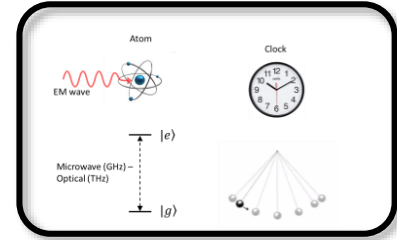


Enhanced Continuous magic-wavelength trap loading



Summary

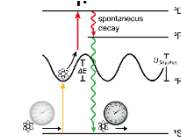
- Introduction
- Two-level system
 - Atom light interaction
 - Highly accurate two level systems – mHz forbidden optical transitions
- How optical atomic clock works:
 - Atomic frequency standards
 - Physics of optical atomic clocks – (optical lattice clock)
- Current research directions
 - How to improve state of the art atomic clocks (optical lattice clock) ?
 - Next generation optical clocks ?



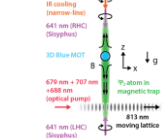
Cryogenic clock



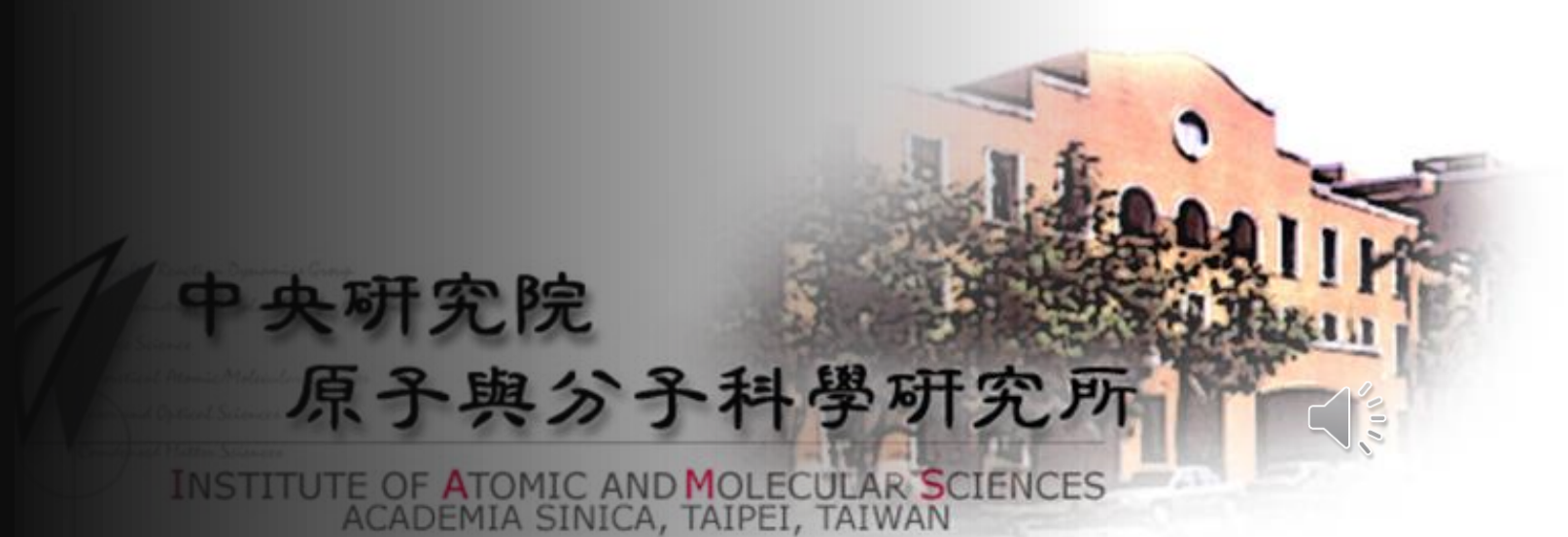
Clock-line Sisyphus

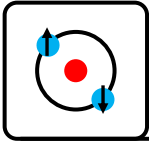


Continuous - Clock

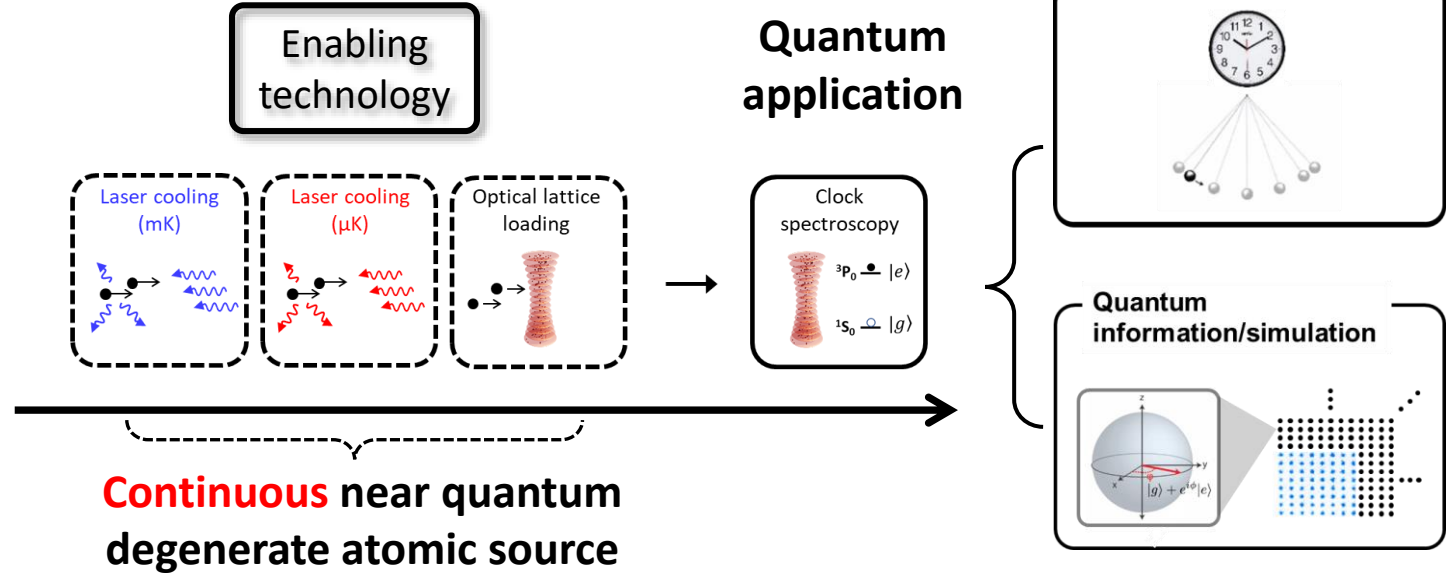
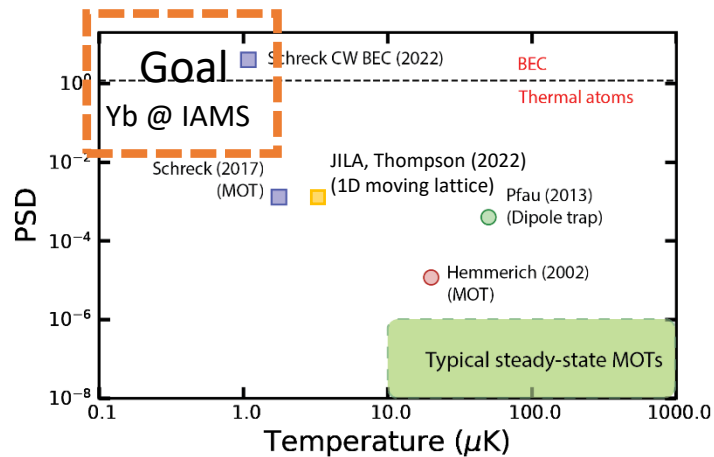
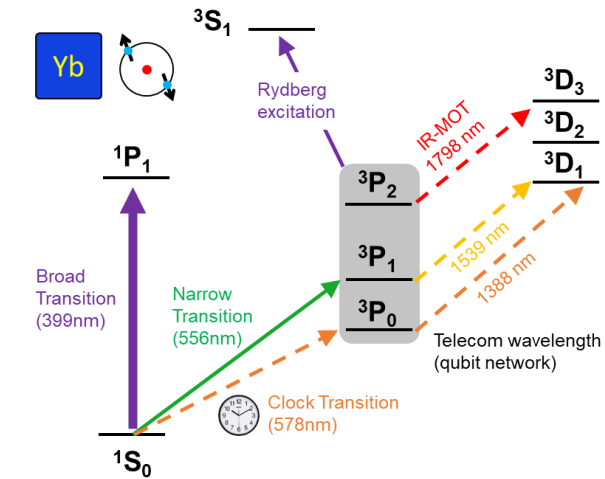


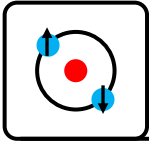
Current efforts and outlook at IAMS





Towards **Continuous Yb Optical clock** and **large array quantum registers**





徵才



中研院(台大校本部) 原子與分子科學所 原子-奈米光子複合量子技術實驗室 / 量子精密感測實驗室 誠徵博士生，碩士生，大學部專題生，研究助理

我們的團隊專注於為中性原子（鹼土金屬原子）開發創新的原子控制技術，目標是實現下一代代的量子科技應用，例如量子模擬與量子資訊處理的實驗平台與次世代超精密原子鐘

聯絡資訊：

聯絡人：陳俊嘉 博士 / 肖恩 博士

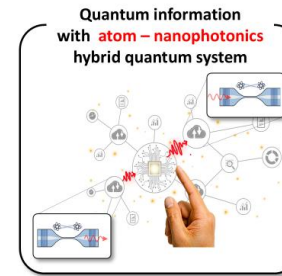
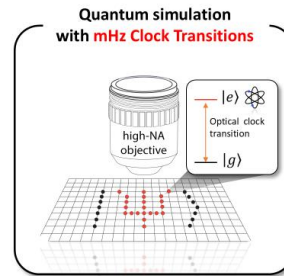
地址：中央研究院 原子與分子科學研究所110實驗室（台大校園內）

Email：chunchia.chen@g.iam.s.sinica.edu.tw / s.p.bennetts@g.iam.s.sinica.edu.tw

Website：https://iamsquantum.github.io/



SCAN ME



實驗工作內容：（許多工作內容具高度研究性，未來可銜接碩士或博士學位進修發展）

- 參與中性原子量子控制實驗與高穩定度雷射技術的開發
- 設計搭建原子光譜光學架設及雷射穩頻系統
- 搭建超高真空（UHV）實驗系統
- 開發控制系統與資料擷取系統（包含電子電路設計，電子儀器製造）
- 奈米光纖製備與測試

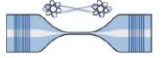
應徵資格：

1. 對於實驗物理技術開發與量子科技應用具高度興趣與研究熱忱
2. 對動手做、解決問題有興趣，具團隊合作精神、積極主動之工作態度
3. 歡迎有志留學或攻讀碩士學位者加入，累積實驗經驗與研究實績

工作地點：中央研究院原子與分子科學研究所 Lab 110（位於台大校園內）



Quantum Metrology Lab



Hybrid Atom-Nanophotonics Lab

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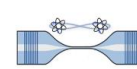
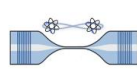
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